

THE EYE

VOLUME 6

Infrastructure Architecture

The deployment, compute, network, state, runtime, trust, reliability, continuity, delivery, and operating architecture for The Eye

OBSERVE / UNDERSTAND / PREDICT / SIMULATE / DECIDE / LEARN

CONTROLLED INFRASTRUCTURE ARCHITECTURE BASELINE / V1.0

Inherited from Volume 0 - Product Constitution, Volume 3 - Technical Architecture, Volume 4 - Engineering Specification, and Volume 5 - AI Architecture

1 August 2026

Document Control

Volume 6 is the controlled infrastructure architecture of The Eye. It governs how the frozen Strategic Intelligence Operating System is deployed, isolated, accelerated, connected, persisted, secured, observed, recovered, changed, accepted, and retired.

CONTROL	VALUE
Product	The Eye
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Status	Controlled Infrastructure Architecture Baseline
Binding sources	Volume 0 - Product Constitution v1.0; Volume 3 - Technical Architecture v1.0; Volume 4 - Engineering Specification v1.0; Volume 5 - AI Architecture v1.0
Primary audience	CTO, CIO, CISO, principal infrastructure and platform architects, SRE, security engineering, data and AI platform engineering, cloud and on-premise engineering, operations, procurement, governance, and customer technical authorities
Normative scope	Deployment profiles; topology; compute; accelerators; network; storage; messaging; runtime platforms; identity; policy; cryptography; sovereignty; observability; SLOs; continuity; delivery; fleet; economics; acceptance; governance
Downstream inheritance	Volume 7 - Data Platform; environment designs; landing zones; infrastructure modules; workload manifests; operational readiness plans; disaster-recovery plans; runbooks; implementation repositories

Normative language

SHALL, MUST, SHALL NOT, and MUST NOT are binding. MAY permits an implementation choice inside a frozen semantic, authority, state, isolation, failure, and evidence boundary. Requirements carry stable IA-CC-NNN identities. Conformance is scoped to exact artifacts, configuration, topology, customer domain, deployment profile, workload, consequence class, and period.

CONTROLLING RULE

Volume 0 controls constitutional meaning. Volume 3 controls product architecture. Volume 4 controls engineering realization. Volume 5 controls AI architecture. Volume 6 controls infrastructure beneath those authorities. Infrastructure may host and enforce the system; it may not redefine truth, human authority, product semantics, or canonical ownership.

CONSTITUTIONAL INHERITANCE / C-003 · C-004 · C-008 · C-036 · C-049 · C-051 · C-052

Infrastructure Architecture Charter

The Eye runs as one governed strategic intelligence system across managed cloud, customer cloud, customer-owned infrastructure, disconnected environments, and certified air gaps. Infrastructure exists to preserve semantic parity, contain failure, enforce trust, protect customer control, and produce operating evidence.

INFRASTRUCTURE COMMITMENT	BINDING ARCHITECTURE CONSEQUENCE
One product across deployments	SaaS, Private Cloud, and On-Premise preserve the same objects, workflows, authority, evidence, degraded states, replay, and lifecycle semantics.
Cell-oriented containment	Tenant, domain, release, capacity, and failure scope are bounded through intelligence cells and explicit shared services.
State before technology	Every store, queue, cache, index, checkpoint, and backup implements a declared authority, consistency, locality, durability, and recovery class.
Trust outside workloads	Identity, policy, cryptography, capabilities, network, artifact integrity, admission, audit, and stop controls do not depend on application or model compliance.
Failure is designed state	Partial, stale, denied, isolated, recovering, unreconciled, and terminal conditions are explicit, observable, and testable.
Recovery preserves meaning	Failover and disaster recovery preserve tenant, time, policy, evidence, sovereignty, human authority, and version semantics.
Change is controlled execution	Infrastructure plans, approvals, cohorts, observations, stop, rollback, effect reconciliation, and closure are durable evidence.
Customers retain control	Hosting choice cannot silently transfer responsibility, administrator authority, key control, residency, incident ownership, or acceptance.

Definition of infrastructure architecture complete

- Every component, runtime unit, state class, network zone, workload, dependency, failure domain, control, SLO, degraded mode, recovery path, release, and owner has stable identity.
- Every deployment profile can install, operate, diagnose, secure, fail, restore, upgrade, export, and retire the same strategic product contract.
- Every workload runs with authenticated identity, policy, resource, placement, network, state, secret, artifact, telemetry, failure, and lifecycle boundaries.
- Every critical service demonstrates capacity, availability, backup, restore, disaster recovery, incident, change, and security evidence for its exact released scope.
- No successful process, healthy host, reachable endpoint, or available model is represented as strategic success when evidence, authority, state, policy, or product quality is incomplete.

CONSTITUTIONAL INHERITANCE / C-001 · C-004 · C-007 · C-008 · C-035 · C-036 · C-041 · C-049 · C-052

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Native Word heading styles provide navigation. Chapter, requirement, component, runtime, network, state, control, failure, metric, recovery, and decision identifiers are stable references for architecture, engineering, assurance, operations, customers, and downstream volumes.

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Executive Infrastructure Architecture Statement

Volume 6 defines the physical and operational system that allows The Eye to remain one Strategic Intelligence Operating System across hyperscale managed service, customer-controlled cloud, customer-owned infrastructure, disconnected sites, and certified air gaps.

The infrastructure is organized around isolated intelligence cells, canonical state and evidence services, shared control capabilities, governed compute and acceleration, explicit network and state contracts, externally enforced trust, and evidence-driven operations. Technology and provider selections remain replaceable behind stable infrastructure component, runtime unit, state class, workload, and deployment-profile contracts. Human authority, temporal truth, provenance, uncertainty, sovereignty, correction, replay, and strategic meaning do not move with the hosting model.

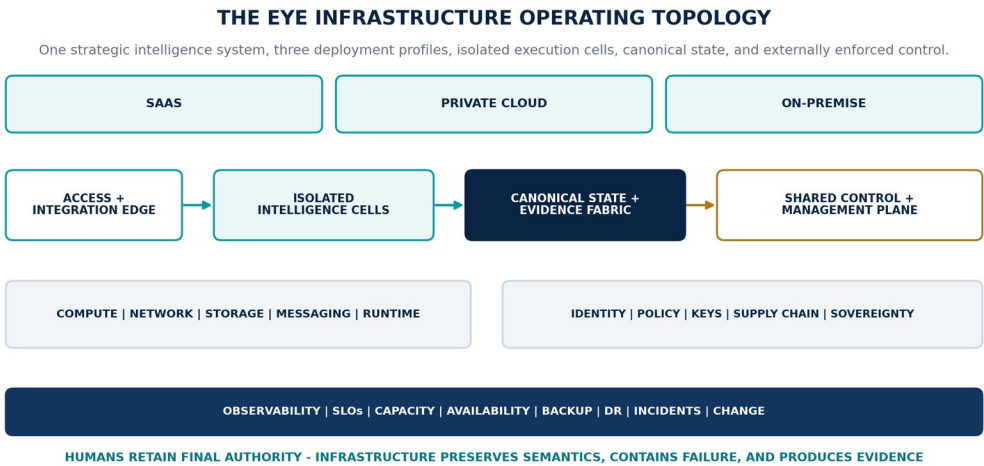


Figure 1. The Eye infrastructure operating topology. SaaS, Private Cloud, and On-Premise implement the same strategic intelligence semantics through access and integration edges, isolated intelligence cells, canonical state and evidence, shared control services, governed compute, network, storage, runtime, trust, observability, continuity, and human authority.

INFRASTRUCTURE PLANE	ARCHITECTURE OBLIGATION
Deployment and topology plane	Realize certified SaaS, Private Cloud, On-Premise, disconnected, and air-gapped profiles through regions, zones, sites, cells, responsibility boundaries, and management-plane relationships.
Compute and network plane	Schedule CPU, accelerators, edge runtimes, service connectivity, ingress, egress, private links, naming, and time under isolation, capacity, and sovereignty constraints.
State and messaging plane	Implement authoritative, evidence, graph, twin, index, queue, log, cache, checkpoint, backup, and recovery classes with declared semantics.
Platform runtime plane	Host services, workflows, agents, models, retrieval, pipelines, connectors, executive experiences, and contracts as identity-bound governed workloads.
Trust and sovereignty plane	Enforce identity, policy, cryptography, secrets, confidential processing, isolation, artifact integrity, detection, response, residency, and customer control.
Reliability and delivery plane	Measure service levels and capacity; contain failure; restore state; command incidents; govern change, release, fleet, economics, acceptance, and architecture.

INFRASTRUCTURE ARCHITECTURE THESIS

Infrastructure quality is not machine uptime. It is the demonstrated ability to preserve strategic meaning, authority, evidence, isolation, protection, continuity, customer control, and accountable recovery through normal operation, overload, failure, attack, disconnection, change, and retirement.

PART I

Infrastructure Authority

Normative authority, conformance, semantic parity, responsibility, failure domains, and traceability.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

1 Infrastructure Architecture Authority and Scope

Infrastructure Architecture Authority and Scope establishes the infrastructure necessary to translate the frozen constitutional, technical, engineering, and AI baselines into deployable infrastructure without changing product meaning. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

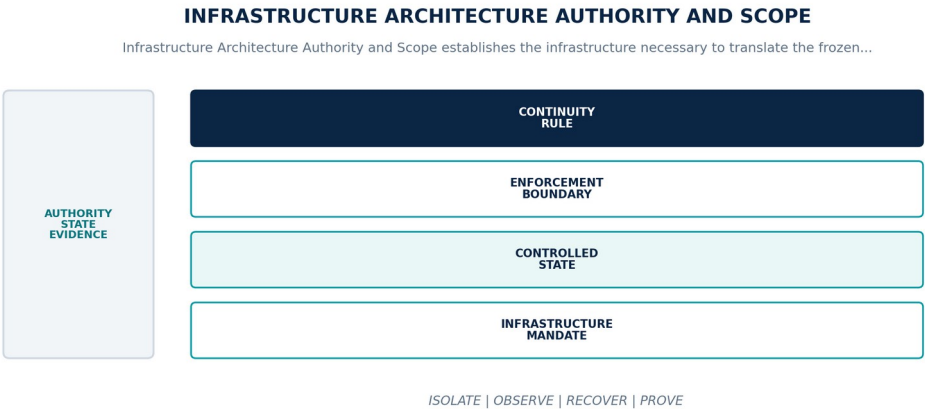


Figure 2. Infrastructure Architecture Authority and Scope. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Translate the frozen constitutional, technical, engineering, and AI baselines into deployable infrastructure without changing product meaning.
Controlled state	Infrastructure component identity, deployment profile, owner, lifecycle state, control evidence, and conformance disposition.
Enforcement boundary	Infrastructure may host canonical capabilities but may not own strategic truth, policy meaning, final decisions, or an eleventh product layer.
Continuity rule	When a workload, platform service, or deployment cannot resolve to an approved infrastructure contract, the platform shall deny registration or promotion, preserve the non-conformant state, and route the conflict to architecture governance.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-01-001	The infrastructure realization of Infrastructure Architecture Authority and Scope SHALL translate the frozen constitutional, technical, engineering, and AI baselines into deployable infrastructure without changing product meaning.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-01-002	Every production instance SHALL declare and continuously reconcile infrastructure component identity, deployment profile, owner, lifecycle state, control evidence, and conformance disposition through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-01-003	The implementation SHALL enforce this boundary: infrastructure may host canonical capabilities but may not own strategic truth, policy meaning, final decisions, or an eleventh product layer.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-01-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Infrastructure Architecture Authority and Scope.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-01-005	When a workload, platform service, or deployment cannot resolve to an approved infrastructure contract, the platform SHALL deny registration or promotion, preserve the non-conformant state, and route the conflict to architecture governance; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-01-006	Production admission and continued operation SHALL require complete component inventory, authority trace, cross-deployment conformance, and signed operational acceptance, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A workload, platform service, or deployment cannot resolve to an approved infrastructure contract	Deny registration or promotion, preserve the non-conformant state, and route the conflict to architecture governance.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared infrastructure component identity, deployment profile, owner, lifecycle state, control evidence, and conformance disposition.
- Positive and negative boundary tests prove that infrastructure may host canonical capabilities but may not own strategic truth, policy meaning, final decisions, or an eleventh product layer.
- Fault exercises inject a workload, platform service, or deployment cannot resolve to an approved infrastructure contract and verify that the platform can deny registration or promotion, preserve the non-conformant state, and route the conflict to architecture governance.
- Independent review accepts complete component inventory, authority trace, cross-deployment conformance, and signed operational acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Infrastructure Architecture Authority and Scope is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 1, Ch. 4, Ch. 6, Ch. 7, Ch. 72. Primary Volume 5 AI anchors: Ch. 1, Ch. 2, Ch. 6, Ch. 71, Ch. 72.

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2 Specification Hierarchy and Conformance

Specification Hierarchy and Conformance establishes the infrastructure necessary to enforce the control order from Volume 0 through infrastructure manifests, implementation, verification, and operational evidence. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

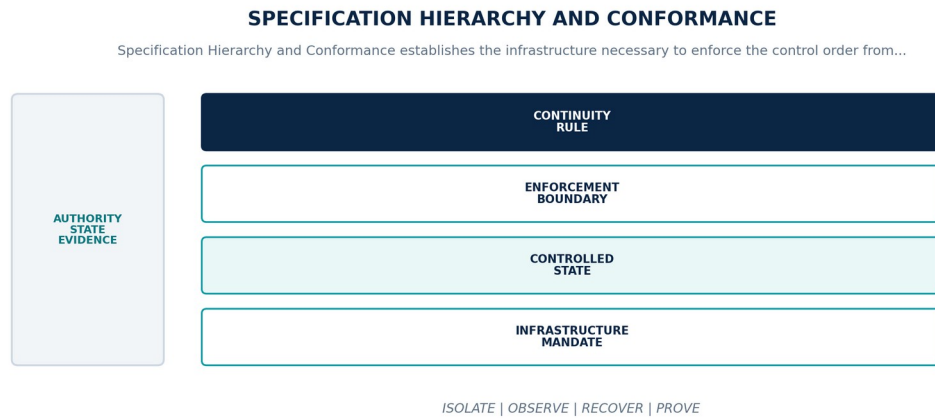


Figure 3. Specification Hierarchy and Conformance. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Enforce the control order from Volume 0 through infrastructure manifests, implementation, verification, and operational evidence.
Controlled state	Normative artifact identity, authority, version, effective time, supersession, exception, and dependent release set.
Enforcement boundary	Lower-level infrastructure specifications may add precision but may not weaken a higher authority or conceal semantic divergence.
Continuity rule	When two controlling artifacts conflict or an exception expires, the platform shall hold affected promotion, select the highest authority, expose dependent releases, and require explicit amendment or closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-02-001	The infrastructure realization of Specification Hierarchy and Conformance SHALL enforce the control order from Volume 0 through infrastructure manifests, implementation, verification, and operational evidence.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-02-002	Every production instance SHALL declare and continuously reconcile normative artifact identity, authority, version, effective time, supersession, exception, and dependent release set through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-02-003	The implementation SHALL enforce this boundary: lower-level infrastructure specifications may add precision but may not weaken a higher authority or conceal semantic divergence.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-02-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Specification Hierarchy and Conformance.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-02-005	When two controlling artifacts conflict or an exception expires, the platform SHALL hold affected promotion, select the highest authority, expose dependent releases, and require explicit amendment or closure; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-02-006	Production admission and continued operation SHALL require machine-readable conformance manifests, conflict tests, exception expiry exercises, and dependency-complete review, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Two controlling artifacts conflict or an exception expires	Hold affected promotion, select the highest authority, expose dependent releases, and require explicit amendment or closure.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared normative artifact identity, authority, version, effective time, supersession, exception, and dependent release set.
- Positive and negative boundary tests prove that lower-level infrastructure specifications may add precision but may not weaken a higher authority or conceal semantic divergence.
- Fault exercises inject two controlling artifacts conflict or an exception expires and verify that the platform can hold affected promotion, select the highest authority, expose dependent releases, and require explicit amendment or closure.
- Independent review accepts machine-readable conformance manifests, conflict tests, exception expiry exercises, and dependency-complete review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Specification Hierarchy and Conformance is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 3, Ch. 6, Ch. 23, Ch. 72. Primary Volume 5 AI anchors: Ch. 2, Ch. 6, Ch. 70, Ch. 71, Ch. 72.

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3 Infrastructure Principles and Semantic Parity

Infrastructure Principles and Semantic Parity establishes the infrastructure necessary to preserve one product contract across SaaS, Private Cloud, and On-Premise while allowing controlled physical variation. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

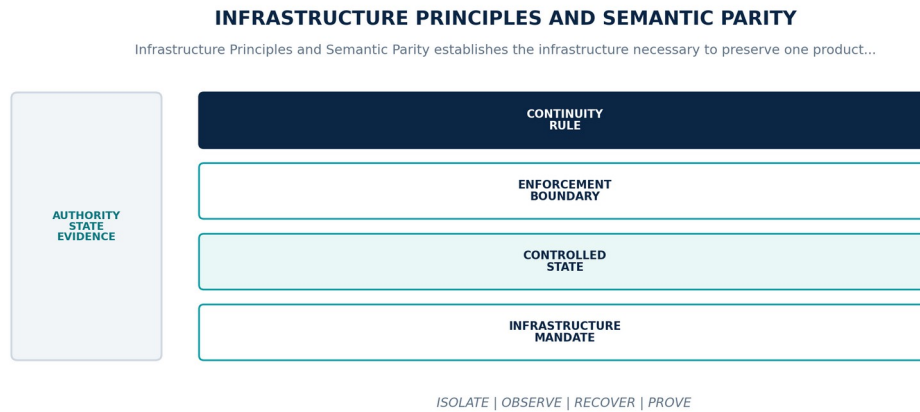


Figure 4. Infrastructure Principles and Semantic Parity. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Preserve one product contract across SaaS, Private Cloud, and On-Premise while allowing controlled physical variation.
Controlled state	Deployment capability profile, semantic parity claim, permitted physical variance, test suite, and evidence validity.
Enforcement boundary	Provider, topology, operating model, and locality may vary; authority, object, workflow, failure, audit, replay, and human-decision semantics may not.
Continuity rule	When a deployment profile cannot pass the common semantic conformance suite, the platform shall mark the capability unavailable or explicitly degraded for that profile and block claims of product parity.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-03-001	The infrastructure realization of Infrastructure Principles and Semantic Parity SHALL preserve one product contract across SaaS, Private Cloud, and On-Premise while allowing controlled physical variation.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-03-002	Every production instance SHALL declare and continuously reconcile deployment capability profile, semantic parity claim, permitted physical variance, test suite, and evidence validity through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-03-003	The implementation SHALL enforce this boundary: provider, topology, operating model, and locality may vary; authority, object, workflow, failure, audit, replay, and human-decision semantics may not.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-03-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Infrastructure Principles and Semantic Parity.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-03-005	When a deployment profile cannot pass the common semantic conformance suite, the platform SHALL mark the capability unavailable or explicitly degraded for that profile and block claims of product parity; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-03-006	Production admission and continued operation SHALL require identical golden fixtures, parity diffs, exception records, and accountable sign-off for every deployment profile, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A deployment profile cannot pass the common semantic conformance suite	Mark the capability unavailable or explicitly degraded for that profile and block claims of product parity.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared deployment capability profile, semantic parity claim, permitted physical variance, test suite, and evidence validity.
- Positive and negative boundary tests prove that provider, topology, operating model, and locality may vary; authority, object, workflow, failure, audit, replay, and human-decision semantics may not.
- Fault exercises inject a deployment profile cannot pass the common semantic conformance suite and verify that the platform can mark the capability unavailable or explicitly degraded for that profile and block claims of product parity.
- Independent review accepts identical golden fixtures, parity diffs, exception records, and accountable sign-off for every deployment profile for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Infrastructure Principles and Semantic Parity is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 5, Ch. 14, Ch. 23, Ch. 64, Ch. 71. Primary Volume 5 AI anchors: Ch. 7, Ch. 9, Ch. 20, Ch. 71.

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4 Shared Responsibility and Customer Control

Shared Responsibility and Customer Control establishes the infrastructure necessary to make operational ownership, customer authority, platform duties, support access, and evidence obligations explicit for every deployment profile. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

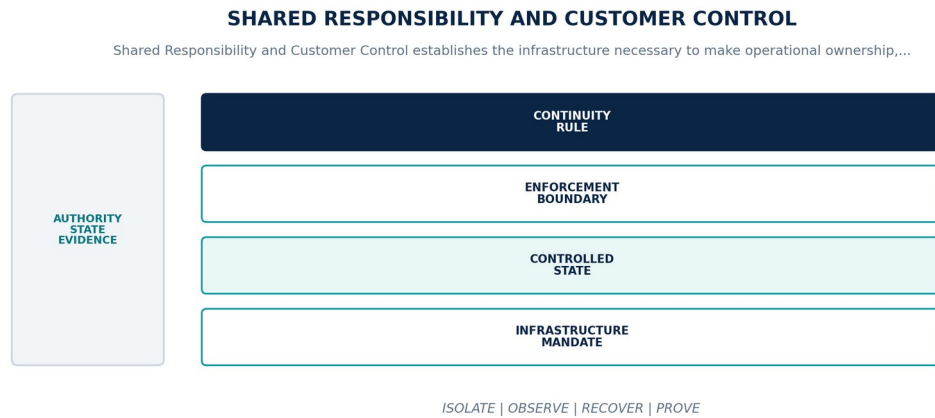


Figure 5. Shared Responsibility and Customer Control. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Make operational ownership, customer authority, platform duties, support access, and evidence obligations explicit for every deployment profile.
Controlled state	Responsibility assignment, control owner, operator role, customer-held authority, access path, evidence producer, and escalation path.
Enforcement boundary	No responsibility may be silently transferred through hosting choice, support action, managed service, or contractual shorthand.
Continuity rule	When an incident or control activity has no unambiguous accountable owner, the platform shall stop the affected operation when safe, invoke the recorded escalation path, and keep the responsibility gap visible until accepted.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-04-001	The infrastructure realization of Shared Responsibility and Customer Control SHALL make operational ownership, customer authority, platform duties, support access, and evidence obligations explicit for every deployment profile.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-04-002	Every production instance SHALL declare and continuously reconcile responsibility assignment, control owner, operator role, customer-held authority, access path, evidence producer, and escalation path through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-04-003	The implementation SHALL enforce this boundary: no responsibility may be silently transferred through hosting choice, support action, managed service, or contractual shorthand.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-04-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Shared Responsibility and Customer Control.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-04-005	When an incident or control activity has no unambiguous accountable owner, the platform SHALL stop the affected operation when safe, invoke the recorded escalation path, and keep the responsibility gap visible until accepted; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-04-006	Production admission and continued operation SHALL require signed responsibility matrices, privileged-access records, customer-control tests, and joint operating exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
An incident or control activity has no unambiguous accountable owner	Stop the affected operation when safe, invoke the recorded escalation path, and keep the responsibility gap visible until accepted.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared responsibility assignment, control owner, operator role, customer-held authority, access path, evidence producer, and escalation path.
- Positive and negative boundary tests prove that no responsibility may be silently transferred through hosting choice, support action, managed service, or contractual shorthand.
- Fault exercises inject an incident or control activity has no unambiguous accountable owner and verify that the platform can stop the affected operation when safe, invoke the recorded escalation path, and keep the responsibility gap visible until accepted.
- Independent review accepts signed responsibility matrices, privileged-access records, customer-control tests, and joint operating exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Shared Responsibility and Customer Control is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 10, Ch. 49, Ch. 63, Ch. 72. Primary Volume 5 AI anchors: Ch. 3, Ch. 37, Ch. 65, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-004 · C-006 · C-028 · C-036 · C-049

5 Failure-Domain and Blast-Radius Doctrine

Failure-Domain and Blast-Radius Doctrine establishes the infrastructure necessary to bound correlated failure by tenant, intelligence domain, cell, zone, region, site, service, state class, and release cohort. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

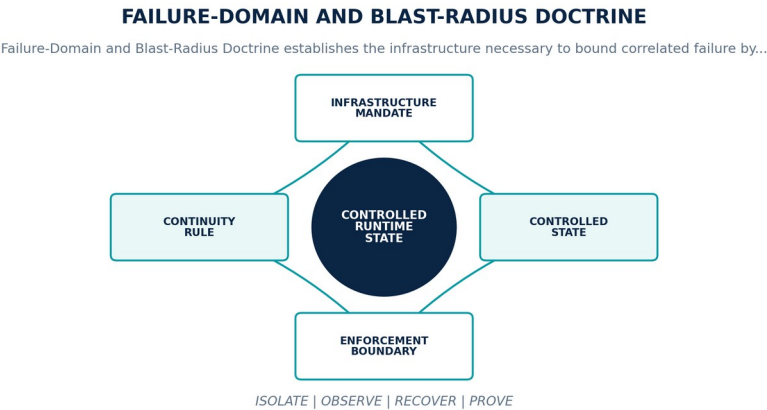


Figure 6. Failure-Domain and Blast-Radius Doctrine. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Bound correlated failure by tenant, intelligence domain, cell, zone, region, site, service, state class, and release cohort.
Controlled state	Failure-domain identity, dependency graph, isolation boundary, recovery order, maximum affected scope, and tested containment state.
Enforcement boundary	Shared infrastructure may aggregate efficiency only where failure, data, policy, and change blast radius remain measurable and enforceable.
Continuity rule	When a fault crosses its declared failure domain or the actual blast radius cannot be established, the platform shall isolate the suspected domain, suspend unsafe automation, preserve evidence, and recover in dependency order under incident command.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-05-001	The infrastructure realization of Failure-Domain and Blast-Radius Doctrine SHALL bound correlated failure by tenant, intelligence domain, cell, zone, region, site, service, state class, and release cohort.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-05-002	Every production instance SHALL declare and continuously reconcile failure-domain identity, dependency graph, isolation boundary, recovery order, maximum affected scope, and tested containment state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-05-003	The implementation SHALL enforce this boundary: shared infrastructure may aggregate efficiency only where failure, data, policy, and change blast radius remain measurable and enforceable.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-05-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Failure-Domain and Blast-Radius Doctrine.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-05-005	When a fault crosses its declared failure domain or the actual blast radius cannot be established, the platform SHALL isolate the suspected domain, suspend unsafe automation, preserve evidence, and recover in dependency order under incident command; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-05-006	Production admission and continued operation SHALL require fault-injection traces, dependency closure, blast-radius measurements, isolation proofs, and recovery exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A fault crosses its declared failure domain or the actual blast radius cannot be established	Isolate the suspected domain, suspend unsafe automation, preserve evidence, and recover in dependency order under incident command.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared failure-domain identity, dependency graph, isolation boundary, recovery order, maximum affected scope, and tested containment state.
- Positive and negative boundary tests prove that shared infrastructure may aggregate efficiency only where failure, data, policy, and change blast radius remain measurable and enforceable.
- Fault exercises inject a fault crosses its declared failure domain or the actual blast radius cannot be established and verify that the platform can isolate the suspected domain, suspend unsafe automation, preserve evidence, and recover in dependency order under incident command.
- Independent review accepts fault-injection traces, dependency closure, blast-radius measurements, isolation proofs, and recovery exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Failure-Domain and Blast-Radius Doctrine is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 9, Ch. 11, Ch. 59, Ch. 61. Primary Volume 5 AI anchors: Ch. 31, Ch. 34, Ch. 38, Ch. 68, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-008 · C-034 · C-038 · C-041 · C-049

6 Infrastructure Traceability and Documentation Inheritance

Infrastructure Traceability and Documentation Inheritance establishes the infrastructure necessary to connect every infrastructure requirement to constitutional invariants, logical components, engineering owners, AI workloads, manifests, tests, releases, and operations. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

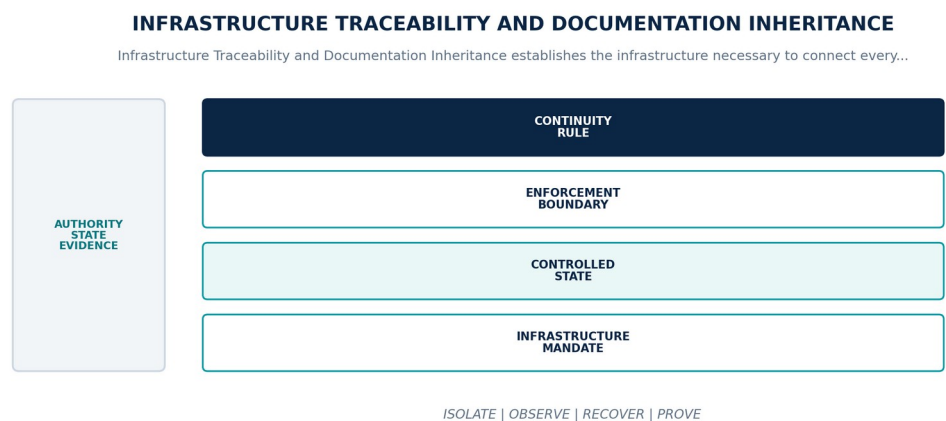


Figure 7. Infrastructure Traceability and Documentation Inheritance. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Connect every infrastructure requirement to constitutional invariants, logical components, engineering owners, AI workloads, manifests, tests, releases, and operations.
Controlled state	Requirement identity, source authority, implementation owner, component, workload, evidence, release, exception, and retirement linkage.
Enforcement boundary	Traceability must cross document, repository, build, deployment, runtime, incident, and retirement boundaries without relying on prose similarity.
Continuity rule	When a production artifact has incomplete, stale, circular, or unverifiable inheritance, the platform shall deny promotion or mark the running artifact non-conformant, identify affected decisions and services, and restore a closed evidence chain.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-06-001	The infrastructure realization of Infrastructure Traceability and Documentation Inheritance SHALL connect every infrastructure requirement to constitutional invariants, logical components, engineering owners, AI workloads, manifests, tests, releases, and operations.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-06-002	Every production instance SHALL declare and continuously reconcile requirement identity, source authority, implementation owner, component, workload, evidence, release, exception, and retirement linkage through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-06-003	The implementation SHALL enforce this boundary: traceability must cross document, repository, build, deployment, runtime, incident, and retirement boundaries without relying on prose similarity.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-06-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Infrastructure Traceability and Documentation Inheritance.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-06-005	When a production artifact has incomplete, stale, circular, or unverifiable inheritance, the platform SHALL deny promotion or mark the running artifact non-conformant, identify affected decisions and services, and restore a closed evidence chain; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-06-006	Production admission and continued operation SHALL require bidirectional trace queries, orphan detection, release-manifest closure, runtime attestation, and retirement proofs, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A production artifact has incomplete, stale, circular, or unverifiable inheritance	Deny promotion or mark the running artifact non-conformant, identify affected decisions and services, and restore a closed evidence chain.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared requirement identity, source authority, implementation owner, component, workload, evidence, release, exception, and retirement linkage.
- Positive and negative boundary tests prove that traceability must cross document, repository, build, deployment, runtime, incident, and retirement boundaries without relying on prose similarity.
- Fault exercises inject a production artifact has incomplete, stale, circular, or unverifiable inheritance and verify that the platform can deny promotion or mark the running artifact non-conformant, identify affected decisions and services, and restore a closed evidence chain.
- Independent review accepts bidirectional trace queries, orphan detection, release-manifest closure, runtime attestation, and retirement proofs for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Infrastructure Traceability and Documentation Inheritance is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 6, Ch. 23, Ch. 66, Ch. 69, Ch. 72. Primary Volume 5 AI anchors: Ch. 6, Ch. 15, Ch. 30, Ch. 70, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-003 · C-008 · C-043 · C-050 · C-052

PART II

Deployment and Runtime Foundation

Deployment profiles, disconnected operation, topology, isolation, and management-plane architecture.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

7 System Context and Deployment Models

System Context and Deployment Models establishes the infrastructure necessary to define the physical system context connecting users, customer systems, external sources, intelligence cells, canonical state, and shared control services. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

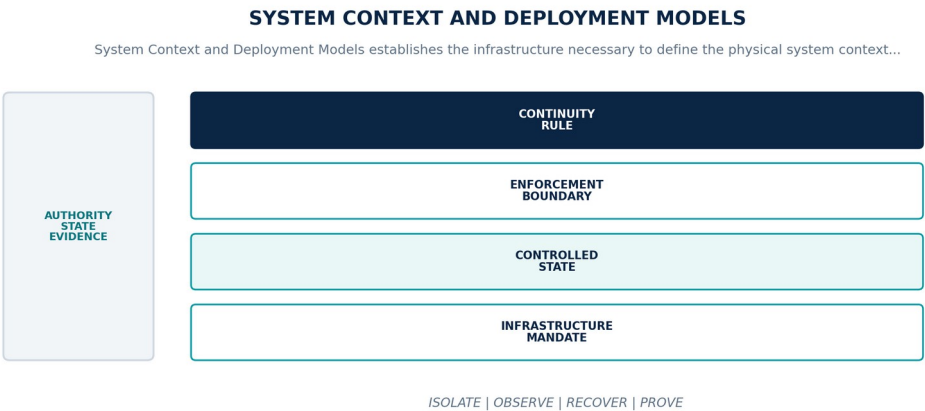


Figure 8. System Context and Deployment Models. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Define the physical system context connecting users, customer systems, external sources, intelligence cells, canonical state, and shared control services.
Controlled state	Deployment instance, customer domain, region or site, cell set, trust boundary, connected systems, and control-plane relationship.
Enforcement boundary	The infrastructure context must preserve canonical layer ownership and distinguish customer, platform, provider, source, and external-effect boundaries.
Continuity rule	When a dependency or trust boundary is absent from the deployed context, the platform shall treat connectivity as denied or indeterminate, prevent implicit trust, and require context reconciliation before service admission.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-07-001	The infrastructure realization of System Context and Deployment Models SHALL define the physical system context connecting users, customer systems, external sources, intelligence cells, canonical state, and shared control services.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-07-002	Every production instance SHALL declare and continuously reconcile deployment instance, customer domain, region or site, cell set, trust boundary, connected systems, and control-plane relationship through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-07-003	The implementation SHALL enforce this boundary: the infrastructure context must preserve canonical layer ownership and distinguish customer, platform, provider, source, and external-effect boundaries.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-07-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for System Context and Deployment Models.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-07-005	When a dependency or trust boundary is absent from the deployed context, the platform SHALL treat connectivity as denied or indeterminate, prevent implicit trust, and require context reconciliation before service admission; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-07-006	Production admission and continued operation SHALL require deployed topology inventory, boundary probes, dependency tests, data-flow evidence, and customer acceptance, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A dependency or trust boundary is absent from the deployed context	Treat connectivity as denied or indeterminate, prevent implicit trust, and require context reconciliation before service admission.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared deployment instance, customer domain, region or site, cell set, trust boundary, connected systems, and control-plane relationship.
- Positive and negative boundary tests prove that the infrastructure context must preserve canonical layer ownership and distinguish customer, platform, provider, source, and external-effect boundaries.
- Fault exercises inject a dependency or trust boundary is absent from the deployed context and verify that the platform can treat connectivity as denied or indeterminate, prevent implicit trust, and require context reconciliation before service admission.
- Independent review accepts deployed topology inventory, boundary probes, dependency tests, data-flow evidence, and customer acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

System Context and Deployment Models is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 8, Ch. 10, Ch. 14, Ch. 71. Primary Volume 5 AI anchors: Ch. 7, Ch. 8, Ch. 9, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-001 · C-007 · C-036 · C-045 · C-049

8 SaaS Reference Architecture

SaaS Reference Architecture establishes the infrastructure necessary to operate a multi-region managed service with cell isolation, tenant control, governed shared services, and continuous upgradeability. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

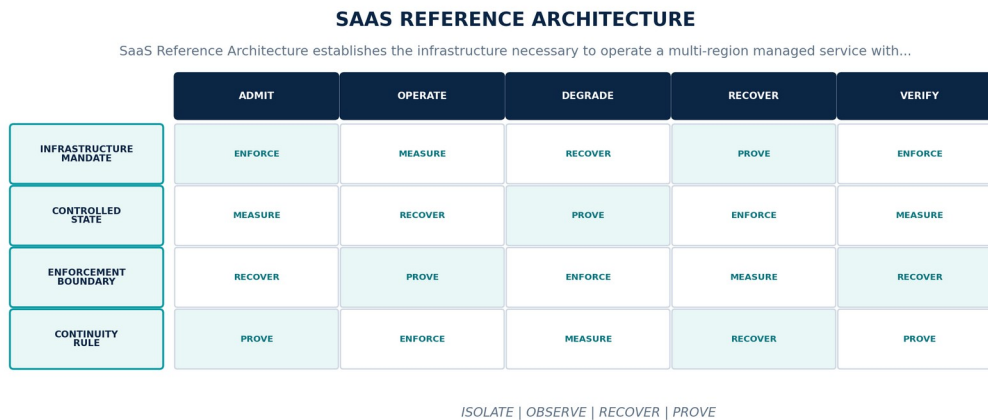


Figure 9. SaaS Reference Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Operate a multi-region managed service with cell isolation, tenant control, governed shared services, and continuous upgradeability.
Controlled state	SaaS environment, region, zone, cell, tenant placement, shared-service binding, release cohort, and operational state.
Enforcement boundary	Multi-tenancy may share physical capacity only where identity, policy, state, telemetry, encryption, quotas, and failures remain tenant-scoped.
Continuity rule	When a region, cell, shared service, or tenant isolation control becomes unavailable or suspect, the platform shall route or contain within approved residency and consistency limits, enter explicit degraded mode, and preserve tenant evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-08-001	The infrastructure realization of SaaS Reference Architecture SHALL operate a multi-region managed service with cell isolation, tenant control, governed shared services, and continuous upgradeability.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-08-002	Every production instance SHALL declare and continuously reconcile SaaS environment, region, zone, cell, tenant placement, shared-service binding, release cohort, and operational state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-08-003	The implementation SHALL enforce this boundary: multi-tenancy may share physical capacity only where identity, policy, state, telemetry, encryption, quotas, and failures remain tenant-scoped.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-08-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for SaaS Reference Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-08-005	When a region, cell, shared service, or tenant isolation control becomes unavailable or suspect, the platform SHALL route or contain within approved residency and consistency limits, enter explicit degraded mode, and preserve tenant evidence; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-08-006	Production admission and continued operation SHALL require multi-region failover, noisy-neighbor, isolation, upgrade, rollback, residency, and shared-responsibility evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A region, cell, shared service, or tenant isolation control becomes unavailable or suspect	Route or contain within approved residency and consistency limits, enter explicit degraded mode, and preserve tenant evidence.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared SaaS environment, region, zone, cell, tenant placement, shared-service binding, release cohort, and operational state.
- Positive and negative boundary tests prove that multi-tenancy may share physical capacity only where identity, policy, state, telemetry, encryption, quotas, and failures remain tenant-scoped.
- Fault exercises inject a region, cell, shared service, or tenant isolation control becomes unavailable or suspect and verify that the platform can route or contain within approved residency and consistency limits, enter explicit degraded mode, and preserve tenant evidence.
- Independent review accepts multi-region failover, noisy-neighbor, isolation, upgrade, rollback, residency, and shared-responsibility evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

SaaS Reference Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 8, Ch. 9, Ch. 10, Ch. 14, Ch. 61. Primary Volume 5 AI anchors: Ch. 7, Ch. 10, Ch. 12, Ch. 68, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-006 · C-034 · C-036 · C-038 · C-045

9 Private Cloud Reference Architecture

Private Cloud Reference Architecture establishes the infrastructure necessary to deploy the full product into a customer-controlled cloud boundary while preserving The Eye's control contracts and supportability. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

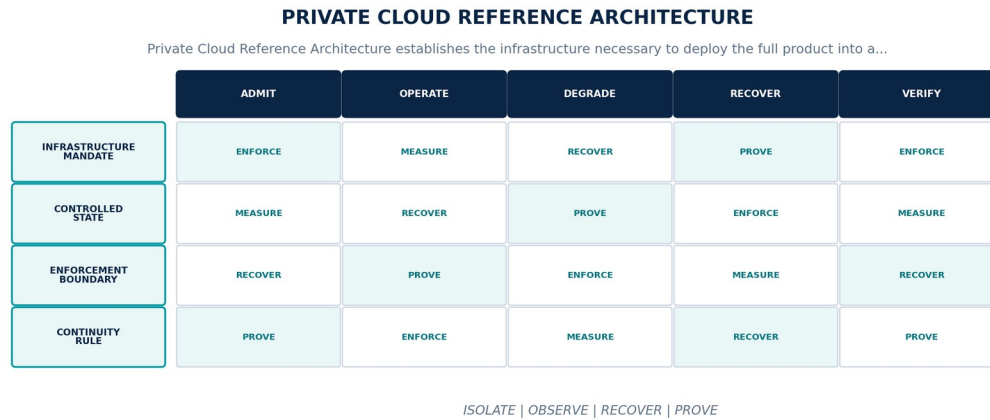


Figure 10. Private Cloud Reference Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Deploy the full product into a customer-controlled cloud boundary while preserving The Eye's control contracts and supportability.
Controlled state	Customer cloud account or subscription, regions, network attachments, cell set, customer-managed dependencies, and release channel.
Enforcement boundary	Customer control of cloud resources may change responsibility but not semantic behavior, evidence quality, upgrade discipline, or human authority.
Continuity rule	When a required customer-managed service, permission, quota, or network path violates the certified profile, the platform shall surface the exact dependency state, disable affected capabilities, and operate only within the last verified boundary.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-09-001	The infrastructure realization of Private Cloud Reference Architecture SHALL deploy the full product into a customer-controlled cloud boundary while preserving The Eye's control contracts and supportability.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-09-002	Every production instance SHALL declare and continuously reconcile customer cloud account or subscription, regions, network attachments, cell set, customer-managed dependencies, and release channel through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-09-003	The implementation SHALL enforce this boundary: customer control of cloud resources may change responsibility but not semantic behavior, evidence quality, upgrade discipline, or human authority.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.
IA-09-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Private Cloud Reference Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-09-005	When a required customer-managed service, permission, quota, or network path violates the certified profile, the platform SHALL surface the exact dependency state, disable affected capabilities, and operate only within the last verified boundary; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-09-006	Production admission and continued operation SHALL require landing-zone certification, dependency compatibility, customer-key tests, disconnected support procedure, and parity evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A required customer-managed service, permission, quota, or network path violates the certified profile	Surface the exact dependency state, disable affected capabilities, and operate only within the last verified boundary.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared customer cloud account or subscription, regions, network attachments, cell set, customer-managed dependencies, and release channel.
- Positive and negative boundary tests prove that customer control of cloud resources may change responsibility but not semantic behavior, evidence quality, upgrade discipline, or human authority.
- Fault exercises inject a required customer-managed service, permission, quota, or network path violates the certified profile and verify that the platform can surface the exact dependency state, disable affected capabilities, and operate only within the last verified boundary.
- Independent review accepts landing-zone certification, dependency compatibility, customer-key tests, disconnected support procedure, and parity evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Private Cloud Reference Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 8, Ch. 10, Ch. 13, Ch. 14, Ch. 71. Primary Volume 5 AI anchors: Ch. 7, Ch. 17, Ch. 21, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-006 · C-036 · C-045 · C-046 · C-049

10 On-Premise Reference Architecture

On-Premise Reference Architecture establishes the infrastructure necessary to operate the complete platform on customer-owned infrastructure with deterministic packaging, local control, bounded dependencies, and recoverable lifecycle management. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

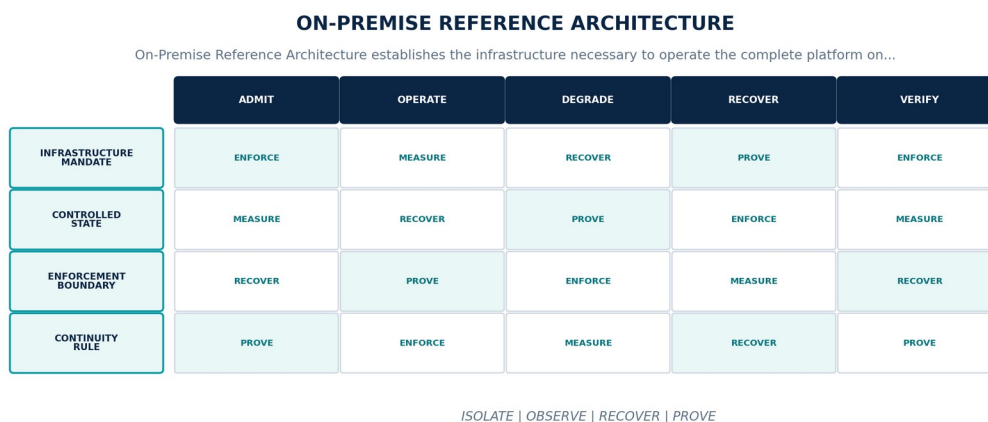


Figure 11. On-Premise Reference Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Operate the complete platform on customer-owned infrastructure with deterministic packaging, local control, bounded dependencies, and recoverable lifecycle management.
Controlled state	Site, cluster, appliance or node set, certified dependency versions, local control services, data domains, and installed release state.
Enforcement boundary	The installation boundary must remain operable, diagnosable, secure, upgradeable, exportable, and recoverable without hidden SaaS dependencies.
Continuity rule	When local infrastructure drifts beyond the certified compatibility envelope or loses vendor connectivity, the platform shall continue within the declared support window, block unsafe promotion, expose drift, and use signed offline maintenance artifacts.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-10-001	The infrastructure realization of On-Premise Reference Architecture SHALL operate the complete platform on customer-owned infrastructure with deterministic packaging, local control, bounded dependencies, and recoverable lifecycle management.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-10-002	Every production instance SHALL declare and continuously reconcile site, cluster, appliance or node set, certified dependency versions, local control services, data domains, and installed release state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-10-003	The implementation SHALL enforce this boundary: the installation boundary must remain operable, diagnosable, secure, upgradeable, exportable, and recoverable without hidden SaaS dependencies.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-10-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for On-Premise Reference Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-10-005	When local infrastructure drifts beyond the certified compatibility envelope or loses vendor connectivity, the platform SHALL continue within the declared support window, block unsafe promotion, expose drift, and use signed offline maintenance artifacts; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-10-006	Production admission and continued operation SHALL require site acceptance, dependency bill of materials, offline install, upgrade and rollback, backup and restore, and parity evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Local infrastructure drifts beyond the certified compatibility envelope or loses vendor connectivity	Continue within the declared support window, block unsafe promotion, expose drift, and use signed offline maintenance artifacts.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared site, cluster, appliance or node set, certified dependency versions, local control services, data domains, and installed release state.
- Positive and negative boundary tests prove that the installation boundary must remain operable, diagnosable, secure, upgradeable, exportable, and recoverable without hidden SaaS dependencies.
- Fault exercises inject local infrastructure drifts beyond the certified compatibility envelope or loses vendor connectivity and verify that the platform can continue within the declared support window, block unsafe promotion, expose drift, and use signed offline maintenance artifacts.
- Independent review accepts site acceptance, dependency bill of materials, offline install, upgrade and rollback, backup and restore, and parity evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

On-Premise Reference Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 8, Ch. 14, Ch. 56, Ch. 64, Ch. 71. Primary Volume 5 AI anchors: Ch. 7, Ch. 17, Ch. 22, Ch. 70, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-006 • C-036 • C-045 • C-046 • C-047

11 Disconnected and Air-Gapped Operations

Disconnected and Air-Gapped Operations establishes the infrastructure necessary to preserve bounded strategic operations where external connectivity is denied, intermittent, mediated, or physically separated. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

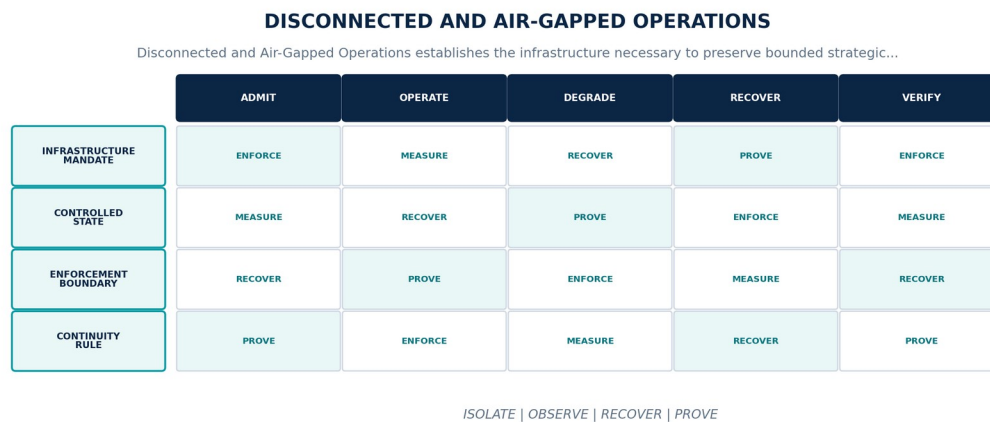


Figure 12. Disconnected and Air-Gapped Operations. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Preserve bounded strategic operations where external connectivity is denied, intermittent, mediated, or physically separated.
Controlled state	Connectivity state, import and export package, transfer authority, content inspection, time basis, update channel, and synchronization ledger.
Enforcement boundary	No runtime path may assume public DNS, external identity, cloud control, telemetry export, model provider, or license service inside an air gap.
Continuity rule	When trusted time, update media, transfer station, or synchronization evidence is unavailable, the platform shall enter the defined local continuity mode, reject unverifiable packages, and preserve pending exchanges for later reconciliation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-11-001	The infrastructure realization of Disconnected and Air-Gapped Operations SHALL preserve bounded strategic operations where external connectivity is denied, intermittent, mediated, or physically separated.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-11-002	Every production instance SHALL declare and continuously reconcile connectivity state, import and export package, transfer authority, content inspection, time basis, update channel, and synchronization ledger through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-11-003	The implementation SHALL enforce this boundary: no runtime path may assume public DNS, external identity, cloud control, telemetry export, model provider, or license service inside an air gap.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-11-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Disconnected and Air-Gapped Operations.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-11-005	When trusted time, update media, transfer station, or synchronization evidence is unavailable, the platform SHALL enter the defined local continuity mode, reject unverifiable packages, and preserve pending exchanges for later reconciliation; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-11-006	Production admission and continued operation SHALL require zero-egress tests, removable-media ceremonies, offline identity, package signature, delayed synchronization, and recovery drills, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Trusted time, update media, transfer station, or synchronization evidence is unavailable	Enter the defined local continuity mode, reject unverifiable packages, and preserve pending exchanges for later reconciliation.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared connectivity state, import and export package, transfer authority, content inspection, time basis, update channel, and synchronization ledger.
- Positive and negative boundary tests prove that no runtime path may assume public DNS, external identity, cloud control, telemetry export, model provider, or license service inside an air gap.
- Fault exercises inject trusted time, update media, transfer station, or synchronization evidence is unavailable and verify that the platform can enter the defined local continuity mode, reject unverifiable packages, and preserve pending exchanges for later reconciliation.
- Independent review accepts zero-egress tests, removable-media ceremonies, offline identity, package signature, delayed synchronization, and recovery drills for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Disconnected and Air-Gapped Operations is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 29, Ch. 54, Ch. 56, Ch. 62. Primary Volume 5 AI anchors: Ch. 21, Ch. 26, Ch. 60, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-007 · C-017 · C-036 · C-046 · C-047

12 Regional, Zonal, Site, and Cell Topology

Regional, Zonal, Site, and Cell Topology establishes the infrastructure necessary to compose locations and cells into explicit availability, latency, sovereignty, capacity, and recovery domains. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

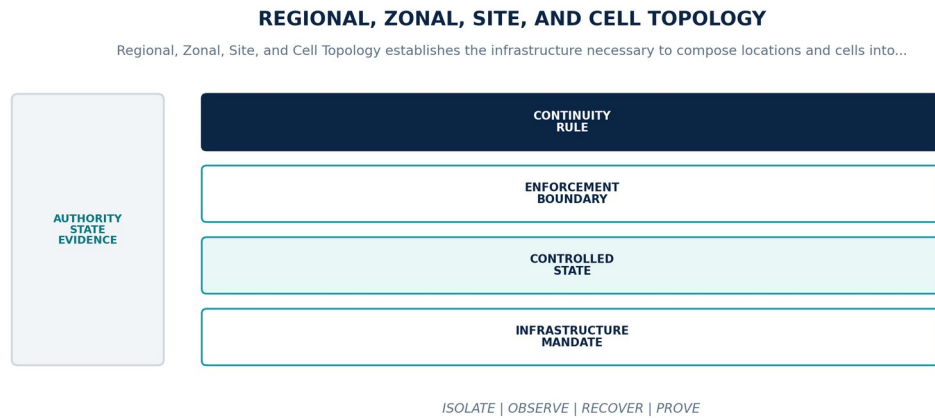


Figure 13. Regional, Zonal, Site, and Cell Topology. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Compose locations and cells into explicit availability, latency, sovereignty, capacity, and recovery domains.
Controlled state	Topology node, parent domain, role, locality, dependencies, capacity, consistency class, health, and evacuation state.
Enforcement boundary	Placement must satisfy residency, latency, replication, quorum, failure-domain, data-class, and customer-control constraints together.
Continuity rule	When topology membership, quorum, locality, or health becomes ambiguous, the platform shall freeze unsafe placement, prefer consistency and sovereignty over convenience, and execute the recorded evacuation or isolation plan.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-12-001	The infrastructure realization of Regional, Zonal, Site, and Cell Topology SHALL compose locations and cells into explicit availability, latency, sovereignty, capacity, and recovery domains.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-12-002	Every production instance SHALL declare and continuously reconcile topology node, parent domain, role, locality, dependencies, capacity, consistency class, health, and evacuation state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-12-003	The implementation SHALL enforce this boundary: placement must satisfy residency, latency, replication, quorum, failure-domain, data-class, and customer-control constraints together.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-12-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Regional, Zonal, Site, and Cell Topology.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-12-005	When topology membership, quorum, locality, or health becomes ambiguous, the platform SHALL freeze unsafe placement, prefer consistency and sovereignty over convenience, and execute the recorded evacuation or isolation plan; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-12-006	Production admission and continued operation SHALL require placement simulation, quorum loss, zone and site evacuation, latency envelopes, and topology-reconciliation evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Topology membership, quorum, locality, or health becomes ambiguous	Freeze unsafe placement, prefer consistency and sovereignty over convenience, and execute the recorded evacuation or isolation plan.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared topology node, parent domain, role, locality, dependencies, capacity, consistency class, health, and evacuation state.
- Positive and negative boundary tests prove that placement must satisfy residency, latency, replication, quorum, failure-domain, data-class, and customer-control constraints together.
- Fault exercises inject topology membership, quorum, locality, or health becomes ambiguous and verify that the platform can freeze unsafe placement, prefer consistency and sovereignty over convenience, and execute the recorded evacuation or isolation plan.
- Independent review accepts placement simulation, quorum loss, zone and site evacuation, latency envelopes, and topology-reconciliation evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Regional, Zonal, Site, and Cell Topology is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 9, Ch. 12, Ch. 59, Ch. 61. Primary Volume 5 AI anchors: Ch. 12, Ch. 19, Ch. 22, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-034 · C-036 · C-038 · C-041 · C-045

13 Tenant, Domain, and Intelligence Cell Isolation

Tenant, Domain, and Intelligence Cell Isolation establishes the infrastructure necessary to make customer, intelligence-domain, workload, state, network, key, policy, quota, telemetry, and failure isolation enforceable at runtime. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

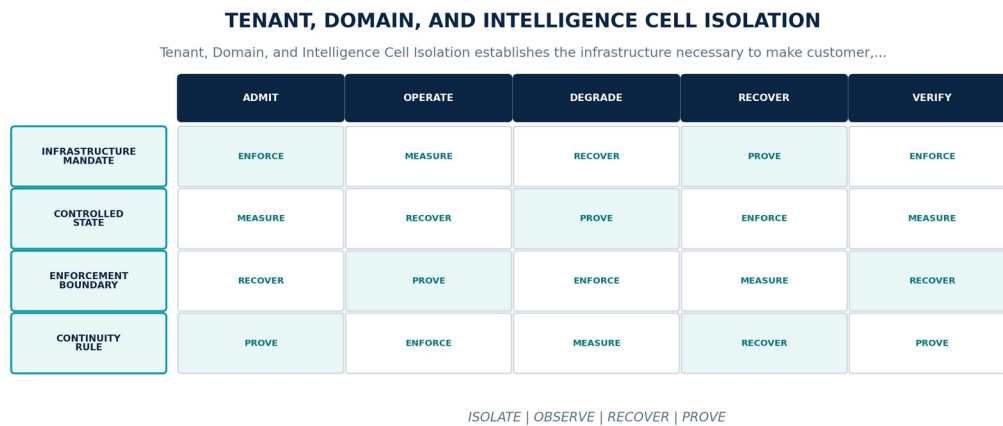


Figure 14. Tenant, Domain, and Intelligence Cell Isolation. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Make customer, intelligence-domain, workload, state, network, key, policy, quota, telemetry, and failure isolation enforceable at runtime.
Controlled state	Tenant and domain identity, cell placement, namespace, key scope, network policy, quota, state binding, and isolation posture.
Enforcement boundary	Shared services may receive requests only through authenticated tenant and purpose context and may not infer scope from network location.
Continuity rule	When tenant context is missing, conflicting, or leakage is suspected, the platform shall fail closed, quarantine affected paths and evidence, rotate or revoke exposed authority, and initiate cross-tenant impact analysis.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-13-001	The infrastructure realization of Tenant, Domain, and Intelligence Cell Isolation SHALL make customer, intelligence-domain, workload, state, network, key, policy, quota, telemetry, and failure isolation enforceable at runtime.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-13-002	Every production instance SHALL declare and continuously reconcile tenant and domain identity, cell placement, namespace, key scope, network policy, quota, state binding, and isolation posture through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-13-003	The implementation SHALL enforce this boundary: shared services may receive requests only through authenticated tenant and purpose context and may not infer scope from network location.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-13-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Tenant, Domain, and Intelligence Cell Isolation.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-13-005	When tenant context is missing, conflicting, or leakage is suspected, the platform SHALL fail closed, quarantine affected paths and evidence, rotate or revoke exposed authority, and initiate cross-tenant impact analysis; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-13-006	Production admission and continued operation SHALL require cross-tenant negative tests, namespace and key proofs, policy traces, noisy-neighbor tests, and forensic exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Tenant context is missing, conflicting, or leakage is suspected	Fail closed, quarantine affected paths and evidence, rotate or revoke exposed authority, and initiate cross-tenant impact analysis.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared tenant and domain identity, cell placement, namespace, key scope, network policy, quota, state binding, and isolation posture.
- Positive and negative boundary tests prove that shared services may receive requests only through authenticated tenant and purpose context and may not infer scope from network location.
- Fault exercises inject tenant context is missing, conflicting, or leakage is suspected and verify that the platform can fail closed, quarantine affected paths and evidence, rotate or revoke exposed authority, and initiate cross-tenant impact analysis.
- Independent review accepts cross-tenant negative tests, namespace and key proofs, policy traces, noisy-neighbor tests, and forensic exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Tenant, Domain, and Intelligence Cell Isolation is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 8, Ch. 9, Ch. 13, Ch. 25, Ch. 51. Primary Volume 5 AI anchors: Ch. 7, Ch. 12, Ch. 19, Ch. 61.

CONSTITUTIONAL INHERITANCE / C-006 · C-007 · C-030 · C-034 · C-049

14 Shared Control Services and Management Plane

Shared Control Services and Management Plane establishes the infrastructure necessary to operate identity, policy, registry, configuration, keys, telemetry, release, licensing, and fleet control without placing customer data truth in the management plane. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

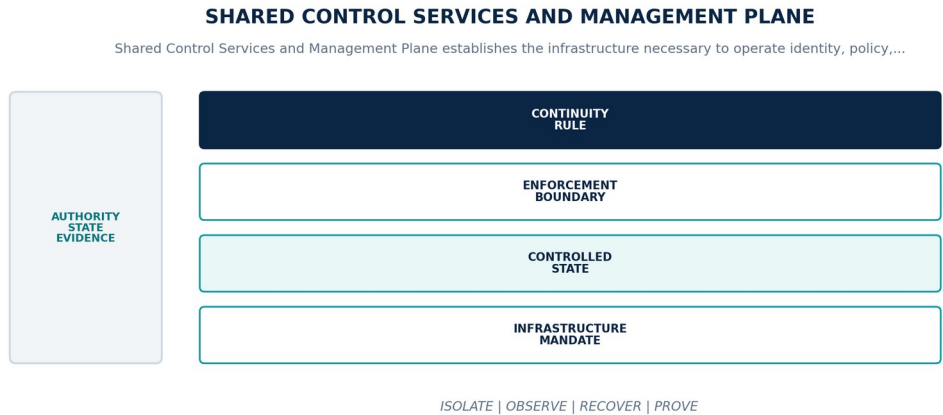


Figure 15. Shared Control Services and Management Plane. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Operate identity, policy, registry, configuration, keys, telemetry, release, licensing, and fleet control without placing customer data truth in the management plane.
Controlled state	Control-service identity, desired state, policy and registry versions, fleet inventory, command state, evidence, and operator action.
Enforcement boundary	Management-plane reachability must be one-way or mediated where required and cannot become an unbounded data-plane backdoor.
Continuity rule	When the management plane is unavailable, compromised, partitioned, or inconsistent, the platform shall freeze unsafe change, continue last-known approved local control where permitted, revoke suspect authority, and reconcile before resuming.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-14-001	The infrastructure realization of Shared Control Services and Management Plane SHALL operate identity, policy, registry, configuration, keys, telemetry, release, licensing, and fleet control without placing customer data truth in the management plane.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-14-002	Every production instance SHALL declare and continuously reconcile control-service identity, desired state, policy and registry versions, fleet inventory, command state, evidence, and operator action through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-14-003	The implementation SHALL enforce this boundary: management-plane reachability must be one-way or mediated where required and cannot become an unbounded data-plane backdoor.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-14-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Shared Control Services and Management Plane.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-14-005	When the management plane is unavailable, compromised, partitioned, or inconsistent, the platform SHALL freeze unsafe change, continue last-known approved local control where permitted, revoke suspect authority, and reconcile before resuming; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-14-006	Production admission and continued operation SHALL require plane-separation tests, break-glass exercises, offline control continuity, command audit, and compromise recovery, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
The management plane is unavailable, compromised, partitioned, or inconsistent	Freeze unsafe change, continue last-known approved local control where permitted, revoke suspect authority, and reconcile before resuming.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared control-service identity, desired state, policy and registry versions, fleet inventory, command state, evidence, and operator action.
- Positive and negative boundary tests prove that management-plane reachability must be one-way or mediated where required and cannot become an unbounded data-plane backdoor.
- Fault exercises inject the management plane is unavailable, compromised, partitioned, or inconsistent and verify that the platform can freeze unsafe change, continue last-known approved local control where permitted, revoke suspect authority, and reconcile before resuming.
- Independent review accepts plane-separation tests, break-glass exercises, offline control continuity, command audit, and compromise recovery for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Shared Control Services and Management Plane is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 10, Ch. 13, Ch. 49, Ch. 54, Ch. 63. Primary Volume 5 AI anchors: Ch. 10, Ch. 11, Ch. 14, Ch. 65, Ch. 71.

PART III

Compute and Acceleration

CPU, GPU, model serving, agents, processing, edge execution, scheduling, and resource governance.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

15 Compute Platform Architecture

Compute Platform Architecture establishes the infrastructure necessary to provide portable, policy-governed compute pools for stateless, stateful, interactive, batch, streaming, AI, and control workloads. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

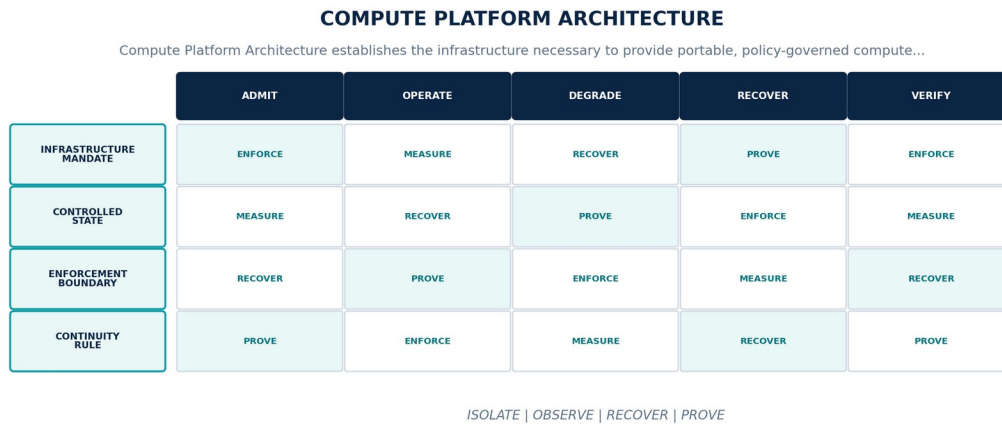


Figure 16. Compute Platform Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Provide portable, policy-governed compute pools for stateless, stateful, interactive, batch, streaming, AI, and control workloads.
Controlled state	Compute pool, node class, architecture, placement capability, health, allocatable capacity, reservation, and maintenance state.
Enforcement boundary	Workloads declare resource, isolation, locality, accelerator, latency, disruption, and data-class needs; schedulers may not guess them.
Continuity rule	When capacity, node health, scheduler state, or placement guarantees fall below the admitted workload contract, the platform shall shed or queue by consequence and priority, preserve state, and prevent uncontrolled oversubscription or silent relocation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-15-001	The infrastructure realization of Compute Platform Architecture SHALL provide portable, policy-governed compute pools for stateless, stateful, interactive, batch, streaming, AI, and control workloads.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-15-002	Every production instance SHALL declare and continuously reconcile compute pool, node class, architecture, placement capability, health, allocatable capacity, reservation, and maintenance state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-15-003	The implementation SHALL enforce this boundary: workloads declare resource, isolation, locality, accelerator, latency, disruption, and data-class needs; schedulers may not guess them.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-15-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Compute Platform Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-15-005	When capacity, node health, scheduler state, or placement guarantees fall below the admitted workload contract, the platform SHALL shed or queue by consequence and priority, preserve state, and prevent uncontrolled oversubscription or silent relocation; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-15-006	Production admission and continued operation SHALL require capacity models, scheduling fixtures, node-loss tests, heterogeneous-platform conformance, and workload SLO evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Capacity, node health, scheduler state, or placement guarantees fall below the admitted workload contract	Shed or queue by consequence and priority, preserve state, and prevent uncontrolled oversubscription or silent relocation.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared compute pool, node class, architecture, placement capability, health, allocatable capacity, reservation, and maintenance state.
- Positive and negative boundary tests prove that workloads declare resource, isolation, locality, accelerator, latency, disruption, and data-class needs; schedulers may not guess them.
- Fault exercises inject capacity, node health, scheduler state, or placement guarantees fall below the admitted workload contract and verify that the platform can shed or queue by consequence and priority, preserve state, and prevent uncontrolled oversubscription or silent relocation.
- Independent review accepts capacity models, scheduling fixtures, node-loss tests, heterogeneous-platform conformance, and workload SLO evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Compute Platform Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 9, Ch. 11, Ch. 58, Ch. 59. Primary Volume 5 AI anchors: Ch. 22, Ch. 31, Ch. 68, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-008 · C-034 · C-038 · C-041 · C-045

16 CPU Workload Architecture

CPU Workload Architecture establishes the infrastructure necessary to run general-purpose services, graph and data processing, orchestration, policy, and deterministic analytics on certified CPU profiles. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

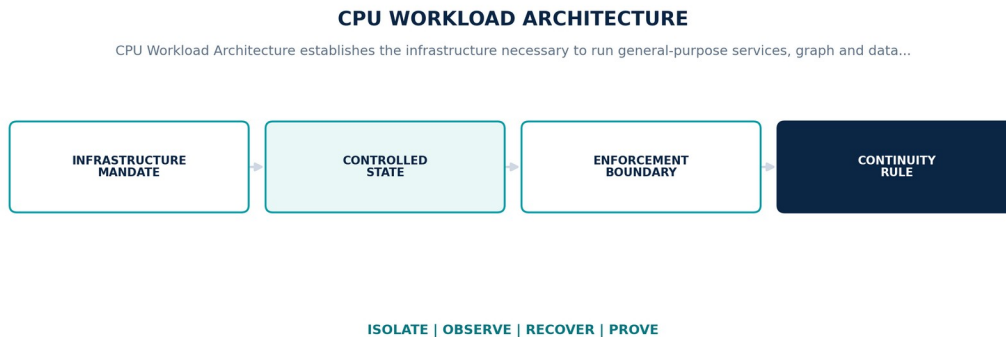


Figure 17. CPU Workload Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Run general-purpose services, graph and data processing, orchestration, policy, and deterministic analytics on certified CPU profiles.
Controlled state	CPU architecture, core and memory profile, instruction-set capabilities, runtime class, reservation, throttling, and performance evidence.
Enforcement boundary	Portable workloads must define compatible architectures and deterministic behavior; architecture-specific optimization remains an explicit variant.
Continuity rule	When CPU saturation, memory pressure, thermal throttling, or architecture incompatibility threatens the workload contract, the platform shall throttle, reschedule, scale, or degrade according to priority while preserving deadlines, checkpoints, and observable partial state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-16-001	The infrastructure realization of CPU Workload Architecture SHALL run general-purpose services, graph and data processing, orchestration, policy, and deterministic analytics on certified CPU profiles.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-16-002	Every production instance SHALL declare and continuously reconcile CPU architecture, core and memory profile, instruction-set capabilities, runtime class, reservation, throttling, and performance evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-16-003	The implementation SHALL enforce this boundary: portable workloads must define compatible architectures and deterministic behavior; architecture-specific optimization remains an explicit variant.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-16-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for CPU Workload Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-16-005	When CPU saturation, memory pressure, thermal throttling, or architecture incompatibility threatens the workload contract, the platform SHALL throttle, reschedule, scale, or degrade according to priority while preserving deadlines, checkpoints, and observable partial state; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-16-006	Production admission and continued operation SHALL require cross-architecture tests, saturation curves, memory-pressure exercises, deterministic-output checks, and cost profiles, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
CPU saturation, memory pressure, thermal throttling, or architecture incompatibility threatens the workload contract	Throttle, reschedule, scale, or degrade according to priority while preserving deadlines, checkpoints, and observable partial state.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared CPU architecture, core and memory profile, instruction-set capabilities, runtime class, reservation, throttling, and performance evidence.
- Positive and negative boundary tests prove that portable workloads must define compatible architectures and deterministic behavior; architecture-specific optimization remains an explicit variant.
- Fault exercises inject CPU saturation, memory pressure, thermal throttling, or architecture incompatibility threatens the workload contract and verify that the platform can throttle, reschedule, scale, or degrade according to priority while preserving deadlines, checkpoints, and observable partial state.
- Independent review accepts cross-architecture tests, saturation curves, memory-pressure exercises, deterministic-output checks, and cost profiles for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

CPU Workload Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 11, Ch. 41, Ch. 58, Ch. 59. Primary Volume 5 AI anchors: Ch. 22, Ch. 34, Ch. 66, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-008 · C-035 · C-038 · C-041

17 GPU and Accelerator Architecture

GPU and Accelerator Architecture establishes the infrastructure necessary to govern scarce accelerators as isolated, attestable, schedulable, observable resources for training, inference, simulation, and high-performance analytics. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

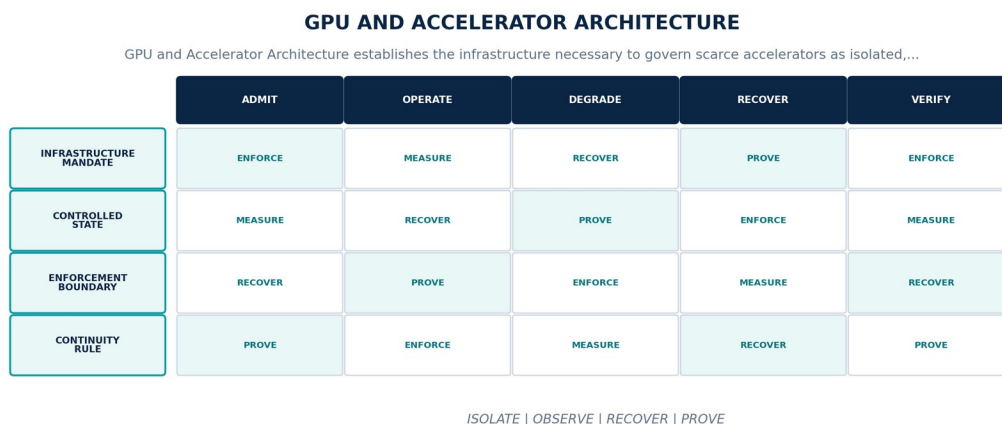


Figure 18. GPU and Accelerator Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Govern scarce accelerators as isolated, attestable, schedulable, observable resources for training, inference, simulation, and high-performance analytics.
Controlled state	Accelerator model, firmware and driver stack, partition, topology, memory, health, allocation, tenancy, and attestation.
Enforcement boundary	Accelerator sharing must preserve memory isolation, workload identity, timing controls, supply-chain integrity, and bounded side channels.
Continuity rule	When device health, memory integrity, driver compatibility, partition isolation, or capacity becomes unsafe, the platform shall drain the device, invalidate affected work, preserve checkpoints, route only to an equivalent certified profile, and open hardware evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-17-001	The infrastructure realization of GPU and Accelerator Architecture SHALL govern scarce accelerators as isolated, attestable, schedulable, observable resources for training, inference, simulation, and high-performance analytics.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.
IA-17-002	Every production instance SHALL declare and continuously reconcile accelerator model, firmware and driver stack, partition, topology, memory, health, allocation, tenancy, and attestation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-17-003	The implementation SHALL enforce this boundary: accelerator sharing must preserve memory isolation, workload identity, timing controls, supply-chain integrity, and bounded side channels.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-17-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for GPU and Accelerator Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-17-005	When device health, memory integrity, driver compatibility, partition isolation, or capacity becomes unsafe, the platform SHALL drain the device, invalidate affected work, preserve checkpoints, route only to an equivalent certified profile, and open hardware evidence; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-17-006	Production admission and continued operation SHALL require device qualification, isolation tests, topology benchmarks, fault injection, attestation, and workload-quality comparisons, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Device health, memory integrity, driver compatibility, partition isolation, or capacity becomes unsafe	Drain the device, invalidate affected work, preserve checkpoints, route only to an equivalent certified profile, and open hardware evidence.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared accelerator model, firmware and driver stack, partition, topology, memory, health, allocation, tenancy, and attestation.
- Positive and negative boundary tests prove that accelerator sharing must preserve memory isolation, workload identity, timing controls, supply-chain integrity, and bounded side channels.
- Fault exercises inject device health, memory integrity, driver compatibility, partition isolation, or capacity becomes unsafe and verify that the platform can drain the device, invalidate affected work, preserve checkpoints, route only to an equivalent certified profile, and open hardware evidence.
- Independent review accepts device qualification, isolation tests, topology benchmarks, fault injection, attestation, and workload-quality comparisons for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

GPU and Accelerator Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 11, Ch. 53, Ch. 56, Ch. 58, Ch. 59. Primary Volume 5 AI anchors: Ch. 15, Ch. 16, Ch. 22, Ch. 62, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-007 · C-034 · C-038 · C-045 · C-048

18 Model Serving Infrastructure

Model Serving Infrastructure establishes the infrastructure necessary to serve approved model artifacts through stable gateways, immutable runtime images, bounded context, autoscaling, and evidence-rich inference endpoints. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

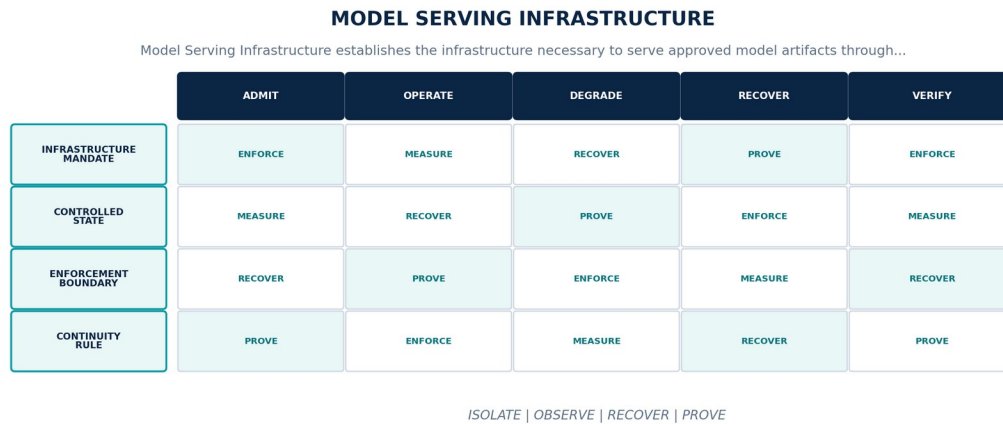


Figure 19. Model Serving Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Serve approved model artifacts through stable gateways, immutable runtime images, bounded context, autoscaling, and evidence-rich inference endpoints.
Controlled state	Serving deployment, model and adapter digest, runtime, hardware profile, endpoint, rollout, capacity, fitness, and revocation state.
Enforcement boundary	Serving infrastructure may execute only registry-resolved artifacts and may not alter prompts, context, outputs, or safety controls outside the release manifest.
Continuity rule	When artifact resolution, serving health, calibration, capacity, or safety posture becomes invalid, the platform shall stop new admissions, drain or revoke the affected revision, invoke the declared fallback, and preserve exact inference evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-18-001	The infrastructure realization of Model Serving Infrastructure SHALL serve approved model artifacts through stable gateways, immutable runtime images, bounded context, autoscaling, and evidence-rich inference endpoints.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-18-002	Every production instance SHALL declare and continuously reconcile serving deployment, model and adapter digest, runtime, hardware profile, endpoint, rollout, capacity, fitness, and revocation state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-18-003	The implementation SHALL enforce this boundary: serving infrastructure may execute only registry-resolved artifacts and may not alter prompts, context, outputs, or safety controls outside the release manifest.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, contained, and audited.
IA-18-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Model Serving Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-18-005	When artifact resolution, serving health, calibration, capacity, or safety posture becomes invalid, the platform SHALL stop new admissions, drain or revoke the affected revision, invoke the declared fallback, and preserve exact inference evidence; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-18-006	Production admission and continued operation SHALL require artifact attestation, golden inference fixtures, load and failover tests, canary analysis, revocation propagation, and replay, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Artifact resolution, serving health, calibration, capacity, or safety posture becomes invalid	Stop new admissions, drain or revoke the affected revision, invoke the declared fallback, and preserve exact inference evidence.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared serving deployment, model and adapter digest, runtime, hardware profile, endpoint, rollout, capacity, fitness, and revocation state.
- Positive and negative boundary tests prove that serving infrastructure may execute only registry-resolved artifacts and may not alter prompts, context, outputs, or safety controls outside the release manifest.
- Fault exercises inject artifact resolution, serving health, calibration, capacity, or safety posture becomes invalid and verify that the platform can stop new admissions, drain or revoke the affected revision, invoke the declared fallback, and preserve exact inference evidence.
- Independent review accepts artifact attestation, golden inference fixtures, load and failover tests, canary analysis, revocation propagation, and replay for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Model Serving Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 43, Ch. 44, Ch. 52, Ch. 58, Ch. 61. Primary Volume 5 AI anchors: Ch. 15, Ch. 16, Ch. 17, Ch. 20, Ch. 22, Ch. 70.

19 Agent and Workflow Runtime Infrastructure

Agent and Workflow Runtime Infrastructure establishes the infrastructure necessary to execute agents and durable workflows as identity-bound, resource-bounded, checkpointed, supervised, and recoverable runtime units. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

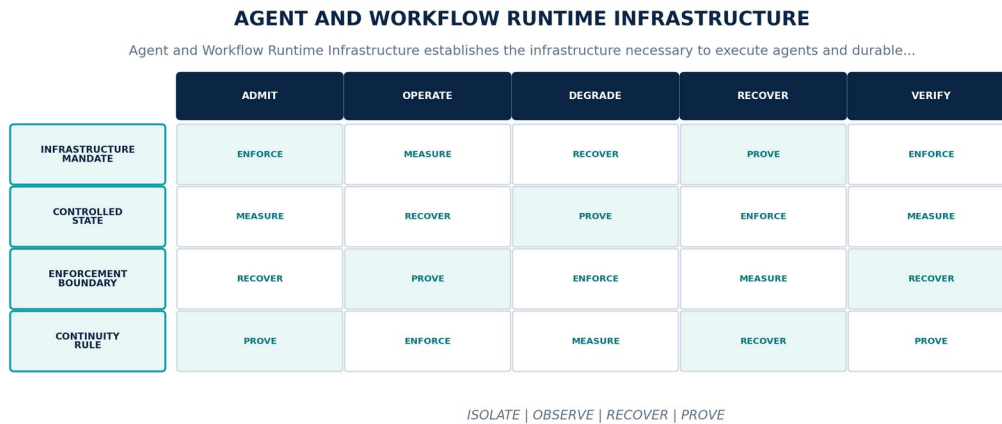


Figure 20. Agent and Workflow Runtime Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Execute agents and durable workflows as identity-bound, resource-bounded, checkpointed, supervised, and recoverable runtime units.
Controlled state	Workflow instance, agent package, execution envelope, sandbox, capability grants, checkpoint, queue state, and terminal disposition.
Enforcement boundary	Agent processes may request capabilities but infrastructure enforces network, file, secret, tool, budget, deadline, and side-effect boundaries.
Continuity rule	When a worker, supervisor, queue, checkpoint, or capability service fails during material work, the platform shall suspend or reassign from committed state, prevent duplicate effects, preserve causation, and escalate unresolved terminal state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-19-001	The infrastructure realization of Agent and Workflow Runtime Infrastructure SHALL execute agents and durable workflows as identity-bound, resource-bounded, checkpointed, supervised, and recoverable runtime units.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-19-002	Every production instance SHALL declare and continuously reconcile workflow instance, agent package, execution envelope, sandbox, capability grants, checkpoint, queue state, and terminal disposition through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-19-003	The implementation SHALL enforce this boundary: agent processes may request capabilities but infrastructure enforces network, file, secret, tool, budget, deadline, and side-effect boundaries.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-19-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Agent and Workflow Runtime Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-19-005	When a worker, supervisor, queue, checkpoint, or capability service fails during material work, the platform SHALL suspend or reassign from committed state, prevent duplicate effects, preserve causation, and escalate unresolved terminal state; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-19-006	Production admission and continued operation SHALL require worker-loss, duplicate-delivery, checkpoint-restore, capability-denial, human-gate, and replay tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A worker, supervisor, queue, checkpoint, or capability service fails during material work	Suspend or reassign from committed state, prevent duplicate effects, preserve causation, and escalate unresolved terminal state.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared workflow instance, agent package, execution envelope, sandbox, capability grants, checkpoint, queue state, and terminal disposition.
- Positive and negative boundary tests prove that agent processes may request capabilities but infrastructure enforces network, file, secret, tool, budget, deadline, and side-effect boundaries.
- Fault exercises inject a worker, supervisor, queue, checkpoint, or capability service fails during material work and verify that the platform can suspend or reassign from committed state, prevent duplicate effects, preserve causation, and escalate unresolved terminal state.
- Independent review accepts worker-loss, duplicate-delivery, checkpoint-restore, capability-denial, human-gate, and replay tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Agent and Workflow Runtime Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 41, Ch. 42, Ch. 43, Ch. 49, Ch. 59. Primary Volume 5 AI anchors: Ch. 30, Ch. 31, Ch. 33, Ch. 34, Ch. 36, Ch. 38.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-028 · C-035

20 Batch and Stream Processing Infrastructure

Batch and Stream Processing Infrastructure establishes the infrastructure necessary to process bounded datasets and continuous event flows with explicit identity, lineage, watermarks, checkpoints, backpressure, and quality gates. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

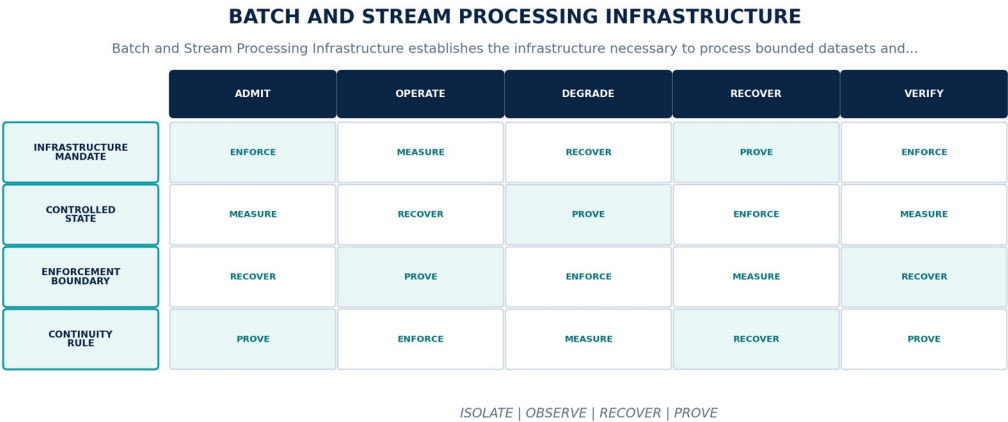


Figure 21. Batch and Stream Processing Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Process bounded datasets and continuous event flows with explicit identity, lineage, watermarks, checkpoints, backpressure, and quality gates.
Controlled state	Job or stream identity, input revisions, partition set, watermark, checkpoint, output commit, quality state, and replay cursor.
Enforcement boundary	Processing infrastructure must separate compute completion from canonical output admission and preserve deterministic or declared nondeterministic behavior.
Continuity rule	When partitions lag, reorder, duplicate, corrupt, or become unavailable beyond the declared window, the platform shall apply backpressure, quarantine affected partitions, resume from committed state, and mark output incomplete or stale until reconciled.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-20-001	The infrastructure realization of Batch and Stream Processing Infrastructure SHALL process bounded datasets and continuous event flows with explicit identity, lineage, watermarks, checkpoints, backpressure, and quality gates.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-20-002	Every production instance SHALL declare and continuously reconcile job or stream identity, input revisions, partition set, watermark, checkpoint, output commit, quality state, and replay cursor through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-20-003	The implementation SHALL enforce this boundary: processing infrastructure must separate compute completion from canonical output admission and preserve deterministic or declared nondeterministic behavior.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.
IA-20-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Batch and Stream Processing Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-20-005	When partitions lag, reorder, duplicate, corrupt, or become unavailable beyond the declared window, the platform SHALL apply backpressure, quarantine affected partitions, resume from committed state, and mark output incomplete or stale until reconciled; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-20-006	Production admission and continued operation SHALL require replay equivalence, watermark and late-data tests, partition loss, backpressure, lineage closure, and output admission evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Partitions lag, reorder, duplicate, corrupt, or become unavailable beyond the declared window	Apply backpressure, quarantine affected partitions, resume from committed state, and mark output incomplete or stale until reconciled.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared job or stream identity, input revisions, partition set, watermark, checkpoint, output commit, quality state, and replay cursor.
- Positive and negative boundary tests prove that processing infrastructure must separate compute completion from canonical output admission and preserve deterministic or declared nondeterministic behavior.
- Fault exercises inject partitions lag, reorder, duplicate, corrupt, or become unavailable beyond the declared window and verify that the platform can apply backpressure, quarantine affected partitions, resume from committed state, and mark output incomplete or stale until reconciled.
- Independent review accepts replay equivalence, watermark and late-data tests, partition loss, backpressure, lineage closure, and output admission evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Batch and Stream Processing Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 22, Ch. 28, Ch. 41, Ch. 57. Primary Volume 5 AI anchors: Ch. 19, Ch. 23, Ch. 26, Ch. 66, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-011 · C-016 · C-035 · C-037 · C-041

21 Edge, IoT, and Remote Collection Infrastructure

Edge, IoT, and Remote Collection Infrastructure establishes the infrastructure necessary to collect and process remote signals under intermittent connectivity, constrained power, hostile networks, device diversity, and delayed custody transfer. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

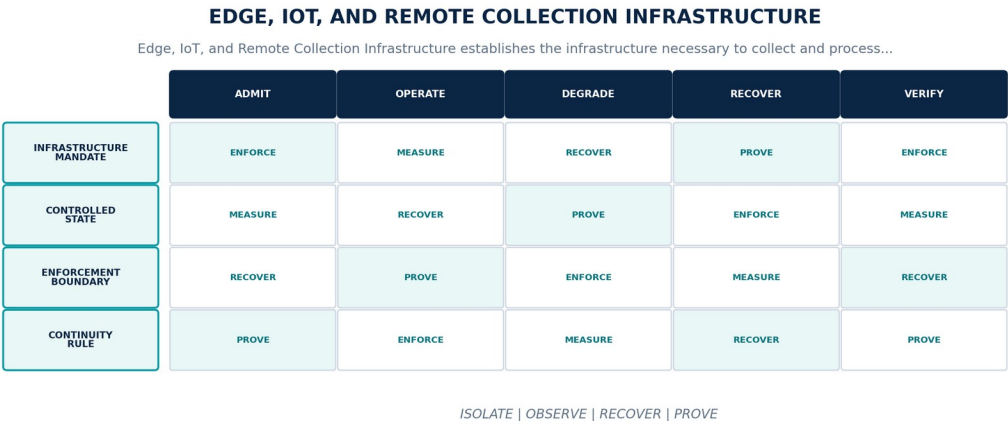


Figure 22. Edge, IoT, and Remote Collection Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Collect and process remote signals under intermittent connectivity, constrained power, hostile networks, device diversity, and delayed custody transfer.
Controlled state	Device identity, firmware, sensor profile, locality, clock, collection cursor, buffer, custody state, health, and revocation.
Enforcement boundary	Edge nodes may filter or summarize only through signed policy and must retain mapping to source observations and transfer evidence.
Continuity rule	When device trust, clock, connectivity, local storage, or custody evidence is lost, the platform shall stop or quarantine collection as required, preserve buffered evidence, expose gaps, and reconcile after trust is restored.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-21-001	The infrastructure realization of Edge, IoT, and Remote Collection Infrastructure SHALL collect and process remote signals under intermittent connectivity, constrained power, hostile networks, device diversity, and delayed custody transfer.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-21-002	Every production instance SHALL declare and continuously reconcile device identity, firmware, sensor profile, locality, clock, collection cursor, buffer, custody state, health, and revocation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-21-003	The implementation SHALL enforce this boundary: edge nodes may filter or summarize only through signed policy and must retain mapping to source observations and transfer evidence.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-21-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Edge, IoT, and Remote Collection Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-21-005	When device trust, clock, connectivity, local storage, or custody evidence is lost, the platform SHALL stop or quarantine collection as required, preserve buffered evidence, expose gaps, and reconcile after trust is restored; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-21-006	Production admission and continued operation SHALL require device attestation, offline buffering, clock-drift, tamper, delayed upload, duplicate suppression, and revocation exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Device trust, clock, connectivity, local storage, or custody evidence is lost	Stop or quarantine collection as required, preserve buffered evidence, expose gaps, and reconcile after trust is restored.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared device identity, firmware, sensor profile, locality, clock, collection cursor, buffer, custody state, health, and revocation.
- Positive and negative boundary tests prove that edge nodes may filter or summarize only through signed policy and must retain mapping to source observations and transfer evidence.
- Fault exercises inject device trust, clock, connectivity, local storage, or custody evidence is lost and verify that the platform can stop or quarantine collection as required, preserve buffered evidence, expose gaps, and reconcile after trust is restored.
- Independent review accepts device attestation, offline buffering, clock-drift, tamper, delayed upload, duplicate suppression, and revocation exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Edge, IoT, and Remote Collection Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 25, Ch. 26, Ch. 28, Ch. 56. Primary Volume 5 AI anchors: Ch. 7, Ch. 21, Ch. 26, Ch. 39, Ch. 60.

22 Resource Scheduling, Quotas, and Admission

Resource Scheduling, Quotas, and Admission establishes the infrastructure necessary to admit infrastructure work according to consequence, priority, deadline, resource fit, tenant entitlement, fairness, and current system risk. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

RESOURCE SCHEDULING, QUOTAS, AND ADMISSION

Resource Scheduling, Quotas, and Admission establishes the infrastructure necessary to admit infrastructure work...

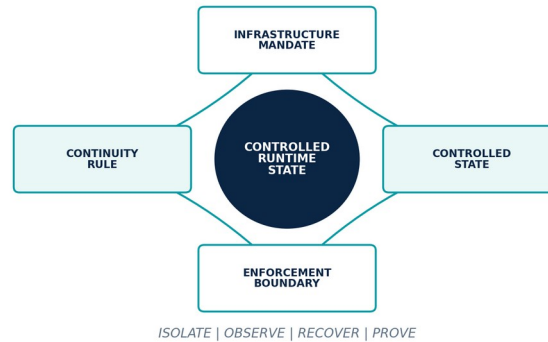


Figure 23. Resource Scheduling, Quotas, and Admission. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Admit infrastructure work according to consequence, priority, deadline, resource fit, tenant entitlement, fairness, and current system risk.
Controlled state	Admission decision, workload class, priority, quota, reservation, budget, preemption policy, queue position, and reason.
Enforcement boundary	Schedulers must not use customer sensitivity, tenant identity, or strategic outcome as an undocumented optimization signal.
Continuity rule	When demand exceeds safe capacity or quota, fairness, deadline, and priority objectives conflict, the platform shall apply the declared admission hierarchy, preserve rejected and queued states, preempt only eligible work, and notify accountable owners.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-22-001	The infrastructure realization of Resource Scheduling, Quotas, and Admission SHALL admit infrastructure work according to consequence, priority, deadline, resource fit, tenant entitlement, fairness, and current system risk.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-22-002	Every production instance SHALL declare and continuously reconcile admission decision, workload class, priority, quota, reservation, budget, preemption policy, queue position, and reason through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-22-003	The implementation SHALL enforce this boundary: schedulers must not use customer sensitivity, tenant identity, or strategic outcome as an undocumented optimization signal.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-22-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Resource Scheduling, Quotas, and Admission.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-22-005	When demand exceeds safe capacity or quota, fairness, deadline, and priority objectives conflict, the platform SHALL apply the declared admission hierarchy, preserve rejected and queued states, preempt only eligible work, and notify accountable owners; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-22-006	Production admission and continued operation SHALL require overload games, fairness and starvation analysis, quota isolation, priority inversion tests, and signed admission-policy review, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Demand exceeds safe capacity or quota, fairness, deadline, and priority objectives conflict	Apply the declared admission hierarchy, preserve rejected and queued states, preempt only eligible work, and notify accountable owners.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared admission decision, workload class, priority, quota, reservation, budget, preemption policy, queue position, and reason.
- Positive and negative boundary tests prove that schedulers must not use customer sensitivity, tenant identity, or strategic outcome as an undocumented optimization signal.
- Fault exercises inject demand exceeds safe capacity or quota, fairness, deadline, and priority objectives conflict and verify that the platform can apply the declared admission hierarchy, preserve rejected and queued states, preempt only eligible work, and notify accountable owners.
- Independent review accepts overload games, fairness and starvation analysis, quota isolation, priority inversion tests, and signed admission-policy review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Resource Scheduling, Quotas, and Admission is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 9, Ch. 13, Ch. 21, Ch. 58, Ch. 59. Primary Volume 5 AI anchors: Ch. 19, Ch. 22, Ch. 31, Ch. 63, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-004 · C-006 · C-035 · C-038 · C-041

PART IV

Network and Connectivity

Segmentation, ingress, service connectivity, plane separation, naming, egress, private connectivity, and resilience.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

23 Network Architecture and Segmentation

Network Architecture and Segmentation establishes the infrastructure necessary to partition connectivity by trust, tenant, domain, plane, environment, workload class, data sensitivity, and external reachability. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

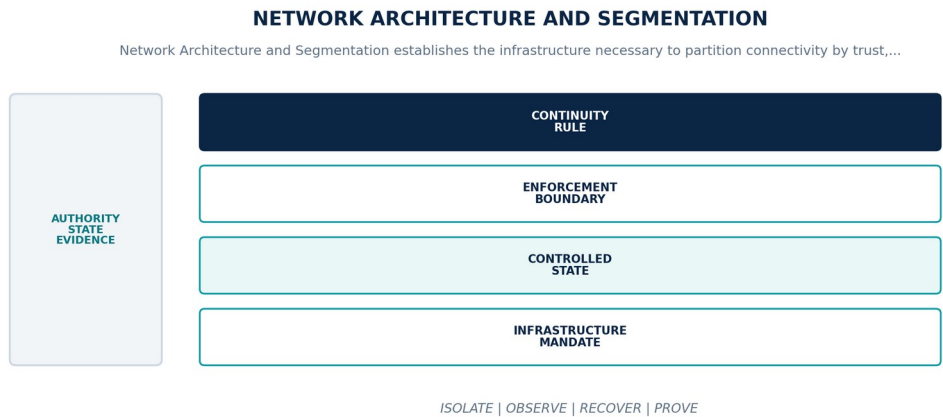


Figure 24. Network Architecture and Segmentation. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Partition connectivity by trust, tenant, domain, plane, environment, workload class, data sensitivity, and external reachability.
Controlled state	Network zone, segment, route domain, policy, attachment, flow identity, inspection state, and approved exception.
Enforcement boundary	Network location is never sufficient identity; every permitted flow requires authenticated workload context and policy.
Continuity rule	When segmentation state diverges, an unknown route appears, or east-west exposure exceeds the declared graph, the platform shall deny or isolate suspect flows, preserve flow evidence, reconcile desired and actual state, and assess lateral impact.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-23-001	The infrastructure realization of Network Architecture and Segmentation SHALL partition connectivity by trust, tenant, domain, plane, environment, workload class, data sensitivity, and external reachability.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-23-002	Every production instance SHALL declare and continuously reconcile network zone, segment, route domain, policy, attachment, flow identity, inspection state, and approved exception through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-23-003	The implementation SHALL enforce this boundary: network location is never sufficient identity; every permitted flow requires authenticated workload context and policy.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-23-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Network Architecture and Segmentation.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-23-005	When segmentation state diverges, an unknown route appears, or east-west exposure exceeds the declared graph, the platform SHALL deny or isolate suspect flows, preserve flow evidence, reconcile desired and actual state, and assess lateral impact; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-23-006	Production admission and continued operation SHALL require reachability matrices, deny-by-default probes, route-drift detection, lateral-movement exercises, and policy attestation, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Segmentation state diverges, an unknown route appears, or east-west exposure exceeds the declared graph	Deny or isolate suspect flows, preserve flow evidence, reconcile desired and actual state, and assess lateral impact.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared network zone, segment, route domain, policy, attachment, flow identity, inspection state, and approved exception.
- Positive and negative boundary tests prove that network location is never sufficient identity; every permitted flow requires authenticated workload context and policy.
- Fault exercises inject segmentation state diverges, an unknown route appears, or east-west exposure exceeds the declared graph and verify that the platform can deny or isolate suspect flows, preserve flow evidence, reconcile desired and actual state, and assess lateral impact.
- Independent review accepts reachability matrices, deny-by-default probes, route-drift detection, lateral-movement exercises, and policy attestation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Network Architecture and Segmentation is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 13, Ch. 49, Ch. 53, Ch. 54, Ch. 60. Primary Volume 5 AI anchors: Ch. 21, Ch. 36, Ch. 60, Ch. 65.

CONSTITUTIONAL INHERITANCE / C-007 · C-030 · C-034 · C-036 · C-049

24 Ingress, API, and Access Edge

Ingress, API, and Access Edge establishes the infrastructure necessary to terminate external access through authenticated, rate-governed, policy-enforced, observable, and resilient ingress points. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

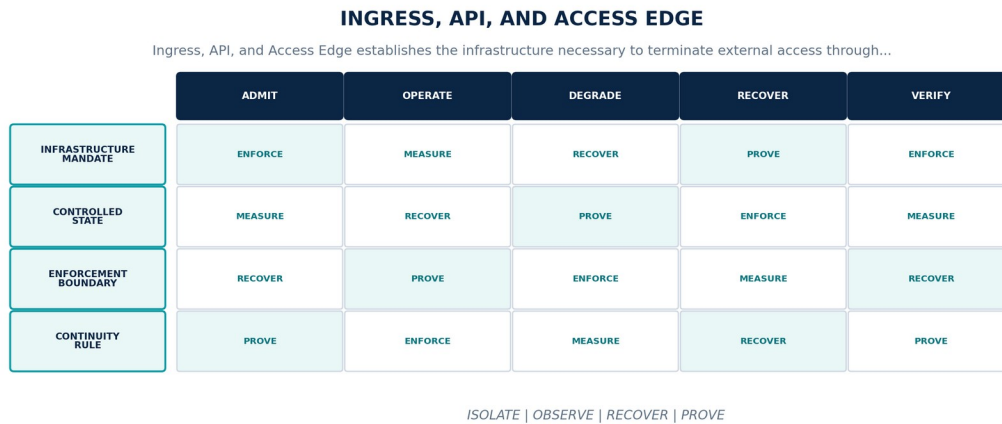


Figure 25. Ingress, API, and Access Edge. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Terminate external access through authenticated, rate-governed, policy-enforced, observable, and resilient ingress points.
Controlled state	Edge endpoint, certificate, route, API contract, identity method, policy, rate class, protection profile, and health.
Enforcement boundary	Ingress may validate and route but cannot reinterpret canonical contracts, authorize final decisions, or conceal downstream partial failure.
Continuity rule	When identity, certificate, gateway, protection, or downstream route becomes unavailable or suspect, the platform shall fail closed for protected access, preserve idempotency and request evidence, and expose retrievable, denied, or degraded state explicitly.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-24-001	The infrastructure realization of Ingress, API, and Access Edge SHALL terminate external access through authenticated, rate-governed, policy-enforced, observable, and resilient ingress points.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-24-002	Every production instance SHALL declare and continuously reconcile edge endpoint, certificate, route, API contract, identity method, policy, rate class, protection profile, and health through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-24-003	The implementation SHALL enforce this boundary: ingress may validate and route but cannot reinterpret canonical contracts, authorize final decisions, or conceal downstream partial failure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-24-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Ingress, API, and Access Edge.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-24-005	When identity, certificate, gateway, protection, or downstream route becomes unavailable or suspect, the platform SHALL fail closed for protected access, preserve idempotency and request evidence, and expose retrievable, denied, or degraded state explicitly; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-24-006	Production admission and continued operation SHALL require protocol conformance, abuse and rate tests, certificate rotation, gateway failover, request replay, and policy traces, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity, certificate, gateway, protection, or downstream route becomes unavailable or suspect	Fail closed for protected access, preserve idempotency and request evidence, and expose retrievable, denied, or degraded state explicitly.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared edge endpoint, certificate, route, API contract, identity method, policy, rate class, protection profile, and health.
- Positive and negative boundary tests prove that ingress may validate and route but cannot reinterpret canonical contracts, authorize final decisions, or conceal downstream partial failure.
- Fault exercises inject identity, certificate, gateway, protection, or downstream route becomes unavailable or suspect and verify that the platform can fail closed for protected access, preserve idempotency and request evidence, and expose retrievable, denied, or degraded state explicitly.
- Independent review accepts protocol conformance, abuse and rate tests, certificate rotation, gateway failover, request replay, and policy traces for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Ingress, API, and Access Edge is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 16, Ch. 20, Ch. 49, Ch. 60. Primary Volume 5 AI anchors: Ch. 7, Ch. 16, Ch. 20, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-007 · C-020 · C-030 · C-035 · C-049

25 Service-to-Service Connectivity

Service-to-Service Connectivity establishes the infrastructure necessary to connect workloads through mutually authenticated, policy-authorized, version-aware, observable service identities. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

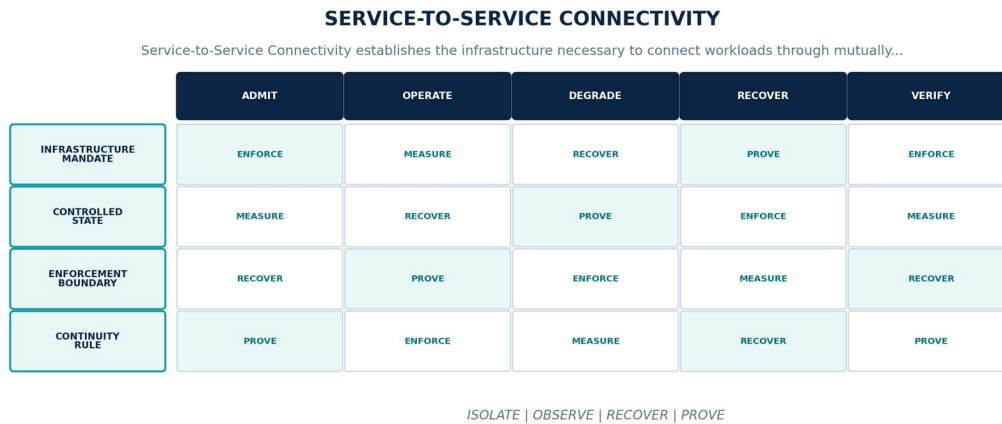


Figure 26. Service-to-Service Connectivity. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Connect workloads through mutually authenticated, policy-authorized, version-aware, observable service identities.
Controlled state	Workload identity, service endpoint, trust domain, contract version, authorization decision, connection, and telemetry state.
Enforcement boundary	Service discovery and encryption establish transport; owning services remain responsible for object, purpose, tenant, and consequence authorization.
Continuity rule	When identity issuance, discovery, trust bundle, policy, or service endpoint becomes inconsistent, the platform shall deny new protected sessions, drain affected connections where safe, use only certified fallbacks, and preserve causal diagnostics.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-25-001	The infrastructure realization of Service-to-Service Connectivity SHALL connect workloads through mutually authenticated, policy-authorized, version-aware, observable service identities.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-25-002	Every production instance SHALL declare and continuously reconcile workload identity, service endpoint, trust domain, contract version, authorization decision, connection, and telemetry state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-25-003	The implementation SHALL enforce this boundary: service discovery and encryption establish transport; owning services remain responsible for object, purpose, tenant, and consequence authorization.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-25-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Service-to-Service Connectivity.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-25-005	When identity issuance, discovery, trust bundle, policy, or service endpoint becomes inconsistent, the platform SHALL deny new protected sessions, drain affected connections where safe, use only certified fallbacks, and preserve causal diagnostics; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-25-006	Production admission and continued operation SHALL require mutual-authentication tests, identity rotation, policy denial, endpoint failover, contract mismatch, and trace continuity, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity issuance, discovery, trust bundle, policy, or service endpoint becomes inconsistent	Deny new protected sessions, drain affected connections where safe, use only certified fallbacks, and preserve causal diagnostics.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared workload identity, service endpoint, trust domain, contract version, authorization decision, connection, and telemetry state.
- Positive and negative boundary tests prove that service discovery and encryption establish transport; owning services remain responsible for object, purpose, tenant, and consequence authorization.
- Fault exercises inject identity issuance, discovery, trust bundle, policy, or service endpoint becomes inconsistent and verify that the platform can deny new protected sessions, drain affected connections where safe, use only certified fallbacks, and preserve causal diagnostics.
- Independent review accepts mutual-authentication tests, identity rotation, policy denial, endpoint failover, contract mismatch, and trace continuity for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Service-to-Service Connectivity is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 11, Ch. 15, Ch. 23, Ch. 49, Ch. 53. Primary Volume 5 AI anchors: Ch. 16, Ch. 17, Ch. 21, Ch. 36.

26 Data Plane and Control Plane Separation

Data Plane and Control Plane Separation establishes the infrastructure necessary to separate customer data processing from policy, identity, configuration, release, telemetry, and fleet control with explicit mediation. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

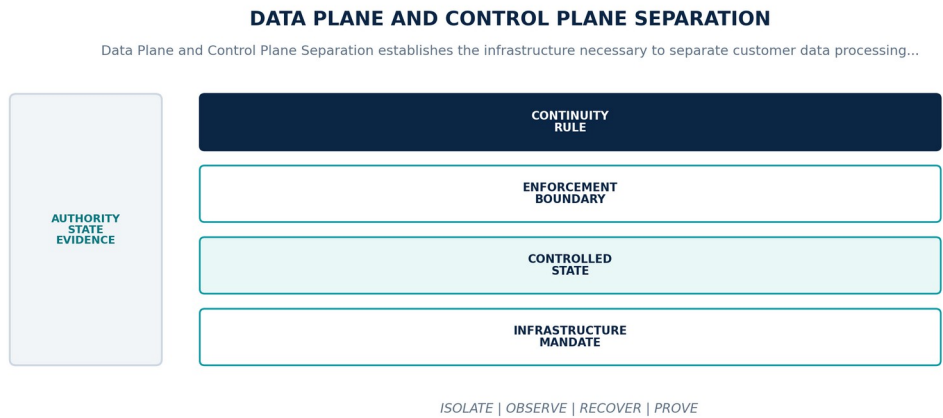


Figure 27. Data Plane and Control Plane Separation. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Separate customer data processing from policy, identity, configuration, release, telemetry, and fleet control with explicit mediation.
Controlled state	Plane identity, endpoint, permitted command or telemetry class, direction, authentication, retention, and emergency state.
Enforcement boundary	Control-plane compromise must not yield unrestricted customer-data access, and data-plane compromise must not rewrite governing control state.
Continuity rule	When plane boundaries blur, a command lacks authority, or telemetry contains prohibited customer content, the platform shall block the channel, revoke affected credentials, preserve evidence, continue approved local policy, and initiate compromise assessment.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-26-001	The infrastructure realization of Data Plane and Control Plane Separation SHALL separate customer data processing from policy, identity, configuration, release, telemetry, and fleet control with explicit mediation.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-26-002	Every production instance SHALL declare and continuously reconcile plane identity, endpoint, permitted command or telemetry class, direction, authentication, retention, and emergency state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-26-003	The implementation SHALL enforce this boundary: control-plane compromise must not yield unrestricted customer-data access, and data-plane compromise must not rewrite governing control state.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-26-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Data Plane and Control Plane Separation.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-26-005	When plane boundaries blur, a command lacks authority, or telemetry contains prohibited customer content, the platform SHALL block the channel, revoke affected credentials, preserve evidence, continue approved local policy, and initiate compromise assessment; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-26-006	Production admission and continued operation SHALL require one-way flow tests, command authorization, telemetry minimization, control-plane outage, and compromise containment, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Plane boundaries blur, a command lacks authority, or telemetry contains prohibited customer content	Block the channel, revoke affected credentials, preserve evidence, continue approved local policy, and initiate compromise assessment.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared plane identity, endpoint, permitted command or telemetry class, direction, authentication, retention, and emergency state.
- Positive and negative boundary tests prove that control-plane compromise must not yield unrestricted customer-data access, and data-plane compromise must not rewrite governing control state.
- Fault exercises inject plane boundaries blur, a command lacks authority, or telemetry contains prohibited customer content and verify that the platform can block the channel, revoke affected credentials, preserve evidence, continue approved local policy, and initiate compromise assessment.
- Independent review accepts one-way flow tests, command authorization, telemetry minimization, control-plane outage, and compromise containment for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Data Plane and Control Plane Separation is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 10, Ch. 13, Ch. 29, Ch. 49, Ch. 54. Primary Volume 5 AI anchors: Ch. 10, Ch. 11, Ch. 12, Ch. 21, Ch. 65.

CONSTITUTIONAL INHERITANCE / C-007 · C-020 · C-030 · C-036 · C-049

27 DNS, Naming, Discovery, and Time

DNS, Naming, Discovery, and Time establishes the infrastructure necessary to provide authoritative names, service discovery, endpoint health, and trusted time as governed infrastructure dependencies. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

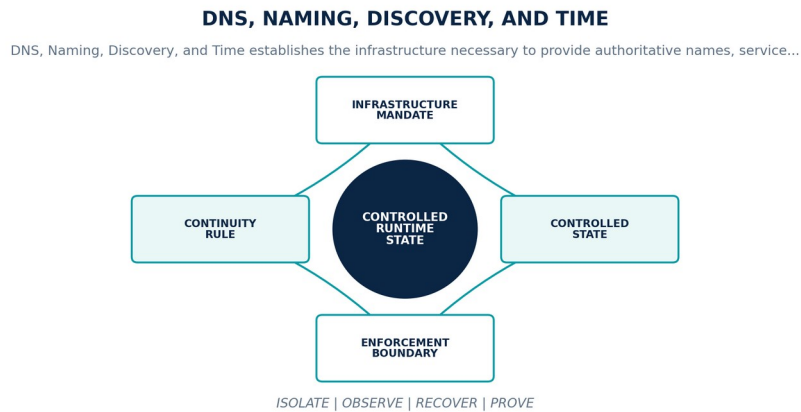


Figure 28. DNS, Naming, Discovery, and Time. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Provide authoritative names, service discovery, endpoint health, and trusted time as governed infrastructure dependencies.
Controlled state	Name, namespace, record version, endpoint set, health, time source, clock quality, lease, and provenance.
Enforcement boundary	Names and clocks must be tenant-aware where required, resistant to spoofing, and reconstructible for historical replay.
Continuity rule	When name resolution, discovery consistency, or trusted time falls outside the declared tolerance, the platform shall deny freshness-sensitive or authority-changing operations, use bounded cached state, expose uncertainty, and reconcile histories.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-27-001	The infrastructure realization of DNS, Naming, Discovery, and Time SHALL provide authoritative names, service discovery, endpoint health, and trusted time as governed infrastructure dependencies.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-27-002	Every production instance SHALL declare and continuously reconcile name, namespace, record version, endpoint set, health, time source, clock quality, lease, and provenance through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-27-003	The implementation SHALL enforce this boundary: names and clocks must be tenant-aware where required, resistant to spoofing, and reconstructible for historical replay.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-27-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for DNS, Naming, Discovery, and Time.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-27-005	When name resolution, discovery consistency, or trusted time falls outside the declared tolerance, the platform SHALL deny freshness-sensitive or authority-changing operations, use bounded cached state, expose uncertainty, and reconcile histories; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-27-006	Production admission and continued operation SHALL require split-horizon tests, poisoning resistance, record rollback, discovery partition, clock-step and drift exercises, and replay, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Name resolution, discovery consistency, or trusted time falls outside the declared tolerance	Deny freshness-sensitive or authority-changing operations, use bounded cached state, expose uncertainty, and reconcile histories.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared name, namespace, record version, endpoint set, health, time source, clock quality, lease, and provenance.
- Positive and negative boundary tests prove that names and clocks must be tenant-aware where required, resistant to spoofing, and reconstructible for historical replay.
- Fault exercises inject name resolution, discovery consistency, or trusted time falls outside the declared tolerance and verify that the platform can deny freshness-sensitive or authority-changing operations, use bounded cached state, expose uncertainty, and reconcile histories.
- Independent review accepts split-horizon tests, poisoning resistance, record rollback, discovery partition, clock-step and drift exercises, and replay for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

DNS, Naming, Discovery, and Time is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 18, Ch. 25, Ch. 26, Ch. 53, Ch. 59. Primary Volume 5 AI anchors: Ch. 15, Ch. 26, Ch. 29, Ch. 65.

CONSTITUTIONAL INHERITANCE / C-007 · C-016 · C-017 · C-035 · C-049

28 Egress, External Integration, and Source Connectivity

Egress, External Integration, and Source Connectivity establishes the infrastructure necessary to mediate outbound connections to sources, providers, customer systems, and external effect endpoints through explicit purpose and data controls. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

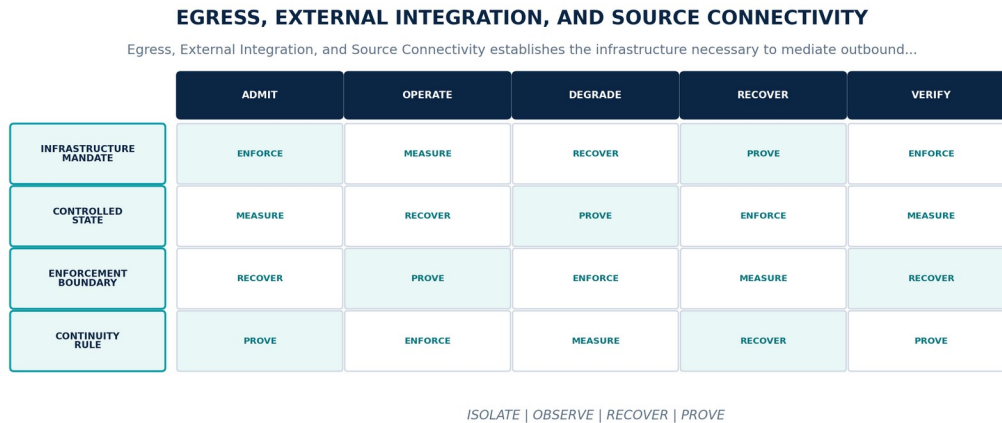


Figure 29. Egress, External Integration, and Source Connectivity. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Mediate outbound connections to sources, providers, customer systems, and external effect endpoints through explicit purpose and data controls.
Controlled state	Destination identity, route, purpose, data class, transfer policy, credential, rate, receipt, and revocation state.
Enforcement boundary	No workload receives generic internet access when a registered source, provider, connector, or capability path can express the intended operation.
Continuity rule	When destination identity changes, content is unsafe, policy denies transfer, or the external endpoint is unavailable, the platform shall block or quarantine the exchange, preserve the request and reason, retry only inside policy, and never fabricate collection or effect success.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-28-001	The infrastructure realization of Egress, External Integration, and Source Connectivity SHALL mediate outbound connections to sources, providers, customer systems, and external effect endpoints through explicit purpose and data controls.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-28-002	Every production instance SHALL declare and continuously reconcile destination identity, route, purpose, data class, transfer policy, credential, rate, receipt, and revocation state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-28-003	The implementation SHALL enforce this boundary: no workload receives generic internet access when a registered source, provider, connector, or capability path can express the intended operation.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-28-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Egress, External Integration, and Source Connectivity.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-28-005	When destination identity changes, content is unsafe, policy denies transfer, or the external endpoint is unavailable, the platform SHALL block or quarantine the exchange, preserve the request and reason, retry only inside policy, and never fabricate collection or effect success; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-28-006	Production admission and continued operation SHALL require destination pinning, exfiltration tests, malware isolation, provider outage, effect receipt, and revocation exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Destination identity changes, content is unsafe, policy denies transfer, or the external endpoint is unavailable	Block or quarantine the exchange, preserve the request and reason, retry only inside policy, and never fabricate collection or effect success.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared destination identity, route, purpose, data class, transfer policy, credential, rate, receipt, and revocation state.
- Positive and negative boundary tests prove that no workload receives generic internet access when a registered source, provider, connector, or capability path can express the intended operation.
- Fault exercises inject destination identity changes, content is unsafe, policy denies transfer, or the external endpoint is unavailable and verify that the platform can block or quarantine the exchange, preserve the request and reason, retry only inside policy, and never fabricate collection or effect success.
- Independent review accepts destination pinning, exfiltration tests, malware isolation, provider outage, effect receipt, and revocation exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Egress, External Integration, and Source Connectivity is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 13, Ch. 19, Ch. 29, Ch. 49, Ch. 60. Primary Volume 5 AI anchors: Ch. 21, Ch. 23, Ch. 36, Ch. 39, Ch. 59, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-007 · C-018 · C-029 · C-030 · C-049

29 Private Connectivity and Customer Networks

Private Connectivity and Customer Networks establishes the infrastructure necessary to join customer and platform networks through explicit routes, overlapping-address controls, identity mediation, capacity, and shared operations. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

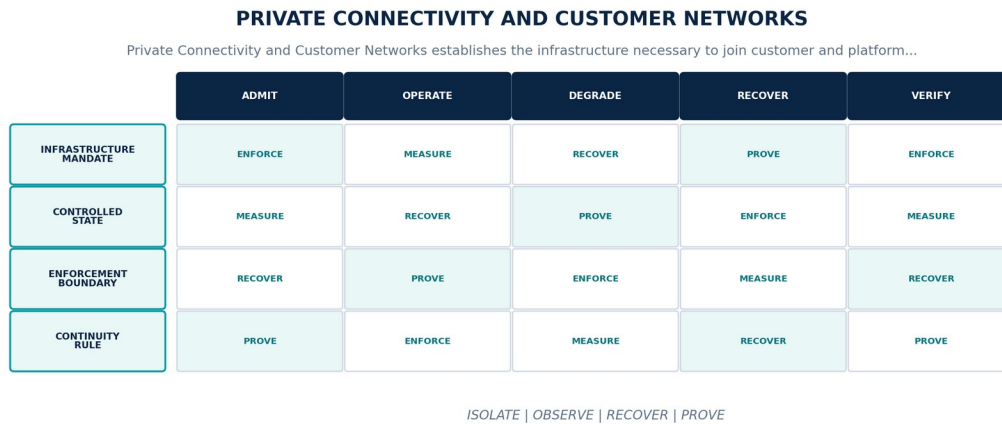


Figure 30. Private Connectivity and Customer Networks. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Join customer and platform networks through explicit routes, overlapping-address controls, identity mediation, capacity, and shared operations.
Controlled state	Connection, customer endpoint, route set, address translation, encryption, service scope, capacity, health, and responsibility owner.
Enforcement boundary	Private connectivity changes reachability but does not create trust, bypass application authorization, or transfer incident ownership.
Continuity rule	When routes overlap, tunnels flap, customer controls change, or the responsibility boundary is disputed, the platform shall limit reachability to the last verified set, expose customer and platform evidence, and execute the joint restoration runbook.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-29-001	The infrastructure realization of Private Connectivity and Customer Networks SHALL join customer and platform networks through explicit routes, overlapping-address controls, identity mediation, capacity, and shared operations.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-29-002	Every production instance SHALL declare and continuously reconcile connection, customer endpoint, route set, address translation, encryption, service scope, capacity, health, and responsibility owner through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-29-003	The implementation SHALL enforce this boundary: private connectivity changes reachability but does not create trust, bypass application authorization, or transfer incident ownership.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-29-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Private Connectivity and Customer Networks.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-29-005	When routes overlap, tunnels flap, customer controls change, or the responsibility boundary is disputed, the platform SHALL limit reachability to the last verified set, expose customer and platform evidence, and execute the joint restoration runbook; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-29-006	Production admission and continued operation SHALL require route and failover tests, overlap simulations, MTU and latency validation, joint incident drills, and responsibility sign-off, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Routes overlap, tunnels flap, customer controls change, or the responsibility boundary is disputed	Limit reachability to the last verified set, expose customer and platform evidence, and execute the joint restoration runbook.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared connection, customer endpoint, route set, address translation, encryption, service scope, capacity, health, and responsibility owner.
- Positive and negative boundary tests prove that private connectivity changes reachability but does not create trust, bypass application authorization, or transfer incident ownership.
- Fault exercises inject routes overlap, tunnels flap, customer controls change, or the responsibility boundary is disputed and verify that the platform can limit reachability to the last verified set, expose customer and platform evidence, and execute the joint restoration runbook.
- Independent review accepts route and failover tests, overlap simulations, MTU and latency validation, joint incident drills, and responsibility sign-off for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Private Connectivity and Customer Networks is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 8, Ch. 10, Ch. 14, Ch. 53, Ch. 63. Primary Volume 5 AI anchors: Ch. 7, Ch. 12, Ch. 21, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-006 · C-007 · C-036 · C-038 · C-049

30 Network Resilience, Performance, and Observability

Network Resilience, Performance, and Observability establishes the infrastructure necessary to engineer connectivity for measured latency, throughput, loss, jitter, availability, convergence, and explainable path state. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

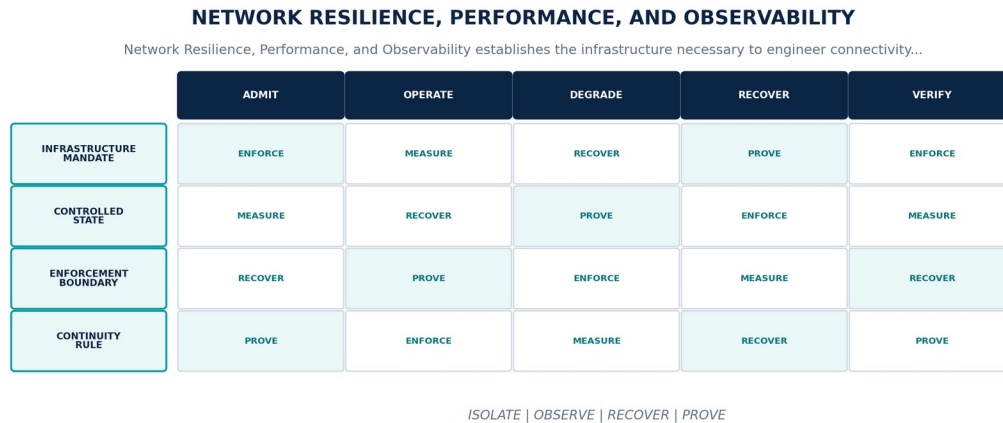


Figure 31. Network Resilience, Performance, and Observability. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Engineer connectivity for measured latency, throughput, loss, jitter, availability, convergence, and explainable path state.
Controlled state	Path identity, SLO class, capacity, latency distribution, loss, dependency, health, change, and incident correlation.
Enforcement boundary	Network optimization may not violate sovereignty, isolation, consistency, evidence capture, or deterministic routing requirements.
Continuity rule	When a path breaches its SLO, routing oscillates, observability gaps appear, or recovery causes unsafe reordering, the platform shall shift eligible traffic, constrain retries, preserve backpressure, expose degraded paths, and diagnose from end-to-end evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-30-001	The infrastructure realization of Network Resilience, Performance, and Observability SHALL engineer connectivity for measured latency, throughput, loss, jitter, availability, convergence, and explainable path state.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.
IA-30-002	Every production instance SHALL declare and continuously reconcile path identity, SLO class, capacity, latency distribution, loss, dependency, health, change, and incident correlation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-30-003	The implementation SHALL enforce this boundary: network optimization may not violate sovereignty, isolation, consistency, evidence capture, or deterministic routing requirements.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-30-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Network Resilience, Performance, and Observability.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-30-005	When a path breaches its SLO, routing oscillates, observability gaps appear, or recovery causes unsafe reordering, the platform SHALL shift eligible traffic, constrain retries, preserve backpressure, expose degraded paths, and diagnose from end-to-end evidence; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-30-006	Production admission and continued operation SHALL require load and failover tests, latency histograms, packet-loss injection, route convergence, trace correlation, and capacity forecasts, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A path breaches its SLO, routing oscillates, observability gaps appear, or recovery causes unsafe reordering	Shift eligible traffic, constrain retries, preserve backpressure, expose degraded paths, and diagnose from end-to-end evidence.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared path identity, SLO class, capacity, latency distribution, loss, dependency, health, change, and incident correlation.
- Positive and negative boundary tests prove that network optimization may not violate sovereignty, isolation, consistency, evidence capture, or deterministic routing requirements.
- Fault exercises inject a path breaches its SLO, routing oscillates, observability gaps appear, or recovery causes unsafe reordering and verify that the platform can shift eligible traffic, constrain retries, preserve backpressure, expose degraded paths, and diagnose from end-to-end evidence.
- Independent review accepts load and failover tests, latency histograms, packet-loss injection, route convergence, trace correlation, and capacity forecasts for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Network Resilience, Performance, and Observability is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 22, Ch. 58, Ch. 59, Ch. 61. Primary Volume 5 AI anchors: Ch. 22, Ch. 31, Ch. 68, Ch. 72.

PART V

Storage, Messaging, and State

State classes, operational stores, evidence, graph, search, twins, streaming, queues, caches, and checkpoints.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

31 Storage Architecture and State Classes

Storage Architecture and State Classes establishes the infrastructure necessary to map every infrastructure state class to an authoritative owner, consistency model, durability, locality, protection, retention, and recovery contract. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

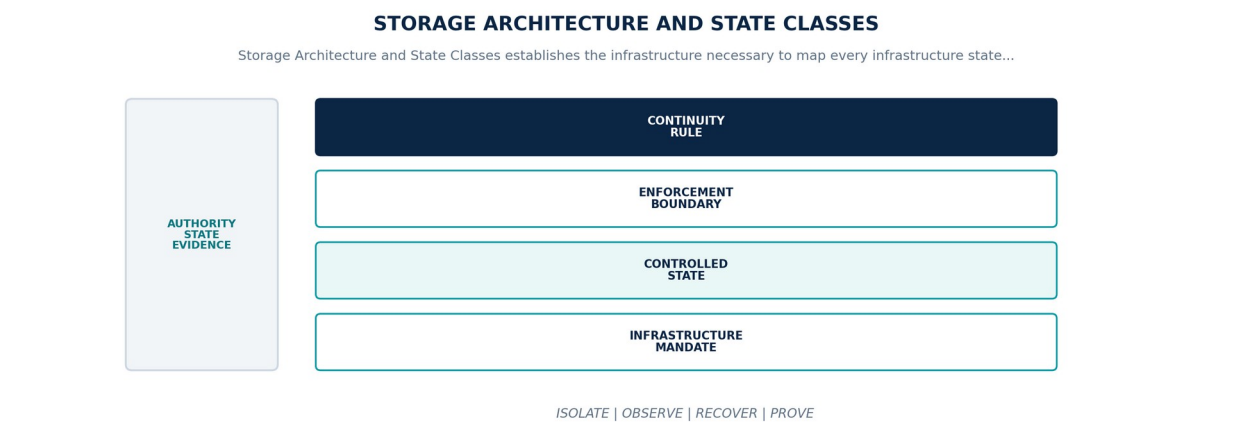


Figure 32. Storage Architecture and State Classes. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Map every infrastructure state class to an authoritative owner, consistency model, durability, locality, protection, retention, and recovery contract.
Controlled state	State-class identity, owner, store, schema, partition key, consistency, replication, retention, backup, recovery, and lifecycle.
Enforcement boundary	Storage technology is selected behind the state contract and may not merge canonical truth, derived projections, cache, evidence, or temporary execution state.
Continuity rule	When state cannot be classified, ownership is ambiguous, or actual durability diverges from the declared class, the platform shall block new writes or admission, preserve available history, expose indeterminate state, and restore ownership before migration.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-31-001	The infrastructure realization of Storage Architecture and State Classes SHALL map every infrastructure state class to an authoritative owner, consistency model, durability, locality, protection, retention, and recovery contract.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-31-002	Every production instance SHALL declare and continuously reconcile state-class identity, owner, store, schema, partition key, consistency, replication, retention, backup, recovery, and lifecycle through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-31-003	The implementation SHALL enforce this boundary: storage technology is selected behind the state contract and may not merge canonical truth, derived projections, cache, evidence, or temporary execution state.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, contained, and audited.
IA-31-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Storage Architecture and State Classes.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-31-005	When state cannot be classified, ownership is ambiguous, or actual durability diverges from the declared class, the platform SHALL block new writes or admission, preserve available history, expose indeterminate state, and restore ownership before migration; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-31-006	Production admission and continued operation SHALL require state inventory, schema and ownership linting, durability tests, restore evidence, residency proofs, and lifecycle review, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
State cannot be classified, ownership is ambiguous, or actual durability diverges from the declared class	Block new writes or admission, preserve available history, expose indeterminate state, and restore ownership before migration.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared state-class identity, owner, store, schema, partition key, consistency, replication, retention, backup, recovery, and lifecycle.
- Positive and negative boundary tests prove that storage technology is selected behind the state contract and may not merge canonical truth, derived projections, cache, evidence, or temporary execution state.
- Fault exercises inject state cannot be classified, ownership is ambiguous, or actual durability diverges from the declared class and verify that the platform can block new writes or admission, preserve available history, expose indeterminate state, and restore ownership before migration.
- Independent review accepts state inventory, schema and ownership linting, durability tests, restore evidence, residency proofs, and lifecycle review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Storage Architecture and State Classes is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 24, Ch. 28, Ch. 29, Ch. 30, Ch. 62. Primary Volume 5 AI anchors: Ch. 9, Ch. 13, Ch. 19, Ch. 25, Ch. 38.

32 Transactional and Operational Datastores

Transactional and Operational Datastores establishes the infrastructure necessary to provide authoritative operational persistence for identities, workflows, decisions, policies, registries, and other consistency-sensitive state. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

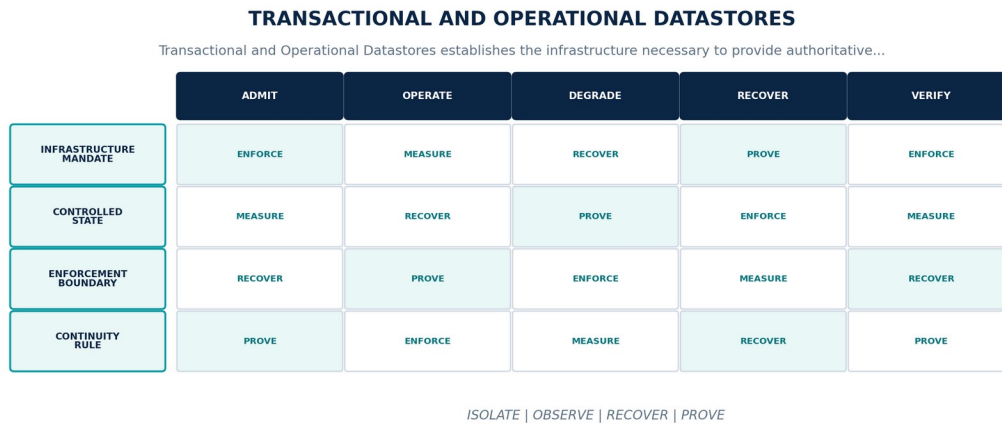


Figure 33. Transactional and Operational Datastores. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Provide authoritative operational persistence for identities, workflows, decisions, policies, registries, and other consistency-sensitive state.
Controlled state	Database cluster, schema revision, transaction boundary, consistency level, replica set, log position, backup, and recovery posture.
Enforcement boundary	Transactions must align with service ownership; cross-service atomicity requires a declared saga, outbox, or equivalent durable protocol.
Continuity rule	When quorum, transaction log, schema compatibility, replica integrity, or capacity becomes unsafe, the platform shall protect write correctness, fail over only inside consistency rules, preserve recovery points, and reconcile consumers before reopening.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-32-001	The infrastructure realization of Transactional and Operational Datastores SHALL provide authoritative operational persistence for identities, workflows, decisions, policies, registries, and other consistency-sensitive state.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-32-002	Every production instance SHALL declare and continuously reconcile database cluster, schema revision, transaction boundary, consistency level, replica set, log position, backup, and recovery posture through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-32-003	The implementation SHALL enforce this boundary: transactions must align with service ownership; cross-service atomicity requires a declared saga, outbox, or equivalent durable protocol.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-32-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Transactional and Operational Datastores.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-32-005	When quorum, transaction log, schema compatibility, replica integrity, or capacity becomes unsafe, the platform SHALL protect write correctness, fail over only inside consistency rules, preserve recovery points, and reconcile consumers before reopening; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-32-006	Production admission and continued operation SHALL require transaction and failover tests, schema migration rehearsals, consistency checks, point-in-time restore, and corruption recovery, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Quorum, transaction log, schema compatibility, replica integrity, or capacity becomes unsafe	Protect write correctness, fail over only inside consistency rules, preserve recovery points, and reconcile consumers before reopening.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared database cluster, schema revision, transaction boundary, consistency level, replica set, log position, backup, and recovery posture.
- Positive and negative boundary tests prove that transactions must align with service ownership; cross-service atomicity requires a declared saga, outbox, or equivalent durable protocol.
- Fault exercises inject quorum, transaction log, schema compatibility, replica integrity, or capacity becomes unsafe and verify that the platform can protect write correctness, fail over only inside consistency rules, preserve recovery points, and reconcile consumers before reopening.
- Independent review accepts transaction and failover tests, schema migration rehearsals, consistency checks, point-in-time restore, and corruption recovery for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Transactional and Operational Datastores is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 18, Ch. 22, Ch. 24, Ch. 30, Ch. 62. Primary Volume 5 AI anchors: Ch. 15, Ch. 20, Ch. 31, Ch. 34, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-016 · C-035 · C-041

33 Object and Evidence Storage

Object and Evidence Storage establishes the infrastructure necessary to retain source artifacts, documents, media, model packages, simulation outputs, evidence bundles, exports, and immutable audit material. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

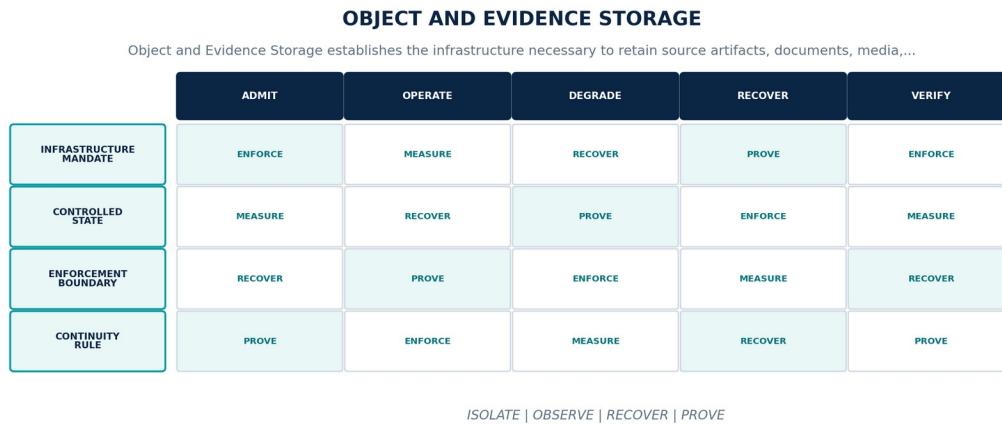


Figure 34. Object and Evidence Storage. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Retain source artifacts, documents, media, model packages, simulation outputs, evidence bundles, exports, and immutable audit material.
Controlled state	Object identity, version, digest, custody, classification, locality, retention, legal hold, replication, and access evidence.
Enforcement boundary	Object stores preserve bytes and metadata; canonical meaning, provenance, admissibility, and policy remain owned by the producing component.
Continuity rule	When digest mismatch, incomplete multipart write, replica loss, malware finding, or retention conflict occurs, the platform shall quarantine affected versions, deny decision-active use, preserve custody, restore from verified replicas, and propagate impact.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-33-001	The infrastructure realization of Object and Evidence Storage SHALL retain source artifacts, documents, media, model packages, simulation outputs, evidence bundles, exports, and immutable audit material.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-33-002	Every production instance SHALL declare and continuously reconcile object identity, version, digest, custody, classification, locality, retention, legal hold, replication, and access evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-33-003	The implementation SHALL enforce this boundary: object stores preserve bytes and metadata; canonical meaning, provenance, admissibility, and policy remain owned by the producing component.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-33-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Object and Evidence Storage.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-33-005	When digest mismatch, incomplete multipart write, replica loss, malware finding, or retention conflict occurs, the platform SHALL quarantine affected versions, deny decision-active use, preserve custody, restore from verified replicas, and propagate impact; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-33-006	Production admission and continued operation SHALL require immutability, integrity, malware, retention, legal-hold, replica-loss, and large-object restore tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Digest mismatch, incomplete multipart write, replica loss, malware finding, or retention conflict occurs	Quarantine affected versions, deny decision-active use, preserve custody, restore from verified replicas, and propagate impact.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared object identity, version, digest, custody, classification, locality, retention, legal hold, replication, and access evidence.
- Positive and negative boundary tests prove that object stores preserve bytes and metadata; canonical meaning, provenance, admissibility, and policy remain owned by the producing component.
- Fault exercises inject digest mismatch, incomplete multipart write, replica loss, malware finding, or retention conflict occurs and verify that the platform can quarantine affected versions, deny decision-active use, preserve custody, restore from verified replicas, and propagate impact.
- Independent review accepts immutability, integrity, malware, retention, legal-hold, replica-loss, and large-object restore tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Object and Evidence Storage is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 28, Ch. 29, Ch. 30, Ch. 54, Ch. 62. Primary Volume 5 AI anchors: Ch. 15, Ch. 21, Ch. 29, Ch. 38, Ch. 65.

34 Knowledge Graph Storage Infrastructure

Knowledge Graph Storage Infrastructure establishes the infrastructure necessary to persist temporal entities, relationships, ontology, evidence links, graph revisions, and derived indexes at strategic scale. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

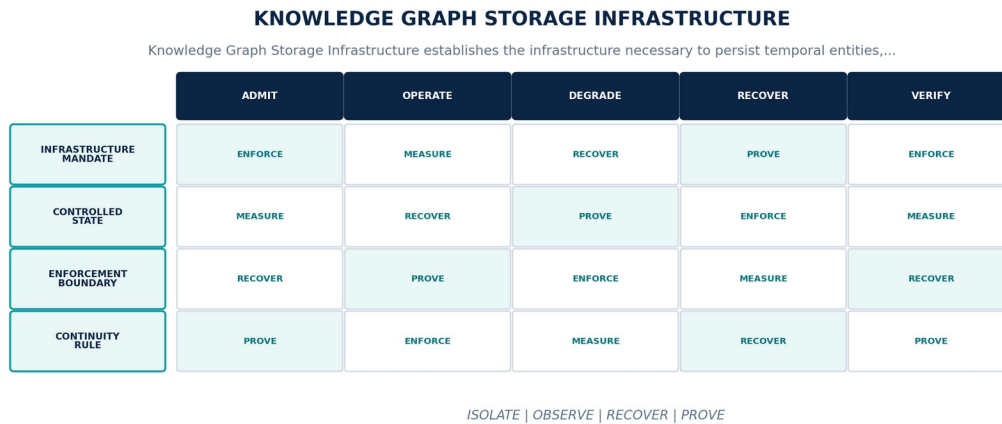


Figure 35. Knowledge Graph Storage Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Persist temporal entities, relationships, ontology, evidence links, graph revisions, and derived indexes at strategic scale.
Controlled state	Graph partition, ontology revision, valid and system time, write log, index revision, consistency scope, and rebuild cursor.
Enforcement boundary	Graph storage must preserve tenant, domain, temporal, provenance, policy, confidence, and correction semantics across traversal and projection.
Continuity rule	When graph partitions diverge, indexes lag, ontology compatibility fails, or lineage cannot close, the platform shall constrain affected traversals, expose revision and staleness, replay authoritative changes, and rebuild projections deterministically.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-34-001	The infrastructure realization of Knowledge Graph Storage Infrastructure SHALL persist temporal entities, relationships, ontology, evidence links, graph revisions, and derived indexes at strategic scale.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-34-002	Every production instance SHALL declare and continuously reconcile graph partition, ontology revision, valid and system time, write log, index revision, consistency scope, and rebuild cursor through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-34-003	The implementation SHALL enforce this boundary: graph storage must preserve tenant, domain, temporal, provenance, policy, confidence, and correction semantics across traversal and projection.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-34-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Knowledge Graph Storage Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-34-005	When graph partitions diverge, indexes lag, ontology compatibility fails, or lineage cannot close, the platform SHALL constrain affected traversals, expose revision and staleness, replay authoritative changes, and rebuild projections deterministically; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-34-006	Production admission and continued operation SHALL require temporal traversal fixtures, partition recovery, index rebuild, ontology migration, lineage closure, and scale benchmarks, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Graph partitions diverge, indexes lag, ontology compatibility fails, or lineage cannot close	Constrain affected traversals, expose revision and staleness, replay authoritative changes, and rebuild projections deterministically.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared graph partition, ontology revision, valid and system time, write log, index revision, consistency scope, and rebuild cursor.
- Positive and negative boundary tests prove that graph storage must preserve tenant, domain, temporal, provenance, policy, confidence, and correction semantics across traversal and projection.
- Fault exercises inject graph partitions diverge, indexes lag, ontology compatibility fails, or lineage cannot close and verify that the platform can constrain affected traversals, expose revision and staleness, replay authoritative changes, and rebuild projections deterministically.
- Independent review accepts temporal traversal fixtures, partition recovery, index rebuild, ontology migration, lineage closure, and scale benchmarks for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Knowledge Graph Storage Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 26, Ch. 27, Ch. 28, Ch. 30, Ch. 34. Primary Volume 5 AI anchors: Ch. 13, Ch. 24, Ch. 27, Ch. 43, Ch. 44, Ch. 45.

CONSTITUTIONAL INHERITANCE / C-014 · C-015 · C-016 · C-033 · C-041

35 Vector, Search, and Index Infrastructure

Vector, Search, and Index Infrastructure establishes the infrastructure necessary to serve lexical, vector, structured, geospatial, temporal, and multimodal discovery from rebuildable, policy-filtered projections. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

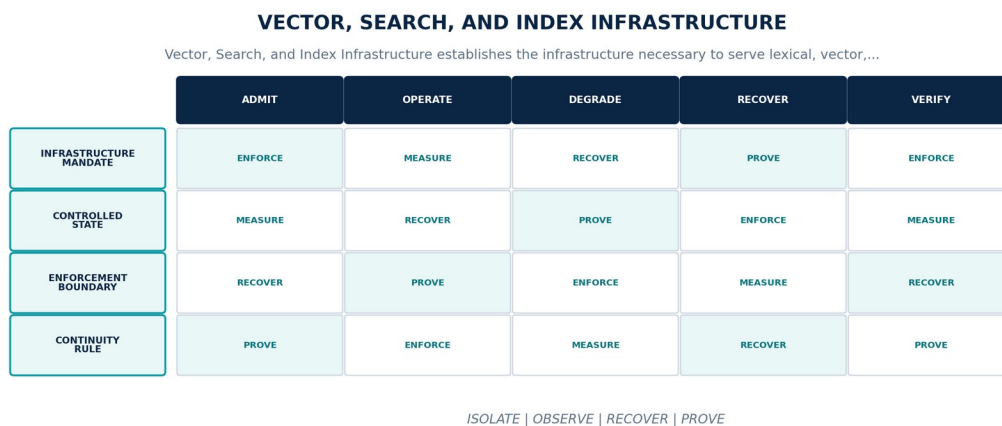


Figure 36. Vector, Search, and Index Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Serve lexical, vector, structured, geospatial, temporal, and multimodal discovery from rebuildable, policy-filtered projections.
Controlled state	Index identity, source revision, embedding or analyzer version, shard, policy scope, freshness, quality, and rebuild state.
Enforcement boundary	Indexes are derived acceleration structures and may not become unversioned truth or bypass source authorization.
Continuity rule	When index freshness, shard availability, policy filtering, embedding compatibility, or recall quality falls below threshold, the platform shall mark results degraded or unavailable, fall back only to policy-equivalent methods, and rebuild from authoritative sources.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-35-001	The infrastructure realization of Vector, Search, and Index Infrastructure SHALL serve lexical, vector, structured, geospatial, temporal, and multimodal discovery from rebuildable, policy-filtered projections.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-35-002	Every production instance SHALL declare and continuously reconcile index identity, source revision, embedding or analyzer version, shard, policy scope, freshness, quality, and rebuild state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-35-003	The implementation SHALL enforce this boundary: indexes are derived acceleration structures and may not become unversioned truth or bypass source authorization.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, contained, and audited.
IA-35-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Vector, Search, and Index Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-35-005	When index freshness, shard availability, policy filtering, embedding compatibility, or recall quality falls below threshold, the platform SHALL mark results degraded or unavailable, fall back only to policy-equivalent methods, and rebuild from authoritative sources; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-35-006	Production admission and continued operation SHALL require authorization leakage tests, recall and latency benchmarks, shard loss, embedding migration, deterministic rebuild, and staleness controls, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Index freshness, shard availability, policy filtering, embedding compatibility, or recall quality falls below threshold	Mark results degraded or unavailable, fall back only to policy-equivalent methods, and rebuild from authoritative sources.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared index identity, source revision, embedding or analyzer version, shard, policy scope, freshness, quality, and rebuild state.
- Positive and negative boundary tests prove that indexes are derived acceleration structures and may not become unversioned truth or bypass source authorization.
- Fault exercises inject index freshness, shard availability, policy filtering, embedding compatibility, or recall quality falls below threshold and verify that the platform can mark results degraded or unavailable, fall back only to policy-equivalent methods, and rebuild from authoritative sources.
- Independent review accepts authorization leakage tests, recall and latency benchmarks, shard loss, embedding migration, deterministic rebuild, and staleness controls for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Vector, Search, and Index Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 18, Ch. 23, Ch. 28, Ch. 30, Ch. 57. Primary Volume 5 AI anchors: Ch. 19, Ch. 23, Ch. 26, Ch. 27, Ch. 29, Ch. 40.

CONSTITUTIONAL INHERITANCE / C-011 · C-016 · C-027 · C-033 · C-035

36 Digital Twin and Simulation State Infrastructure

Digital Twin and Simulation State Infrastructure establishes the infrastructure necessary to persist twin definitions, observed state, assumptions, branches, interventions, checkpoints, simulation runs, and comparison results. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

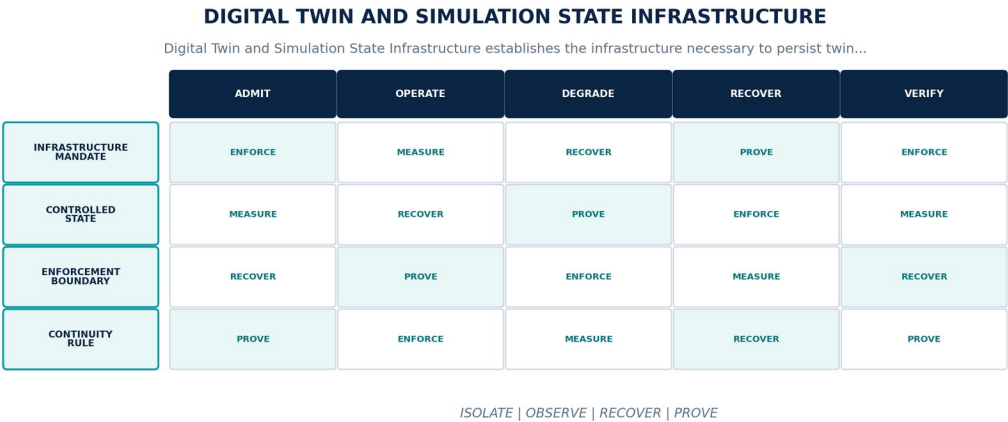


Figure 37. Digital Twin and Simulation State Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Persist twin definitions, observed state, assumptions, branches, interventions, checkpoints, simulation runs, and comparison results.
Controlled state	Twin identity, definition revision, branch, snapshot, temporal boundary, assumption set, run identity, checkpoint, and result lineage.
Enforcement boundary	Observed, inferred, forecast, scenario, and simulated state remain distinct and no branch may silently overwrite the operational twin.
Continuity rule	When branch identity, snapshot compatibility, solver checkpoint, or assumption lineage becomes incomplete, the platform shall suspend affected runs, preserve last committed state, label partial results, and require explicit branch reconciliation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-36-001	The infrastructure realization of Digital Twin and Simulation State Infrastructure SHALL persist twin definitions, observed state, assumptions, branches, interventions, checkpoints, simulation runs, and comparison results.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-36-002	Every production instance SHALL declare and continuously reconcile twin identity, definition revision, branch, snapshot, temporal boundary, assumption set, run identity, checkpoint, and result lineage through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-36-003	The implementation SHALL enforce this boundary: observed, inferred, forecast, scenario, and simulated state remain distinct and no branch may silently overwrite the operational twin.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-36-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Digital Twin and Simulation State Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-36-005	When branch identity, snapshot compatibility, solver checkpoint, or assumption lineage becomes incomplete, the platform SHALL suspend affected runs, preserve last committed state, label partial results, and require explicit branch reconciliation; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-36-006	Production admission and continued operation SHALL require branch isolation, snapshot restore, deterministic replay, solver failover, assumption lineage, and result-comparison tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Branch identity, snapshot compatibility, solver checkpoint, or assumption lineage becomes incomplete	Suspend affected runs, preserve last committed state, label partial results, and require explicit branch reconciliation.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared twin identity, definition revision, branch, snapshot, temporal boundary, assumption set, run identity, checkpoint, and result lineage.
- Positive and negative boundary tests prove that observed, inferred, forecast, scenario, and simulated state remain distinct and no branch may silently overwrite the operational twin.
- Fault exercises inject branch identity, snapshot compatibility, solver checkpoint, or assumption lineage becomes incomplete and verify that the platform can suspend affected runs, preserve last committed state, label partial results, and require explicit branch reconciliation.
- Independent review accepts branch isolation, snapshot restore, deterministic replay, solver failover, assumption lineage, and result-comparison tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Digital Twin and Simulation State Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 26, Ch. 27, Ch. 30, Ch. 44, Ch. 62. Primary Volume 5 AI anchors: Ch. 13, Ch. 28, Ch. 49, Ch. 50, Ch. 63.

37 Event Streaming, Messaging, and Durable Queues

Event Streaming, Messaging, and Durable Queues establishes the infrastructure necessary to move commands, events, work, evidence, and notifications with declared ordering, delivery, durability, retry, and replay semantics. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

EVENT STREAMING, MESSAGING, AND DURABLE QUEUES

Event Streaming, Messaging, and Durable Queues establishes the infrastructure necessary to move commands, events,...

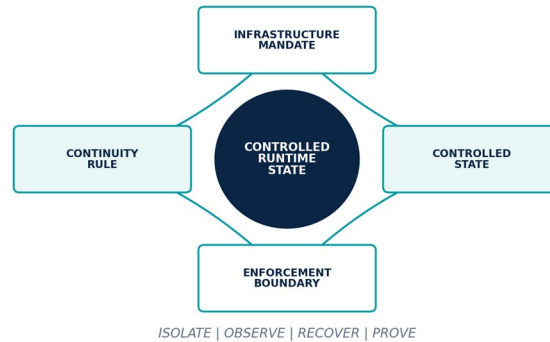


Figure 38. Event Streaming, Messaging, and Durable Queues. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Move commands, events, work, evidence, and notifications with declared ordering, delivery, durability, retry, and replay semantics.
Controlled state	Topic or queue identity, schema, partition, sequence, producer, consumer group, offset, retry, dead-letter, and retention state.
Enforcement boundary	Delivery guarantees are scoped per contract; exactly-once institutional effects require idempotent consumers and committed effect evidence.
Continuity rule	When brokers partition, consumers lag, messages duplicate or reorder, schemas conflict, or dead-letter volume grows unsafe, the platform shall apply backpressure, preserve durable offsets, isolate poison messages, prevent retry storms, and expose incomplete downstream state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-37-001	The infrastructure realization of Event Streaming, Messaging, and Durable Queues SHALL move commands, events, work, evidence, and notifications with declared ordering, delivery, durability, retry, and replay semantics.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-37-002	Every production instance SHALL declare and continuously reconcile topic or queue identity, schema, partition, sequence, producer, consumer group, offset, retry, dead-letter, and retention state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-37-003	The implementation SHALL enforce this boundary: delivery guarantees are scoped per contract; exactly-once institutional effects require idempotent consumers and committed effect evidence.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-37-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Event Streaming, Messaging, and Durable Queues.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-37-005	When brokers partition, consumers lag, messages duplicate or reorder, schemas conflict, or dead-letter volume grows unsafe, the platform SHALL apply backpressure, preserve durable offsets, isolate poison messages, prevent retry storms, and expose incomplete downstream state; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-37-006	Production admission and continued operation SHALL require duplicate and reordering tests, broker loss, consumer replay, schema rejection, dead-letter recovery, and effect reconciliation, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Brokers partition, consumers lag, messages duplicate or reorder, schemas conflict, or dead-letter volume grows unsafe	Apply backpressure, preserve durable offsets, isolate poison messages, prevent retry storms, and expose incomplete downstream state.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared topic or queue identity, schema, partition, sequence, producer, consumer group, offset, retry, dead-letter, and retention state.
- Positive and negative boundary tests prove that delivery guarantees are scoped per contract; exactly-once institutional effects require idempotent consumers and committed effect evidence.
- Fault exercises inject brokers partition, consumers lag, messages duplicate or reorder, schemas conflict, or dead-letter volume grows unsafe and verify that the platform can apply backpressure, preserve durable offsets, isolate poison messages, prevent retry storms, and expose incomplete downstream state.
- Independent review accepts duplicate and reordering tests, broker loss, consumer replay, schema rejection, dead-letter recovery, and effect reconciliation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Event Streaming, Messaging, and Durable Queues is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 20, Ch. 22, Ch. 23, Ch. 41, Ch. 61. Primary Volume 5 AI anchors: Ch. 31, Ch. 34, Ch. 35, Ch. 38, Ch. 68.

38 Cache, Temporary State, and Checkpoint Infrastructure

Cache, Temporary State, and Checkpoint Infrastructure establishes the infrastructure necessary to accelerate reads and recover execution without allowing cache, scratch, session, or checkpoint data to become hidden authority. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

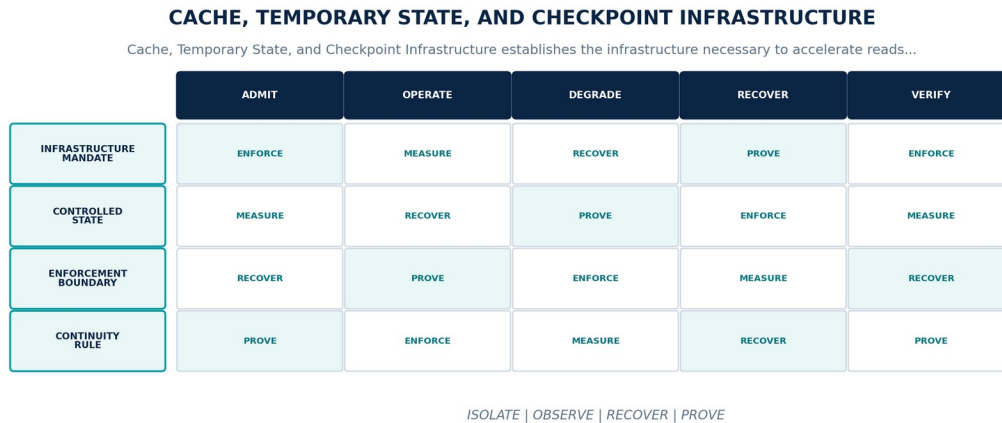


Figure 39. Cache, Temporary State, and Checkpoint Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Accelerate reads and recover execution without allowing cache, scratch, session, or checkpoint data to become hidden authority.
Controlled state	Cache or checkpoint identity, source revision, expiry, invalidation, encryption, locality, owner, recovery compatibility, and deletion state.
Enforcement boundary	Temporary state must be derivable, bounded, classified, tenant-scoped, and excluded from canonical truth unless explicitly committed through its owner.
Continuity rule	When invalidation is missed, checkpoint compatibility fails, cache contents are suspect, or temporary state survives its allowed lifetime, the platform shall bypass or purge affected acceleration state, resume only from verified checkpoints, and make recomputation or loss explicit.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-38-001	The infrastructure realization of Cache, Temporary State, and Checkpoint Infrastructure SHALL accelerate reads and recover execution without allowing cache, scratch, session, or checkpoint data to become hidden authority.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-38-002	Every production instance SHALL declare and continuously reconcile cache or checkpoint identity, source revision, expiry, invalidation, encryption, locality, owner, recovery compatibility, and deletion state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-38-003	The implementation SHALL enforce this boundary: temporary state must be derivable, bounded, classified, tenant-scoped, and excluded from canonical truth unless explicitly committed through its owner.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-38-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Cache, Temporary State, and Checkpoint Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-38-005	When invalidation is missed, checkpoint compatibility fails, cache contents are suspect, or temporary state survives its allowed lifetime, the platform SHALL bypass or purge affected acceleration state, resume only from verified checkpoints, and make recomputation or loss explicit; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-38-006	Production admission and continued operation SHALL require stale-read, invalidation, eviction, checkpoint corruption, restore compatibility, tenant isolation, and deletion tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Invalidation is missed, checkpoint compatibility fails, cache contents are suspect, or temporary state survives its allowed lifetime	Bypass or purge affected acceleration state, resume only from verified checkpoints, and make recomputation or loss explicit.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared cache or checkpoint identity, source revision, expiry, invalidation, encryption, locality, owner, recovery compatibility, and deletion state.
- Positive and negative boundary tests prove that temporary state must be derivable, bounded, classified, tenant-scoped, and excluded from canonical truth unless explicitly committed through its owner.
- Fault exercises inject invalidation is missed, checkpoint compatibility fails, cache contents are suspect, or temporary state survives its allowed lifetime and verify that the platform can bypass or purge affected acceleration state, resume only from verified checkpoints, and make recomputation or loss explicit.
- Independent review accepts stale-read, invalidation, eviction, checkpoint corruption, restore compatibility, tenant isolation, and deletion tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Cache, Temporary State, and Checkpoint Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 21, Ch. 22, Ch. 29, Ch. 30, Ch. 41. Primary Volume 5 AI anchors: Ch. 19, Ch. 26, Ch. 31, Ch. 38, Ch. 69.

CONSTITUTIONAL INHERITANCE / C-011 · C-016 · C-029 · C-033 · C-043

PART VI

Platform Runtime and Intelligence Workloads

Orchestration, developer platform, workflow, agent sandboxing, inference, retrieval, pipelines, experience, and integration.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

39 Container, Orchestration, and Runtime Platform

Container, Orchestration, and Runtime Platform establishes the infrastructure necessary to run deployable units through immutable artifacts, declarative desired state, workload identity, policy, health, placement, and controlled disruption. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

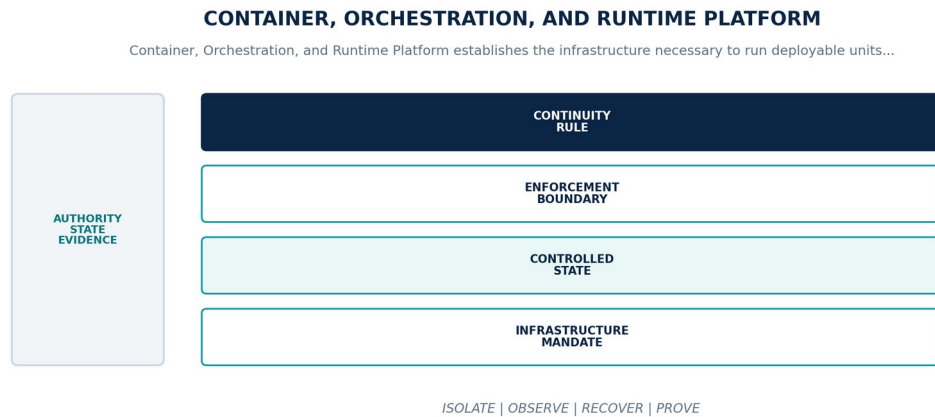


Figure 40. Container, Orchestration, and Runtime Platform. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Run deployable units through immutable artifacts, declarative desired state, workload identity, policy, health, placement, and controlled disruption.
Controlled state	Cluster, namespace, workload, image digest, desired and observed state, placement, identity, policy, health, and rollout.
Enforcement boundary	Orchestration reconciles infrastructure state but cannot define application truth, skip service contracts, or infer safe recovery from process liveness.
Continuity rule	When control quorum, scheduler, node fleet, image pull, admission, or reconciliation becomes impaired, the platform shall freeze unsafe change, maintain eligible running workloads, restore control from verified state, and reconcile before promotion.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-39-001	The infrastructure realization of Container, Orchestration, and Runtime Platform SHALL run deployable units through immutable artifacts, declarative desired state, workload identity, policy, health, placement, and controlled disruption.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-39-002	Every production instance SHALL declare and continuously reconcile cluster, namespace, workload, image digest, desired and observed state, placement, identity, policy, health, and rollout through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-39-003	The implementation SHALL enforce this boundary: orchestration reconciles infrastructure state but cannot define application truth, skip service contracts, or infer safe recovery from process liveness.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-39-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Container, Orchestration, and Runtime Platform.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-39-005	When control quorum, scheduler, node fleet, image pull, admission, or reconciliation becomes impaired, the platform SHALL freeze unsafe change, maintain eligible running workloads, restore control from verified state, and reconcile before promotion; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-39-006	Production admission and continued operation SHALL require control-plane loss, node drain, admission denial, image integrity, rollout and rollback, and multi-version operation tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Control quorum, scheduler, node fleet, image pull, admission, or reconciliation becomes impaired	Freeze unsafe change, maintain eligible running workloads, restore control from verified state, and reconcile before promotion.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared cluster, namespace, workload, image digest, desired and observed state, placement, identity, policy, health, and rollout.
- Positive and negative boundary tests prove that orchestration reconciles infrastructure state but cannot define application truth, skip service contracts, or infer safe recovery from process liveness.
- Fault exercises inject control quorum, scheduler, node fleet, image pull, admission, or reconciliation becomes impaired and verify that the platform can freeze unsafe change, maintain eligible running workloads, restore control from verified state, and reconcile before promotion.
- Independent review accepts control-plane loss, node drain, admission denial, image integrity, rollout and rollback, and multi-version operation tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Container, Orchestration, and Runtime Platform is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 9, Ch. 11, Ch. 53, Ch. 59, Ch. 61. Primary Volume 5 AI anchors: Ch. 15, Ch. 18, Ch. 30, Ch. 68, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-007 · C-008 · C-034 · C-038 · C-048

40 Service Platform and Internal Developer Platform

Service Platform and Internal Developer Platform establishes the infrastructure necessary to provide paved, policy-compliant paths for service creation, contracts, identity, telemetry, delivery, secrets, state, and operational ownership. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

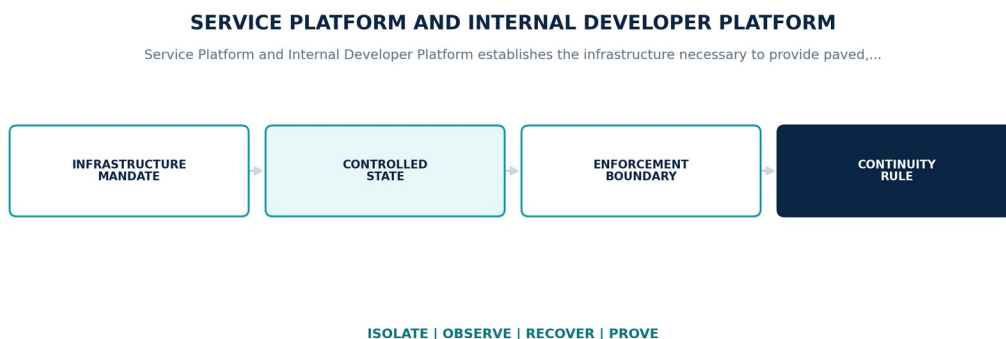


Figure 41. Service Platform and Internal Developer Platform. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Provide paved, policy-compliant paths for service creation, contracts, identity, telemetry, delivery, secrets, state, and operational ownership.
Controlled state	Service registration, template version, owner, contracts, dependencies, SLO, data classes, runtime profile, release, and support state.
Enforcement boundary	Developer convenience may automate controls but may not hide responsibility, weaken conformance, or force one physical technology across deployments.
Continuity rule	When a service bypasses the platform contract or a platform dependency would impose broad correlated failure, the platform shall block registration or isolate the exception, preserve ownership visibility, and provide an approved escape path with equivalent controls.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-40-001	The infrastructure realization of Service Platform and Internal Developer Platform SHALL provide paved, policy-compliant paths for service creation, contracts, identity, telemetry, delivery, secrets, state, and operational ownership.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-40-002	Every production instance SHALL declare and continuously reconcile service registration, template version, owner, contracts, dependencies, SLO, data classes, runtime profile, release, and support state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-40-003	The implementation SHALL enforce this boundary: developer convenience may automate controls but may not hide responsibility, weaken conformance, or force one physical technology across deployments.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-40-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Service Platform and Internal Developer Platform.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-40-005	When a service bypasses the platform contract or a platform dependency would impose broad correlated failure, the platform SHALL block registration or isolate the exception, preserve ownership visibility, and provide an approved escape path with equivalent controls; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-40-006	Production admission and continued operation SHALL require golden-path fixtures, service scorecards, escape-path review, platform outage drills, and adoption and conformance evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A service bypasses the platform contract or a platform dependency would impose broad correlated failure	Block registration or isolate the exception, preserve ownership visibility, and provide an approved escape path with equivalent controls.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared service registration, template version, owner, contracts, dependencies, SLO, data classes, runtime profile, release, and support state.
- Positive and negative boundary tests prove that developer convenience may automate controls but may not hide responsibility, weaken conformance, or force one physical technology across deployments.
- Fault exercises inject a service bypasses the platform contract or a platform dependency would impose broad correlated failure and verify that the platform can block registration or isolate the exception, preserve ownership visibility, and provide an approved escape path with equivalent controls.
- Independent review accepts golden-path fixtures, service scorecards, escape-path review, platform outage drills, and adoption and conformance evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Service Platform and Internal Developer Platform is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 11, Ch. 15, Ch. 23, Ch. 65, Ch. 66, Ch. 72. Primary Volume 5 AI anchors: Ch. 15, Ch. 17, Ch. 30, Ch. 66, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-003 · C-008 · C-035 · C-048 · C-051

41 Workflow Engine Infrastructure

Workflow Engine Infrastructure establishes the infrastructure necessary to host durable, versioned state machines, timers, tasks, compensations, human gates, and terminal outcomes across failures and upgrades. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

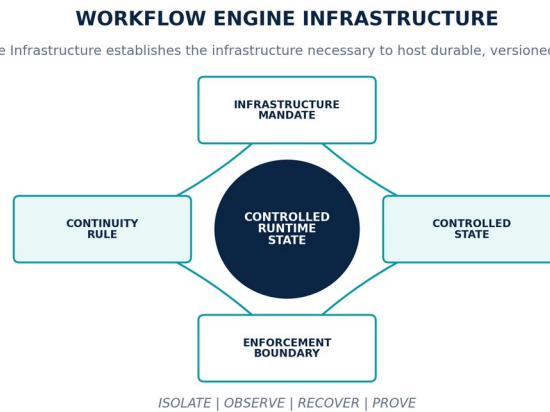


Figure 42. Workflow Engine Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Host durable, versioned state machines, timers, tasks, compensations, human gates, and terminal outcomes across failures and upgrades.
Controlled state	Workflow definition and instance, task lease, transition log, timer, checkpoint, compensation, gate, and terminal state.
Enforcement boundary	Workflow engines persist orchestration state; business truth changes only through owning-service contracts and evidence-bearing effects.
Continuity rule	When workers duplicate, timers drift, leases expire, engine state partitions, or a definition changes during execution, the platform shall resume from committed transitions, suppress duplicate effects, pin versions, and escalate irreconcilable instances.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-41-001	The infrastructure realization of Workflow Engine Infrastructure SHALL host durable, versioned state machines, timers, tasks, compensations, human gates, and terminal outcomes across failures and upgrades.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-41-002	Every production instance SHALL declare and continuously reconcile workflow definition and instance, task lease, transition log, timer, checkpoint, compensation, gate, and terminal state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-41-003	The implementation SHALL enforce this boundary: workflow engines persist orchestration state; business truth changes only through owning-service contracts and evidence-bearing effects.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-41-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Workflow Engine Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-41-005	When workers duplicate, timers drift, leases expire, engine state partitions, or a definition changes during execution, the platform SHALL resume from committed transitions, suppress duplicate effects, pin versions, and escalate irreconcilable instances; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-41-006	Production admission and continued operation SHALL require engine restart, duplicate task, timer and deadline, version migration, compensation, human-gate, and history replay tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Workers duplicate, timers drift, leases expire, engine state partitions, or a definition changes during execution	Resume from committed transitions, suppress duplicate effects, pin versions, and escalate irreconcilable instances.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared workflow definition and instance, task lease, transition log, timer, checkpoint, compensation, gate, and terminal state.
- Positive and negative boundary tests prove that workflow engines persist orchestration state; business truth changes only through owning-service contracts and evidence-bearing effects.
- Fault exercises inject workers duplicate, timers drift, leases expire, engine state partitions, or a definition changes during execution and verify that the platform can resume from committed transitions, suppress duplicate effects, pin versions, and escalate irreconcilable instances.
- Independent review accepts engine restart, duplicate task, timer and deadline, version migration, compensation, human-gate, and history replay tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Workflow Engine Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 22, Ch. 41, Ch. 42, Ch. 44, Ch. 61, Ch. 62. Primary Volume 5 AI anchors: Ch. 31, Ch. 34, Ch. 37, Ch. 38, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-004 · C-011 · C-024 · C-035 · C-041

42 Agent Execution Sandbox Infrastructure

Agent Execution Sandbox Infrastructure establishes the infrastructure necessary to isolate agent code, tools, content, networks, files, secrets, processes, resources, and side effects according to consequence class. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

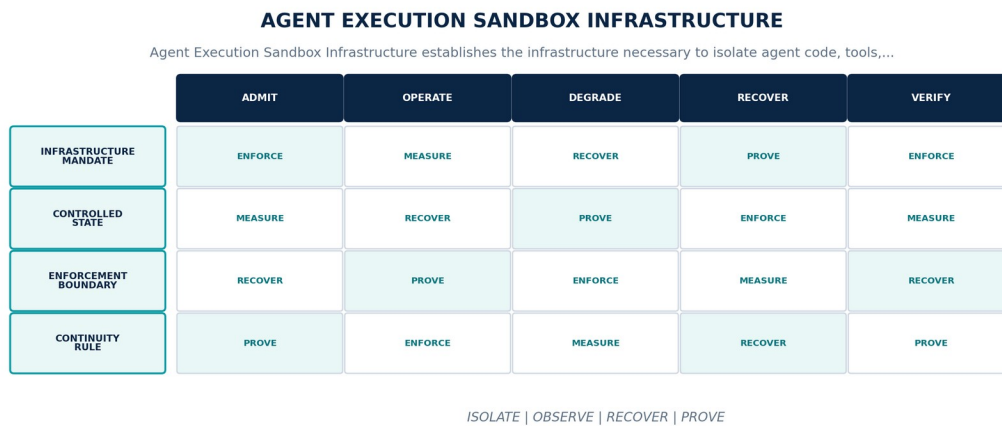


Figure 43. Agent Execution Sandbox Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Isolate agent code, tools, content, networks, files, secrets, processes, resources, and side effects according to consequence class.
Controlled state	Sandbox identity, agent package, runtime image, grants, mounts, network policy, resource budget, tool bridge, and destruction receipt.
Enforcement boundary	Agent instructions cannot expand capabilities; every reachable operation is independently enforced by infrastructure and external policy.
Continuity rule	When sandbox escape, policy bypass, untrusted content execution, secret exposure, or cleanup failure is suspected, the platform shall terminate and quarantine the environment, revoke grants, preserve forensic state, assess reachable effects, and block the package.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-42-001	The infrastructure realization of Agent Execution Sandbox Infrastructure SHALL isolate agent code, tools, content, networks, files, secrets, processes, resources, and side effects according to consequence class.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-42-002	Every production instance SHALL declare and continuously reconcile sandbox identity, agent package, runtime image, grants, mounts, network policy, resource budget, tool bridge, and destruction receipt through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-42-003	The implementation SHALL enforce this boundary: agent instructions cannot expand capabilities; every reachable operation is independently enforced by infrastructure and external policy.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-42-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Agent Execution Sandbox Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-42-005	When sandbox escape, policy bypass, untrusted content execution, secret exposure, or cleanup failure is suspected, the platform SHALL terminate and quarantine the environment, revoke grants, preserve forensic state, assess reachable effects, and block the package; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-42-006	Production admission and continued operation SHALL require escape and injection testing, capability denial, secret and network isolation, resource exhaustion, cleanup, and forensic reconstruction, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Sandbox escape, policy bypass, untrusted content execution, secret exposure, or cleanup failure is suspected	Terminate and quarantine the environment, revoke grants, preserve forensic state, assess reachable effects, and block the package.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared sandbox identity, agent package, runtime image, grants, mounts, network policy, resource budget, tool bridge, and destruction receipt.
- Positive and negative boundary tests prove that agent instructions cannot expand capabilities; every reachable operation is independently enforced by infrastructure and external policy.
- Fault exercises inject sandbox escape, policy bypass, untrusted content execution, secret exposure, or cleanup failure is suspected and verify that the platform can terminate and quarantine the environment, revoke grants, preserve forensic state, assess reachable effects, and block the package.
- Independent review accepts escape and injection testing, capability denial, secret and network isolation, resource exhaustion, cleanup, and forensic reconstruction for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Agent Execution Sandbox Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 43, Ch. 49, Ch. 51, Ch. 54, Ch. 60. Primary Volume 5 AI anchors: Ch. 30, Ch. 31, Ch. 36, Ch. 59, Ch. 60, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-007 · C-024 · C-030

43 Model Gateway and Inference Fabric

Model Gateway and Inference Fabric establishes the infrastructure necessary to mediate model requests across local and external providers with policy, routing, context controls, structured contracts, budgets, and traceability. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

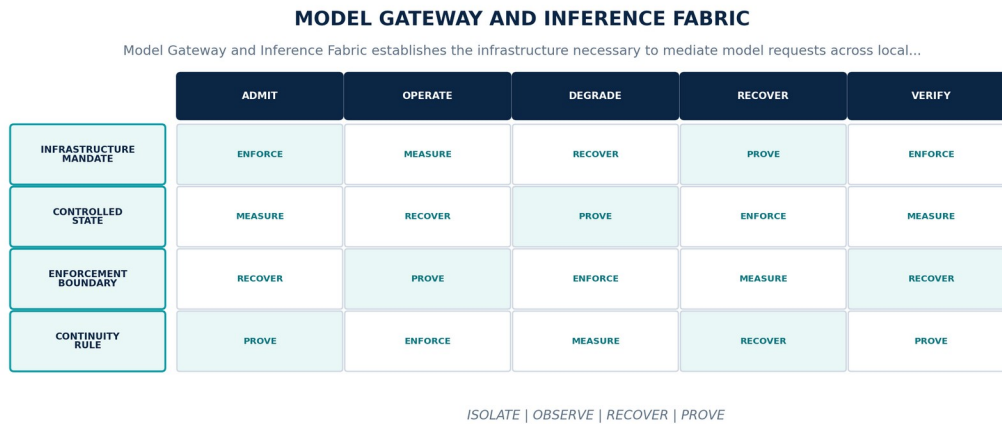


Figure 44. Model Gateway and Inference Fabric. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Mediate model requests across local and external providers with policy, routing, context controls, structured contracts, budgets, and traceability.
Controlled state	Inference request, task contract, model resolution, provider route, context manifest, budget, response, evaluation, and receipt.
Enforcement boundary	The gateway may select approved equivalents but cannot change task meaning, authority, evidence obligations, or consequence controls.
Continuity rule	When provider, route, model fitness, data policy, latency, or budget becomes invalid during execution, the platform shall abort, abstain, or use a declared equivalent fallback, preserve the exact resolution path, and never silently downgrade controls.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-43-001	The infrastructure realization of Model Gateway and Inference Fabric SHALL mediate model requests across local and external providers with policy, routing, context controls, structured contracts, budgets, and traceability.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-43-002	Every production instance SHALL declare and continuously reconcile inference request, task contract, model resolution, provider route, context manifest, budget, response, evaluation, and receipt through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-43-003	The implementation SHALL enforce this boundary: the gateway may select approved equivalents but cannot change task meaning, authority, evidence obligations, or consequence controls.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-43-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Model Gateway and Inference Fabric.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-43-005	When provider, route, model fitness, data policy, latency, or budget becomes invalid during execution, the platform SHALL abort, abstain, or use a declared equivalent fallback, preserve the exact resolution path, and never silently downgrade controls; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-43-006	Production admission and continued operation SHALL require provider failover, routing policy, privacy and egress, structured-output, budget, revocation, and cross-deployment tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Provider, route, model fitness, data policy, latency, or budget becomes invalid during execution	Abort, abstain, or use a declared equivalent fallback, preserve the exact resolution path, and never silently downgrade controls.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared inference request, task contract, model resolution, provider route, context manifest, budget, response, evaluation, and receipt.
- Positive and negative boundary tests prove that the gateway may select approved equivalents but cannot change task meaning, authority, evidence obligations, or consequence controls.
- Fault exercises inject provider, route, model fitness, data policy, latency, or budget becomes invalid during execution and verify that the platform can abort, abstain, or use a declared equivalent fallback, preserve the exact resolution path, and never silently downgrade controls.
- Independent review accepts provider failover, routing policy, privacy and egress, structured-output, budget, revocation, and cross-deployment tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Model Gateway and Inference Fabric is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 20, Ch. 43, Ch. 52, Ch. 58, Ch. 60. Primary Volume 5 AI anchors: Ch. 10, Ch. 15, Ch. 16, Ch. 17, Ch. 20, Ch. 21, Ch. 22.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-007 · C-024 · C-035

44 Retrieval and Context Infrastructure

Retrieval and Context Infrastructure establishes the infrastructure necessary to assemble minimum authorized lexical, vector, graph, temporal, memory, twin, and evidence context through reproducible manifests. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

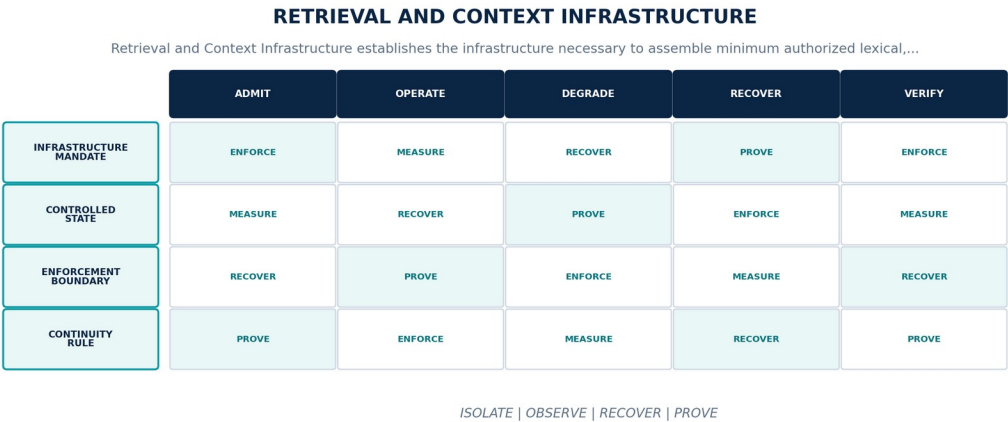


Figure 45. Retrieval and Context Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Assemble minimum authorized lexical, vector, graph, temporal, memory, twin, and evidence context through reproducible manifests.
Controlled state	Retrieval request, identity, purpose, policy, source revisions, queries, candidates, context manifest, expiry, and invalidation.
Enforcement boundary	Retrieval infrastructure filters before content exposure and must preserve source-to-context mapping through ranking, compression, and caching.
Continuity rule	When authorization, source freshness, index quality, context budget, or evidence binding becomes indeterminate, the platform shall exclude the affected material, mark context incomplete, abstain where necessary, and trigger refresh or correction workflows.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-44-001	The infrastructure realization of Retrieval and Context Infrastructure SHALL assemble minimum authorized lexical, vector, graph, temporal, memory, twin, and evidence context through reproducible manifests.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-44-002	Every production instance SHALL declare and continuously reconcile retrieval request, identity, purpose, policy, source revisions, queries, candidates, context manifest, expiry, and invalidation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-44-003	The implementation SHALL enforce this boundary: retrieval infrastructure filters before content exposure and must preserve source-to-context mapping through ranking, compression, and caching.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-44-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Retrieval and Context Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-44-005	When authorization, source freshness, index quality, context budget, or evidence binding becomes indeterminate, the platform SHALL exclude the affected material, mark context incomplete, abstain where necessary, and trigger refresh or correction workflows; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-44-006	Production admission and continued operation SHALL require authorization leakage, hybrid recall, temporal pinning, compression mapping, invalidation, citation, and replay tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Authorization, source freshness, index quality, context budget, or evidence binding becomes indeterminate	Exclude the affected material, mark context incomplete, abstain where necessary, and trigger refresh or correction workflows.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared retrieval request, identity, purpose, policy, source revisions, queries, candidates, context manifest, expiry, and invalidation.
- Positive and negative boundary tests prove that retrieval infrastructure filters before content exposure and must preserve source-to-context mapping through ranking, compression, and caching.
- Fault exercises inject authorization, source freshness, index quality, context budget, or evidence binding becomes indeterminate and verify that the platform can exclude the affected material, mark context incomplete, abstain where necessary, and trigger refresh or correction workflows.
- Independent review accepts authorization leakage, hybrid recall, temporal pinning, compression mapping, invalidation, citation, and replay tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Retrieval and Context Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 18, Ch. 27, Ch. 28, Ch. 29, Ch. 57. Primary Volume 5 AI anchors: Ch. 19, Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 28, Ch. 29.

45 Data and AI Pipeline Runtime

Data and AI Pipeline Runtime establishes the infrastructure necessary to execute ingestion, transformation, feature, evaluation, training, indexing, model, graph, and analytics pipelines as governed releases. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

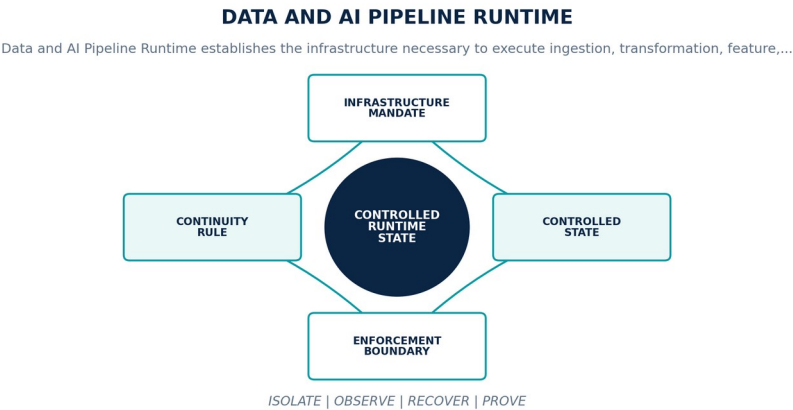


Figure 46. Data and AI Pipeline Runtime. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Execute ingestion, transformation, feature, evaluation, training, indexing, model, graph, and analytics pipelines as governed releases.
Controlled state	Pipeline definition, input revisions, code and environment digest, run, checkpoints, outputs, quality gates, lineage, and promotion.
Enforcement boundary	Pipelines may produce candidates and projections; canonical admission and model fitness remain separate owned decisions.
Continuity rule	When input drift, code mismatch, quality failure, nondeterminism, lineage gap, or partial publish occurs, the platform shall stop promotion, quarantine outputs, preserve run state, reconcile partial effects, and rerun from a known boundary.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-45-001	The infrastructure realization of Data and AI Pipeline Runtime SHALL execute ingestion, transformation, feature, evaluation, training, indexing, model, graph, and analytics pipelines as governed releases.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-45-002	Every production instance SHALL declare and continuously reconcile pipeline definition, input revisions, code and environment digest, run, checkpoints, outputs, quality gates, lineage, and promotion through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-45-003	The implementation SHALL enforce this boundary: pipelines may produce candidates and projections; canonical admission and model fitness remain separate owned decisions.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.
IA-45-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Data and AI Pipeline Runtime.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-45-005	When input drift, code mismatch, quality failure, nondeterminism, lineage gap, or partial publish occurs, the platform SHALL stop promotion, quarantine outputs, preserve run state, reconcile partial effects, and rerun from a known boundary; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-45-006	Production admission and continued operation SHALL require reproducibility, data and code lineage, partial-publish recovery, quality gate, rollback, and deployment-parity tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Input drift, code mismatch, quality failure, nondeterminism, lineage gap, or partial publish occurs	Stop promotion, quarantine outputs, preserve run state, reconcile partial effects, and rerun from a known boundary.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared pipeline definition, input revisions, code and environment digest, run, checkpoints, outputs, quality gates, lineage, and promotion.
- Positive and negative boundary tests prove that pipelines may produce candidates and projections; canonical admission and model fitness remain separate owned decisions.
- Fault exercises inject input drift, code mismatch, quality failure, nondeterminism, lineage gap, or partial publish occurs and verify that the platform can stop promotion, quarantine outputs, preserve run state, reconcile partial effects, and rerun from a known boundary.
- Independent review accepts reproducibility, data and code lineage, partial-publish recovery, quality gate, rollback, and deployment-parity tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Data and AI Pipeline Runtime is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 28, Ch. 30, Ch. 41, Ch. 57, Ch. 68, Ch. 70. Primary Volume 5 AI anchors: Ch. 15, Ch. 20, Ch. 66, Ch. 68, Ch. 69, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-008 · C-011 · C-024 · C-033 · C-044

46 Executive Experience and Collaboration Infrastructure

Executive Experience and Collaboration Infrastructure establishes the infrastructure necessary to deliver executive operating surfaces, briefings, alerts, approvals, collaborative review, and decision records with high availability and explicit trust state. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

EXECUTIVE EXPERIENCE AND COLLABORATION INFRASTRUCTURE

Executive Experience and Collaboration Infrastructure establishes the infrastructure necessary to deliver executive...

	ADMIT	OPERATE	DEGRADE	RECOVER	VERIFY
INFRASTRUCTURE MANDATE	ENFORCE	MEASURE	RECOVER	PROVE	ENFORCE
CONTROLLED STATE	MEASURE	RECOVER	PROVE	ENFORCE	MEASURE
ENFORCEMENT BOUNDARY	RECOVER	PROVE	ENFORCE	MEASURE	RECOVER
CONTINUITY RULE	PROVE	ENFORCE	MEASURE	RECOVER	PROVE

ISOLATE | OBSERVE | RECOVER | PROVE

Figure 47. Executive Experience and Collaboration Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Deliver executive operating surfaces, briefings, alerts, approvals, collaborative review, and decision records with high availability and explicit trust state.
Controlled state	Session, workspace, briefing package, subscription, notification, collaboration channel, approval, and decision-state projection.
Enforcement boundary	Experience infrastructure may compose products but cannot hide provenance, uncertainty, staleness, dissent, policy, or human authority.
Continuity rule	When a channel, session, notification, projection, or collaboration dependency fails or becomes stale, the platform shall preserve decision records, show degraded and last-known state, prevent false acknowledgement, and provide a verified alternate path.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-46-001	The infrastructure realization of Executive Experience and Collaboration Infrastructure SHALL deliver executive operating surfaces, briefings, alerts, approvals, collaborative review, and decision records with high availability and explicit trust state.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-46-002	Every production instance SHALL declare and continuously reconcile session, workspace, briefing package, subscription, notification, collaboration channel, approval, and decision-state projection through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-46-003	The implementation SHALL enforce this boundary: experience infrastructure may compose products but cannot hide provenance, uncertainty, staleness, dissent, policy, or human authority.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-46-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Executive Experience and Collaboration Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-46-005	When a channel, session, notification, projection, or collaboration dependency fails or becomes stale, the platform SHALL preserve decision records, show degraded and last-known state, prevent false acknowledgement, and provide a verified alternate path; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-46-006	Production admission and continued operation SHALL require session and projection consistency, notification delivery, accessibility, approval integrity, offline review, and channel failover, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A channel, session, notification, projection, or collaboration dependency fails or becomes stale	Preserve decision records, show degraded and last-known state, prevent false acknowledgement, and provide a verified alternate path.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared session, workspace, briefing package, subscription, notification, collaboration channel, approval, and decision-state projection.
- Positive and negative boundary tests prove that experience infrastructure may compose products but cannot hide provenance, uncertainty, staleness, dissent, policy, or human authority.
- Fault exercises inject a channel, session, notification, projection, or collaboration dependency fails or becomes stale and verify that the platform can preserve decision records, show degraded and last-known state, prevent false acknowledgement, and provide a verified alternate path.
- Independent review accepts session and projection consistency, notification delivery, accessibility, approval integrity, offline review, and channel failover for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Executive Experience and Collaboration Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 18, Ch. 21, Ch. 42, Ch. 46, Ch. 58. Primary Volume 5 AI anchors: Ch. 9, Ch. 37, Ch. 56, Ch. 57, Ch. 64.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-032 · C-035

47 Integration Runtime and Connector Isolation

Integration Runtime and Connector Isolation establishes the infrastructure necessary to execute source and enterprise connectors as isolated, least-privilege, rate-governed, schema-aware, evidence-producing packages. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

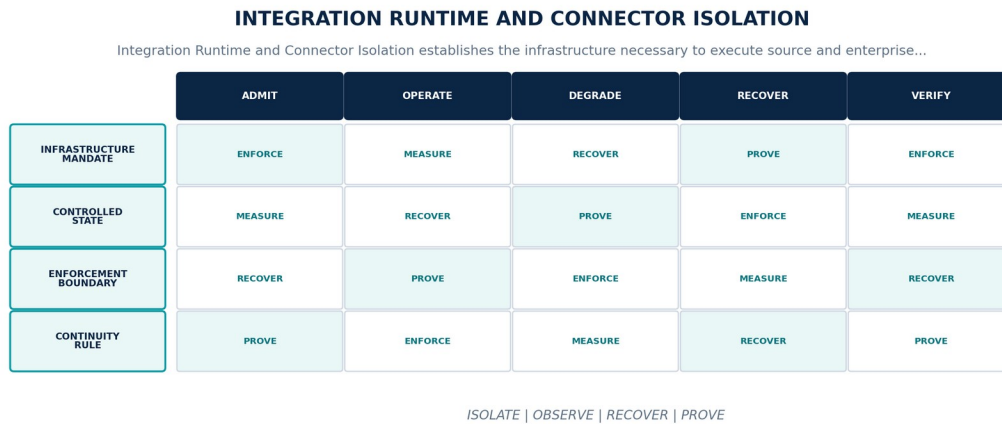


Figure 48. Integration Runtime and Connector Isolation. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Execute source and enterprise connectors as isolated, least-privilege, rate-governed, schema-aware, evidence-producing packages.
Controlled state	Connector package, source identity, credential, cursor, schema, policy, rate, network path, health, and revocation.
Enforcement boundary	Connectors terminate external protocols and may not create canonical truth, conceal source limitations, or share credentials across tenants.
Continuity rule	When credential, schema, source behavior, package integrity, cursor, or content safety becomes invalid, the platform shall suspend the connector, quarantine content, preserve custody and cursor evidence, rotate authority, and require requalification.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-47-001	The infrastructure realization of Integration Runtime and Connector Isolation SHALL execute source and enterprise connectors as isolated, least-privilege, rate-governed, schema-aware, evidence-producing packages.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-47-002	Every production instance SHALL declare and continuously reconcile connector package, source identity, credential, cursor, schema, policy, rate, network path, health, and revocation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-47-003	The implementation SHALL enforce this boundary: connectors terminate external protocols and may not create canonical truth, conceal source limitations, or share credentials across tenants.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-47-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Integration Runtime and Connector Isolation.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-47-005	When credential, schema, source behavior, package integrity, cursor, or content safety becomes invalid, the platform SHALL suspend the connector, quarantine content, preserve custody and cursor evidence, rotate authority, and require requalification; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-47-006	Production admission and continued operation SHALL require connector contract, credential rotation, schema drift, rate limit, malicious content, cursor recovery, and tenant isolation tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Credential, schema, source behavior, package integrity, cursor, or content safety becomes invalid	Suspend the connector, quarantine content, preserve custody and cursor evidence, rotate authority, and require requalification.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared connector package, source identity, credential, cursor, schema, policy, rate, network path, health, and revocation.
- Positive and negative boundary tests prove that connectors terminate external protocols and may not create canonical truth, conceal source limitations, or share credentials across tenants.
- Fault exercises inject credential, schema, source behavior, package integrity, cursor, or content safety becomes invalid and verify that the platform can suspend the connector, quarantine content, preserve custody and cursor evidence, rotate authority, and require requalification.
- Independent review accepts connector contract, credential rotation, schema drift, rate limit, malicious content, cursor recovery, and tenant isolation tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Integration Runtime and Connector Isolation is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 23, Ch. 25, Ch. 28, Ch. 49, Ch. 56. Primary Volume 5 AI anchors: Ch. 21, Ch. 36, Ch. 39, Ch. 40, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-007 · C-010 · C-018 · C-029 · C-030

48 Platform APIs, Events, and Contract Enforcement

Platform APIs, Events, and Contract Enforcement establishes the infrastructure necessary to enforce canonical envelopes, schemas, versions, errors, idempotency, ordering, and policy at every infrastructure-mediated contract boundary. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

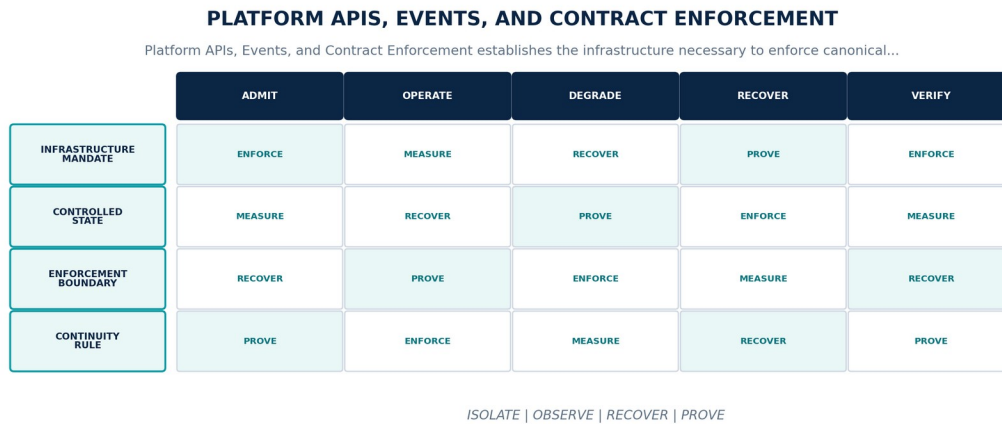


Figure 49. Platform APIs, Events, and Contract Enforcement. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Enforce canonical envelopes, schemas, versions, errors, idempotency, ordering, and policy at every infrastructure-mediated contract boundary.
Controlled state	Contract identity, schema, compatibility range, endpoint or topic, policy, conformance fixtures, consumer set, and retirement state.
Enforcement boundary	Gateways, brokers, runtimes, and SDKs enforce contracts consistently but cannot invent application success or suppress explicit failure states.
Continuity rule	When schema, version, policy, ordering, or consumer compatibility is violated, the platform shall reject or quarantine the exchange, preserve causation and payload evidence, and require compatible replay or migration.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-48-001	The infrastructure realization of Platform APIs, Events, and Contract Enforcement SHALL enforce canonical envelopes, schemas, versions, errors, idempotency, ordering, and policy at every infrastructure-mediated contract boundary.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-48-002	Every production instance SHALL declare and continuously reconcile contract identity, schema, compatibility range, endpoint or topic, policy, conformance fixtures, consumer set, and retirement state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-48-003	The implementation SHALL enforce this boundary: gateways, brokers, runtimes, and SDKs enforce contracts consistently but cannot invent application success or suppress explicit failure states.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-48-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Platform APIs, Events, and Contract Enforcement.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-48-005	When schema, version, policy, ordering, or consumer compatibility is violated, the platform SHALL reject or quarantine the exchange, preserve causation and payload evidence, and require compatible replay or migration; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-48-006	Production admission and continued operation SHALL require producer-consumer fixtures, compatibility matrix, negative schema tests, retry and ordering exercises, and retirement proofs, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Schema, version, policy, ordering, or consumer compatibility is violated	Reject or quarantine the exchange, preserve causation and payload evidence, and require compatible replay or migration.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared contract identity, schema, compatibility range, endpoint or topic, policy, conformance fixtures, consumer set, and retirement state.
- Positive and negative boundary tests prove that gateways, brokers, runtimes, and SDKs enforce contracts consistently but cannot invent application success or suppress explicit failure states.
- Fault exercises inject schema, version, policy, ordering, or consumer compatibility is violated and verify that the platform can reject or quarantine the exchange, preserve causation and payload evidence, and require compatible replay or migration.
- Independent review accepts producer-consumer fixtures, compatibility matrix, negative schema tests, retry and ordering exercises, and retirement proofs for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Platform APIs, Events, and Contract Enforcement is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 19, Ch. 20, Ch. 21, Ch. 22, Ch. 23. Primary Volume 5 AI anchors: Ch. 7, Ch. 9, Ch. 20, Ch. 31, Ch. 35.

CONSTITUTIONAL INHERITANCE / C-003 · C-011 · C-035 · C-037 · C-043

PART VII

Trust, Security, and Sovereignty

Identity, policy, cryptography, secrets, confidential processing, isolation, supply chain, detection, and sovereignty.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

49 Identity, Workload Trust, and Privileged Access

Identity, Workload Trust, and Privileged Access establishes the infrastructure necessary to establish human, workload, device, service, support, and automation identities with strong authentication, short-lived authority, and accountable privilege. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

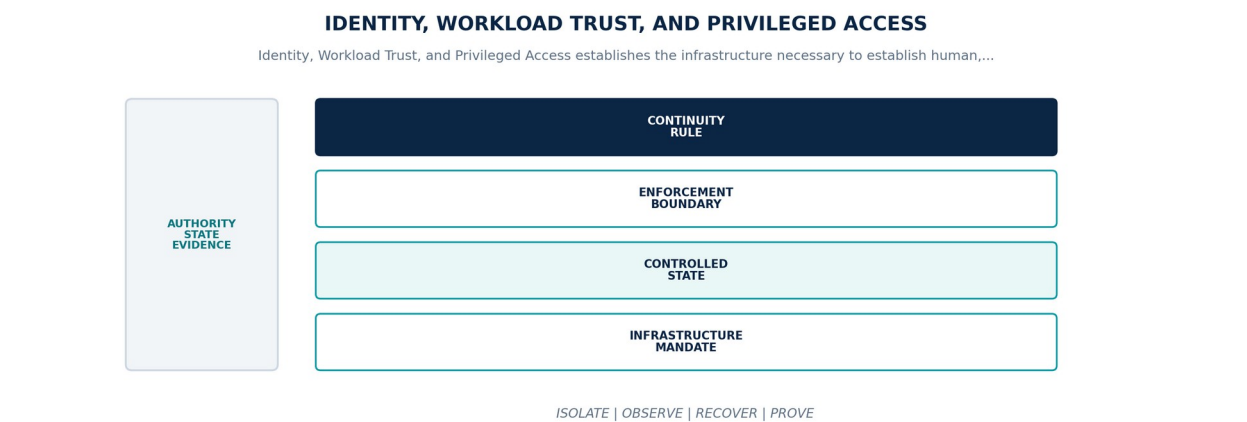


Figure 50. Identity, Workload Trust, and Privileged Access. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Establish human, workload, device, service, support, and automation identities with strong authentication, short-lived authority, and accountable privilege.
Controlled state	Principal identity, trust domain, credential, authentication context, role, delegation, privilege session, approval, and revocation.
Enforcement boundary	Identity proves who or what acts; authorization still evaluates tenant, purpose, object, policy, consequence, and current risk.
Continuity rule	When identity assurance, issuer trust, session context, delegation, or privileged activity becomes suspect, the platform shall deny or terminate access, revoke credentials, contain reachable systems, preserve session evidence, and initiate identity recovery.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-49-001	The infrastructure realization of Identity, Workload Trust, and Privileged Access SHALL establish human, workload, device, service, support, and automation identities with strong authentication, short-lived authority, and accountable privilege.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-49-002	Every production instance SHALL declare and continuously reconcile principal identity, trust domain, credential, authentication context, role, delegation, privilege session, approval, and revocation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-49-003	The implementation SHALL enforce this boundary: identity proves who or what acts; authorization still evaluates tenant, purpose, object, policy, consequence, and current risk.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-49-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Identity, Workload Trust, and Privileged Access.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-49-005	When identity assurance, issuer trust, session context, delegation, or privileged activity becomes suspect, the platform SHALL deny or terminate access, revoke credentials, contain reachable systems, preserve session evidence, and initiate identity recovery; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-49-006	Production admission and continued operation SHALL require authentication and federation tests, workload attestation, privilege escalation denial, session recording, rotation, and revocation, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity assurance, issuer trust, session context, delegation, or privileged activity becomes suspect	Deny or terminate access, revoke credentials, contain reachable systems, preserve session evidence, and initiate identity recovery.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared principal identity, trust domain, credential, authentication context, role, delegation, privilege session, approval, and revocation.
- Positive and negative boundary tests prove that identity proves who or what acts; authorization still evaluates tenant, purpose, object, policy, consequence, and current risk.
- Fault exercises inject identity assurance, issuer trust, session context, delegation, or privileged activity becomes suspect and verify that the platform can deny or terminate access, revoke credentials, contain reachable systems, preserve session evidence, and initiate identity recovery.
- Independent review accepts authentication and federation tests, workload attestation, privilege escalation denial, session recording, rotation, and revocation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Identity, Workload Trust, and Privileged Access is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 25, Ch. 49, Ch. 50, Ch. 55, Ch. 60. Primary Volume 5 AI anchors: Ch. 3, Ch. 36, Ch. 37, Ch. 59, Ch. 65.

CONSTITUTIONAL INHERITANCE / C-004 · C-006 · C-007 · C-020 · C-028

50 Policy Enforcement Infrastructure

Policy Enforcement Infrastructure establishes the infrastructure necessary to evaluate authorization, classification, purpose, retention, residency, egress, model, tool, and consequence policies at material runtime boundaries. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

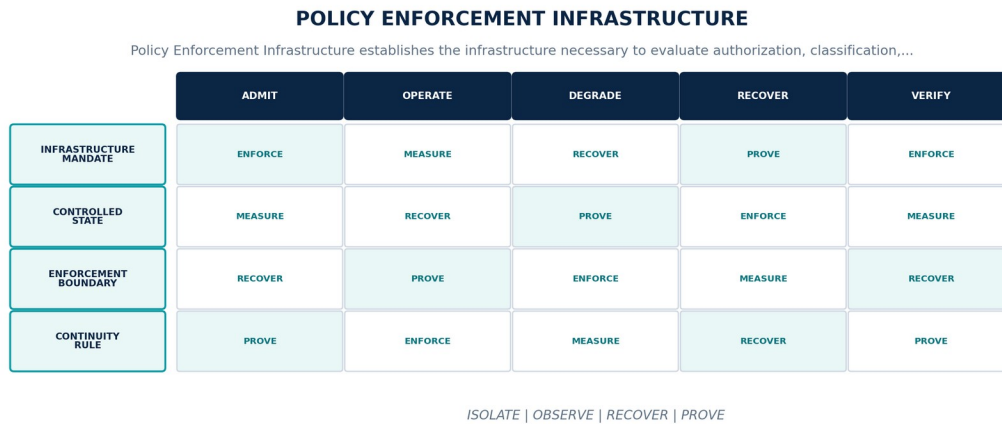


Figure 51. Policy Enforcement Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Evaluate authorization, classification, purpose, retention, residency, egress, model, tool, and consequence policies at material runtime boundaries.
Controlled state	Policy bundle, decision point, enforcement point, input attributes, decision, obligations, version, cache, and evidence.
Enforcement boundary	Policy enforcement remains external to models and agents and must fail according to the protected operation's declared safety posture.
Continuity rule	When policy is unavailable, stale, conflicting, slow, or cannot evaluate required attributes, the platform shall deny protected change and data release, allow only explicitly safe cached decisions, and expose the policy failure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-50-001	The infrastructure realization of Policy Enforcement Infrastructure SHALL evaluate authorization, classification, purpose, retention, residency, egress, model, tool, and consequence policies at material runtime boundaries.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-50-002	Every production instance SHALL declare and continuously reconcile policy bundle, decision point, enforcement point, input attributes, decision, obligations, version, cache, and evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-50-003	The implementation SHALL enforce this boundary: policy enforcement remains external to models and agents and must fail according to the protected operation's declared safety posture.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-50-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Policy Enforcement Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-50-005	When policy is unavailable, stale, conflicting, slow, or cannot evaluate required attributes, the platform SHALL deny protected change and data release, allow only explicitly safe cached decisions, and expose the policy failure; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-50-006	Production admission and continued operation SHALL require decision-table coverage, fail-closed tests, bundle rollback, conflict resolution, cache expiry, and cross-plane traces, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Policy is unavailable, stale, conflicting, slow, or cannot evaluate required attributes	Deny protected change and data release, allow only explicitly safe cached decisions, and expose the policy failure.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared policy bundle, decision point, enforcement point, input attributes, decision, obligations, version, cache, and evidence.
- Positive and negative boundary tests prove that policy enforcement remains external to models and agents and must fail according to the protected operation's declared safety posture.
- Fault exercises inject policy is unavailable, stale, conflicting, slow, or cannot evaluate required attributes and verify that the platform can deny protected change and data release, allow only explicitly safe cached decisions, and expose the policy failure.
- Independent review accepts decision-table coverage, fail-closed tests, bundle rollback, conflict resolution, cache expiry, and cross-plane traces for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Policy Enforcement Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 13, Ch. 29, Ch. 49, Ch. 50, Ch. 60. Primary Volume 5 AI anchors: Ch. 11, Ch. 12, Ch. 21, Ch. 36, Ch. 59, Ch. 65.

CONSTITUTIONAL INHERITANCE / C-004 · C-007 · C-020 · C-024 · C-030

51 Cryptography, Key Management, and Trust Anchors

Cryptography, Key Management, and Trust Anchors establishes the infrastructure necessary to protect data, identity, artifacts, communications, audit, and approvals through governed cryptographic profiles and customer-compatible key authority. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

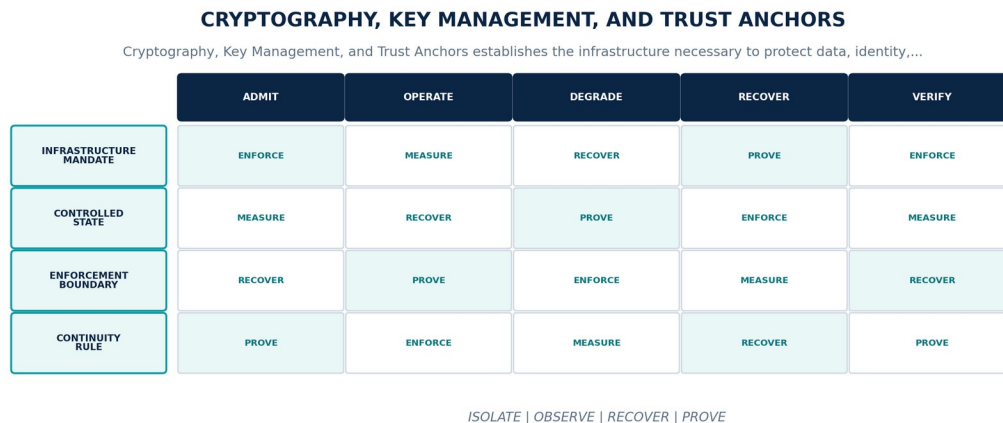


Figure 52. Cryptography, Key Management, and Trust Anchors. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Protect data, identity, artifacts, communications, audit, and approvals through governed cryptographic profiles and customer-compatible key authority.
Controlled state	Key and certificate identity, algorithm profile, owner, purpose, locality, hardware boundary, rotation, revocation, escrow, and evidence.
Enforcement boundary	Keys are separated by tenant, domain, environment, purpose, and consequence where required; no application owns raw long-lived master key material.
Continuity rule	When key service, trust anchor, algorithm, hardware module, rotation, or revocation state becomes unsafe, the platform shall stop affected cryptographic operations, preserve encrypted state, invoke approved continuity keys, and re-establish trust before recovery.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-51-001	The infrastructure realization of Cryptography, Key Management, and Trust Anchors SHALL protect data, identity, artifacts, communications, audit, and approvals through governed cryptographic profiles and customer-compatible key authority.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-51-002	Every production instance SHALL declare and continuously reconcile key and certificate identity, algorithm profile, owner, purpose, locality, hardware boundary, rotation, revocation, escrow, and evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-51-003	The implementation SHALL enforce this boundary: keys are separated by tenant, domain, environment, purpose, and consequence where required; no application owns raw long-lived master key material.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-51-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Cryptography, Key Management, and Trust Anchors.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-51-005	When key service, trust anchor, algorithm, hardware module, rotation, or revocation state becomes unsafe, the platform SHALL stop affected cryptographic operations, preserve encrypted state, invoke approved continuity keys, and re-establish trust before recovery; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-51-006	Production admission and continued operation SHALL require key-generation ceremonies, isolation, rotation, revocation, backup and recovery, algorithm agility, and customer-key tests, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Key service, trust anchor, algorithm, hardware module, rotation, or revocation state becomes unsafe	Stop affected cryptographic operations, preserve encrypted state, invoke approved continuity keys, and re-establish trust before recovery.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared key and certificate identity, algorithm profile, owner, purpose, locality, hardware boundary, rotation, revocation, escrow, and evidence.
- Positive and negative boundary tests prove that keys are separated by tenant, domain, environment, purpose, and consequence where required; no application owns raw long-lived master key material.
- Fault exercises inject key service, trust anchor, algorithm, hardware module, rotation, or revocation state becomes unsafe and verify that the platform can stop affected cryptographic operations, preserve encrypted state, invoke approved continuity keys, and re-establish trust before recovery.
- Independent review accepts key-generation ceremonies, isolation, rotation, revocation, backup and recovery, algorithm agility, and customer-key tests for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Cryptography, Key Management, and Trust Anchors is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 51, Ch. 53, Ch. 54, Ch. 55, Ch. 62. Primary Volume 5 AI anchors: Ch. 15, Ch. 21, Ch. 60, Ch. 65, Ch. 71.

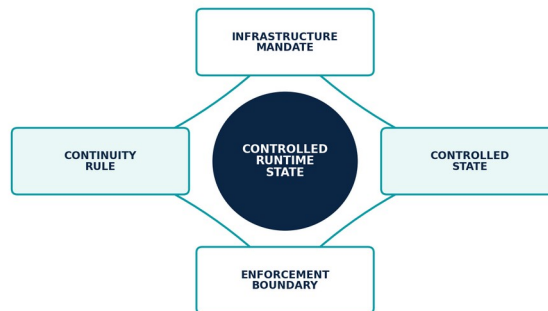
CONSTITUTIONAL INHERITANCE / C-006 · C-007 · C-030 · C-036 · C-049

52 Secrets, Certificates, and Credential Lifecycle

Secrets, Certificates, and Credential Lifecycle establishes the infrastructure necessary to issue, distribute, use, rotate, revoke, and destroy secrets and certificates without embedding durable authority in code, images, logs, or user content. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

SECRETS, CERTIFICATES, AND CREDENTIAL LIFECYCLE

Secrets, Certificates, and Credential Lifecycle establishes the infrastructure necessary to issue, distribute, use,...



ISOLATE | OBSERVE | RECOVER | PROVE

Figure 53. Secrets, Certificates, and Credential Lifecycle. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Issue, distribute, use, rotate, revoke, and destroy secrets and certificates without embedding durable authority in code, images, logs, or user content.
Controlled state	Secret identity, owner, consumer, purpose, scope, issuer, lease, delivery, use evidence, rotation, and revocation.
Enforcement boundary	Secret delivery must bind to attested identity and workload context; shared static credentials require an explicit time-bounded exception.
Continuity rule	When a secret leaks, expires, cannot rotate, resolves to the wrong workload, or remains after revocation, the platform shall revoke and contain immediately, rotate dependent authority, identify all uses and effects, and prove cleanup.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-52-001	The infrastructure realization of Secrets, Certificates, and Credential Lifecycle SHALL issue, distribute, use, rotate, revoke, and destroy secrets and certificates without embedding durable authority in code, images, logs, or user content.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-52-002	Every production instance SHALL declare and continuously reconcile secret identity, owner, consumer, purpose, scope, issuer, lease, delivery, use evidence, rotation, and revocation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-52-003	The implementation SHALL enforce this boundary: secret delivery must bind to attested identity and workload context; shared static credentials require an explicit time-bounded exception.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-52-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Secrets, Certificates, and Credential Lifecycle.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-52-005	When a secret leaks, expires, cannot rotate, resolves to the wrong workload, or remains after revocation, the platform SHALL revoke and contain immediately, rotate dependent authority, identify all uses and effects, and prove cleanup; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-52-006	Production admission and continued operation SHALL require secret scanning, dynamic issuance, rotation under load, revocation propagation, break-glass, and leak-response exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A secret leaks, expires, cannot rotate, resolves to the wrong workload, or remains after revocation	Revoke and contain immediately, rotate dependent authority, identify all uses and effects, and prove cleanup.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared secret identity, owner, consumer, purpose, scope, issuer, lease, delivery, use evidence, rotation, and revocation.
- Positive and negative boundary tests prove that secret delivery must bind to attested identity and workload context; shared static credentials require an explicit time-bounded exception.
- Fault exercises inject a secret leaks, expires, cannot rotate, resolves to the wrong workload, or remains after revocation and verify that the platform can revoke and contain immediately, rotate dependent authority, identify all uses and effects, and prove cleanup.
- Independent review accepts secret scanning, dynamic issuance, rotation under load, revocation propagation, break-glass, and leak-response exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Secrets, Certificates, and Credential Lifecycle is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 49, Ch. 51, Ch. 53, Ch. 55, Ch. 60. Primary Volume 5 AI anchors: Ch. 21, Ch. 36, Ch. 60, Ch. 65, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-007 · C-020 · C-030 · C-043 · C-049

53 Data Protection and Confidential Computing

Data Protection and Confidential Computing establishes the infrastructure necessary to protect strategic data in transit, at rest, in use, in backup, in telemetry, and during external processing according to classification and purpose. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

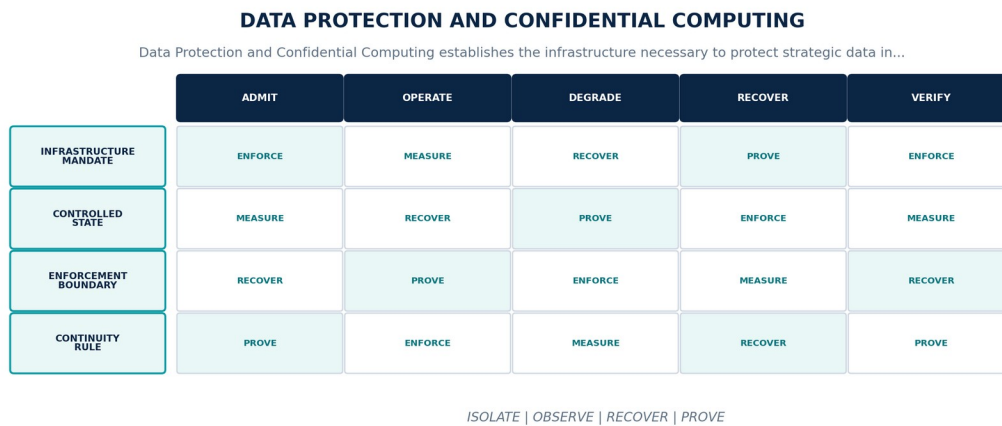


Figure 54. Data Protection and Confidential Computing. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Protect strategic data in transit, at rest, in use, in backup, in telemetry, and during external processing according to classification and purpose.
Controlled state	Protection profile, data class, encryption state, execution trust, privacy transform, locality, exposure path, and attestation.
Enforcement boundary	Confidential processing can reduce infrastructure trust but cannot substitute for authorization, minimization, output control, or human accountability.
Continuity rule	When encryption, attestation, privacy transform, memory isolation, or approved execution location cannot be established, the platform shall deny protected processing or move to an explicitly approved lower-capability mode without exposing the data.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-53-001	The infrastructure realization of Data Protection and Confidential Computing SHALL protect strategic data in transit, at rest, in use, in backup, in telemetry, and during external processing according to classification and purpose.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-53-002	Every production instance SHALL declare and continuously reconcile protection profile, data class, encryption state, execution trust, privacy transform, locality, exposure path, and attestation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-53-003	The implementation SHALL enforce this boundary: confidential processing can reduce infrastructure trust but cannot substitute for authorization, minimization, output control, or human accountability.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-53-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Data Protection and Confidential Computing.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-53-005	When encryption, attestation, privacy transform, memory isolation, or approved execution location cannot be established, the platform SHALL deny protected processing or move to an explicitly approved lower-capability mode without exposing the data; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-53-006	Production admission and continued operation SHALL require encryption and attestation tests, memory and dump controls, privacy leakage evaluation, locality verification, and recovery, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Encryption, attestation, privacy transform, memory isolation, or approved execution location cannot be established	Deny protected processing or move to an explicitly approved lower-capability mode without exposing the data.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared protection profile, data class, encryption state, execution trust, privacy transform, locality, exposure path, and attestation.
- Positive and negative boundary tests prove that confidential processing can reduce infrastructure trust but cannot substitute for authorization, minimization, output control, or human accountability.
- Fault exercises inject encryption, attestation, privacy transform, memory isolation, or approved execution location cannot be established and verify that the platform can deny protected processing or move to an explicitly approved lower-capability mode without exposing the data.
- Independent review accepts encryption and attestation tests, memory and dump controls, privacy leakage evaluation, locality verification, and recovery for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Data Protection and Confidential Computing is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 51, Ch. 53, Ch. 54, Ch. 60. Primary Volume 5 AI anchors: Ch. 21, Ch. 23, Ch. 29, Ch. 61, Ch. 65, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-007 · C-018 · C-029 · C-030 · C-036

54 Tenant, Domain, and Workload Isolation

Tenant, Domain, and Workload Isolation establishes the infrastructure necessary to enforce defense-in-depth isolation across identity, compute, memory, accelerator, network, storage, key, policy, telemetry, backup, and operations. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

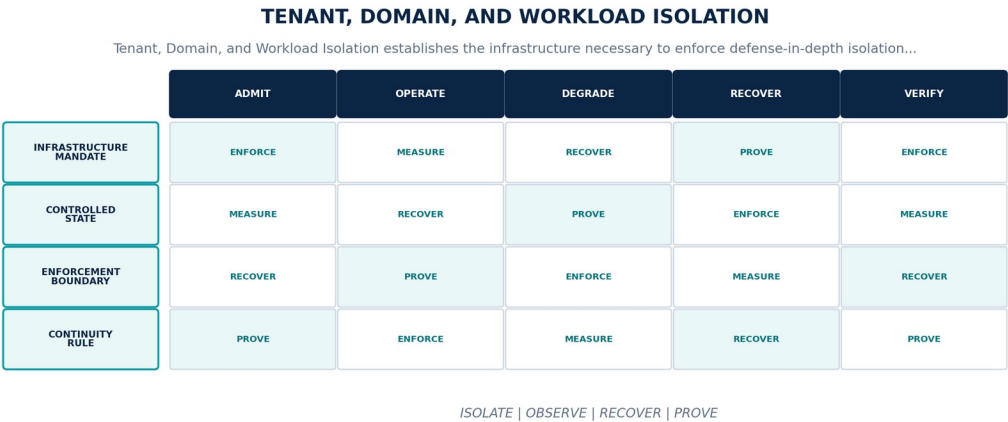


Figure 55. Tenant, Domain, and Workload Isolation. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Enforce defense-in-depth isolation across identity, compute, memory, accelerator, network, storage, key, policy, telemetry, backup, and operations.
Controlled state	Isolation profile, enforcement layers, shared resources, residual channels, tests, exception, monitoring, and incident state.
Enforcement boundary	No single namespace, network, hypervisor, container, application, or contractual control is sufficient isolation for protected workloads.
Continuity rule	When any isolation layer fails, a side channel exceeds tolerance, or tenant scope cannot be proven, the platform shall fail closed at the nearest enforceable boundary, quarantine shared resources, assess cross-domain exposure, and re-establish proof.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-54-001	The infrastructure realization of Tenant, Domain, and Workload Isolation SHALL enforce defense-in-depth isolation across identity, compute, memory, accelerator, network, storage, key, policy, telemetry, backup, and operations.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-54-002	Every production instance SHALL declare and continuously reconcile isolation profile, enforcement layers, shared resources, residual channels, tests, exception, monitoring, and incident state through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-54-003	The implementation SHALL enforce this boundary: no single namespace, network, hypervisor, container, application, or contractual control is sufficient isolation for protected workloads.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-54-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Tenant, Domain, and Workload Isolation.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-54-005	When any isolation layer fails, a side channel exceeds tolerance, or tenant scope cannot be proven, the platform SHALL fail closed at the nearest enforceable boundary, quarantine shared resources, assess cross-domain exposure, and re-establish proof; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-54-006	Production admission and continued operation SHALL require cross-domain negative tests, side-channel assessment, shared-resource failure, backup isolation, operator access, and forensic drills, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Any isolation layer fails, a side channel exceeds tolerance, or tenant scope cannot be proven	Fail closed at the nearest enforceable boundary, quarantine shared resources, assess cross-domain exposure, and re-establish proof.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared isolation profile, enforcement layers, shared resources, residual channels, tests, exception, monitoring, and incident state.
- Positive and negative boundary tests prove that no single namespace, network, hypervisor, container, application, or contractual control is sufficient isolation for protected workloads.
- Fault exercises inject any isolation layer fails, a side channel exceeds tolerance, or tenant scope cannot be proven and verify that the platform can fail closed at the nearest enforceable boundary, quarantine shared resources, assess cross-domain exposure, and re-establish proof.
- Independent review accepts cross-domain negative tests, side-channel assessment, shared-resource failure, backup isolation, operator access, and forensic drills for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Tenant, Domain, and Workload Isolation is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 8, Ch. 9, Ch. 13, Ch. 50, Ch. 54, Ch. 60. Primary Volume 5 AI anchors: Ch. 12, Ch. 21, Ch. 36, Ch. 59, Ch. 61.

CONSTITUTIONAL INHERITANCE / C-006 · C-007 · C-030 · C-034 · C-049

55 Software Supply Chain and Artifact Integrity

Software Supply Chain and Artifact Integrity establishes the infrastructure necessary to admit only traceable, reproducible, signed, scanned, policy-compliant software, models, firmware, infrastructure modules, and configuration. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

SOFTWARE SUPPLY CHAIN AND ARTIFACT INTEGRITY

Software Supply Chain and Artifact Integrity establishes the infrastructure necessary to admit only traceable,...

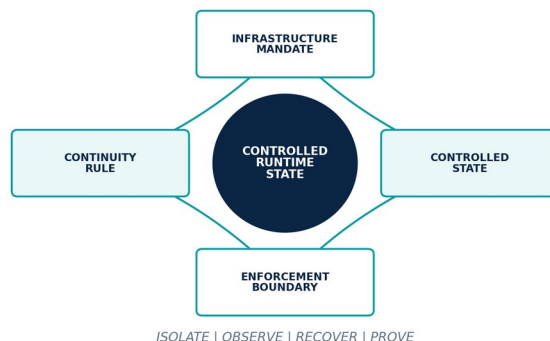


Figure 56. Software Supply Chain and Artifact Integrity. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Admit only traceable, reproducible, signed, scanned, policy-compliant software, models, firmware, infrastructure modules, and configuration.
Controlled state	Source revision, build, dependency and model bill of materials, provenance, signature, scan, policy, release, and revocation.
Enforcement boundary	Artifact trust closes over source, builder, dependencies, base images, packages, models, firmware, configuration, and deployment identity.
Continuity rule	When provenance is missing, a vulnerability or compromise emerges, a signature fails, or dependencies cannot be resolved, the platform shall block promotion, revoke affected artifacts, identify running exposure, execute replacement or rollback, and preserve the investigation chain.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-55-001	The infrastructure realization of Software Supply Chain and Artifact Integrity SHALL admit only traceable, reproducible, signed, scanned, policy-compliant software, models, firmware, infrastructure modules, and configuration.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-55-002	Every production instance SHALL declare and continuously reconcile source revision, build, dependency and model bill of materials, provenance, signature, scan, policy, release, and revocation through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-55-003	The implementation SHALL enforce this boundary: artifact trust closes over source, builder, dependencies, base images, packages, models, firmware, configuration, and deployment identity.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-55-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Software Supply Chain and Artifact Integrity.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-55-005	When provenance is missing, a vulnerability or compromise emerges, a signature fails, or dependencies cannot be resolved, the platform SHALL block promotion, revoke affected artifacts, identify running exposure, execute replacement or rollback, and preserve the investigation chain; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-55-006	Production admission and continued operation SHALL require reproducible-build checks, signature enforcement, dependency substitution, revocation propagation, fleet inventory, and compromise exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Provenance is missing, a vulnerability or compromise emerges, a signature fails, or dependencies cannot be resolved	Block promotion, revoke affected artifacts, identify running exposure, execute replacement or rollback, and preserve the investigation chain.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared source revision, build, dependency and model bill of materials, provenance, signature, scan, policy, release, and revocation.
- Positive and negative boundary tests prove that artifact trust closes over source, builder, dependencies, base images, packages, models, firmware, configuration, and deployment identity.
- Fault exercises inject provenance is missing, a vulnerability or compromise emerges, a signature fails, or dependencies cannot be resolved and verify that the platform can block promotion, revoke affected artifacts, identify running exposure, execute replacement or rollback, and preserve the investigation chain.
- Independent review accepts reproducible-build checks, signature enforcement, dependency substitution, revocation propagation, fleet inventory, and compromise exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Software Supply Chain and Artifact Integrity is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 53, Ch. 56, Ch. 65, Ch. 66, Ch. 68, Ch. 69. Primary Volume 5 AI anchors: Ch. 15, Ch. 18, Ch. 30, Ch. 62, Ch. 67, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-003 · C-007 · C-043 · C-048 · C-050

56 Security Monitoring, Detection, and Response

Security Monitoring, Detection, and Response establishes the infrastructure necessary to detect misuse, intrusion, policy violation, data exposure, supply-chain compromise, unsafe agent behavior, and control degradation across all planes. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

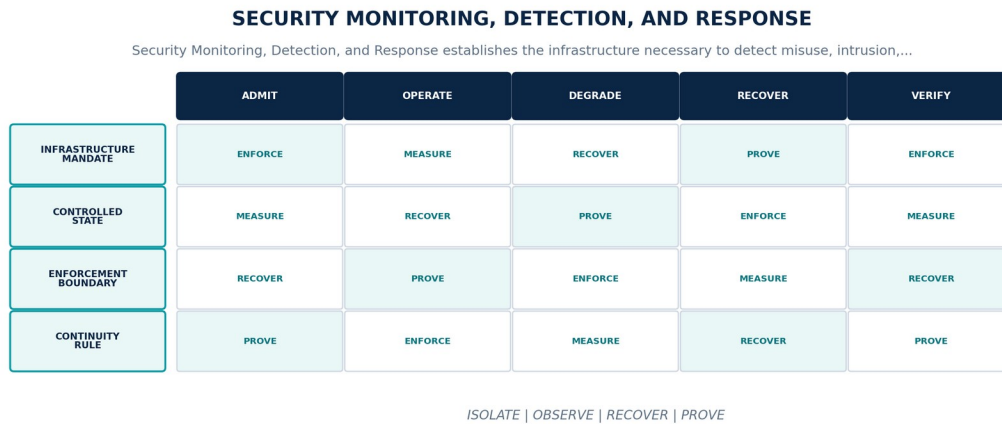


Figure 57. Security Monitoring, Detection, and Response. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Detect misuse, intrusion, policy violation, data exposure, supply-chain compromise, unsafe agent behavior, and control degradation across all planes.
Controlled state	Security signal, detector version, identity and asset context, severity, confidence, case, containment, evidence, and closure.
Enforcement boundary	Security telemetry is access-controlled evidence and must remain separable from product truth and model training unless explicitly governed.
Continuity rule	When critical telemetry is lost, a detector becomes untrusted, or an active threat exceeds automated containment authority, the platform shall enter the declared security-degraded state, constrain high-risk operations, preserve alternate evidence, and escalate to incident command.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-56-001	The infrastructure realization of Security Monitoring, Detection, and Response SHALL detect misuse, intrusion, policy violation, data exposure, supply-chain compromise, unsafe agent behavior, and control degradation across all planes.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-56-002	Every production instance SHALL declare and continuously reconcile security signal, detector version, identity and asset context, severity, confidence, case, containment, evidence, and closure through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-56-003	The implementation SHALL enforce this boundary: security telemetry is access-controlled evidence and must remain separable from product truth and model training unless explicitly governed.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-56-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Security Monitoring, Detection, and Response.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-56-005	When critical telemetry is lost, a detector becomes untrusted, or an active threat exceeds automated containment authority, the platform SHALL enter the declared security-degraded state, constrain high-risk operations, preserve alternate evidence, and escalate to incident command; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-56-006	Production admission and continued operation SHALL require attack simulations, telemetry-loss tests, detector validation, containment exercises, evidence custody, and post-incident verification, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Critical telemetry is lost, a detector becomes untrusted, or an active threat exceeds automated containment authority	Enter the declared security-degraded state, constrain high-risk operations, preserve alternate evidence, and escalate to incident command.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared security signal, detector version, identity and asset context, severity, confidence, case, containment, evidence, and closure.
- Positive and negative boundary tests prove that security telemetry is access-controlled evidence and must remain separable from product truth and model training unless explicitly governed.
- Fault exercises inject critical telemetry is lost, a detector becomes untrusted, or an active threat exceeds automated containment authority and verify that the platform can enter the declared security-degraded state, constrain high-risk operations, preserve alternate evidence, and escalate to incident command.
- Independent review accepts attack simulations, telemetry-loss tests, detector validation, containment exercises, evidence custody, and post-incident verification for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Security Monitoring, Detection, and Response is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 49, Ch. 54, Ch. 55, Ch. 56, Ch. 60, Ch. 63. Primary Volume 5 AI anchors: Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 62, Ch. 65, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-007 · C-030 · C-039 · C-043 · C-049

57 Sovereignty, Residency, and Administrative Control

Sovereignty, Residency, and Administrative Control establishes the infrastructure necessary to bind data, processing, keys, models, telemetry, support, updates, and administrative actions to customer, legal, contractual, and national boundaries. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

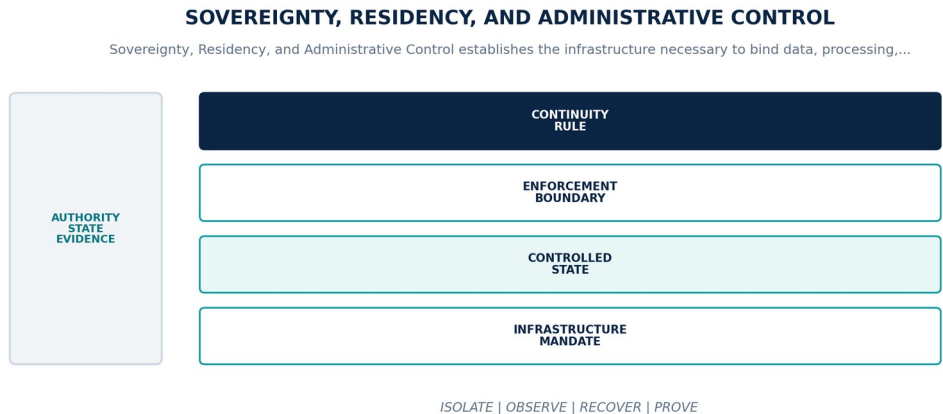


Figure 58. Sovereignty, Residency, and Administrative Control. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Bind data, processing, keys, models, telemetry, support, updates, and administrative actions to customer, legal, contractual, and national boundaries.
Controlled state	Sovereignty profile, data and processing locality, key authority, administrator jurisdiction, transfer path, exception, and evidence.
Enforcement boundary	Residency is enforced end to end across primary, replica, cache, backup, log, support, model provider, and recovery paths.
Continuity rule	When placement, support, recovery, update, or provider behavior would cross an unapproved sovereignty boundary, the platform shall deny the operation, select an approved local path or explicit reduced capability, and preserve the blocked reason.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-57-001	The infrastructure realization of Sovereignty, Residency, and Administrative Control SHALL bind data, processing, keys, models, telemetry, support, updates, and administrative actions to customer, legal, contractual, and national boundaries.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-57-002	Every production instance SHALL declare and continuously reconcile sovereignty profile, data and processing locality, key authority, administrator jurisdiction, transfer path, exception, and evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-57-003	The implementation SHALL enforce this boundary: residency is enforced end to end across primary, replica, cache, backup, log, support, model provider, and recovery paths.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-57-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Sovereignty, Residency, and Administrative Control.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-57-005	When placement, support, recovery, update, or provider behavior would cross an unapproved sovereignty boundary, the platform SHALL deny the operation, select an approved local path or explicit reduced capability, and preserve the blocked reason; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-57-006	Production admission and continued operation SHALL require placement and egress proofs, customer-admin tests, support-access ceremony, backup locality, provider denial, and DR exercises, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Placement, support, recovery, update, or provider behavior would cross an unapproved sovereignty boundary	Deny the operation, select an approved local path or explicit reduced capability, and preserve the blocked reason.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared sovereignty profile, data and processing locality, key authority, administrator jurisdiction, transfer path, exception, and evidence.
- Positive and negative boundary tests prove that residency is enforced end to end across primary, replica, cache, backup, log, support, model provider, and recovery paths.
- Fault exercises inject placement, support, recovery, update, or provider behavior would cross an unapproved sovereignty boundary and verify that the platform can deny the operation, select an approved local path or explicit reduced capability, and preserve the blocked reason.
- Independent review accepts placement and egress proofs, customer-admin tests, support-access ceremony, backup locality, provider denial, and DR exercises for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Sovereignty, Residency, and Administrative Control is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 29, Ch. 49, Ch. 51, Ch. 54, Ch. 62. Primary Volume 5 AI anchors: Ch. 21, Ch. 60, Ch. 61, Ch. 65, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-006 · C-018 · C-029 · C-036 · C-046

PART VIII

Reliability, Continuity, and Operations

Telemetry, service levels, capacity, availability, recovery, incidents, lifecycle operations, and degraded strategic service.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

58 Observability and Telemetry Architecture

Observability and Telemetry Architecture establishes the infrastructure necessary to produce correlated metrics, logs, traces, events, profiles, audit references, topology, and user-impact state without leaking protected content. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

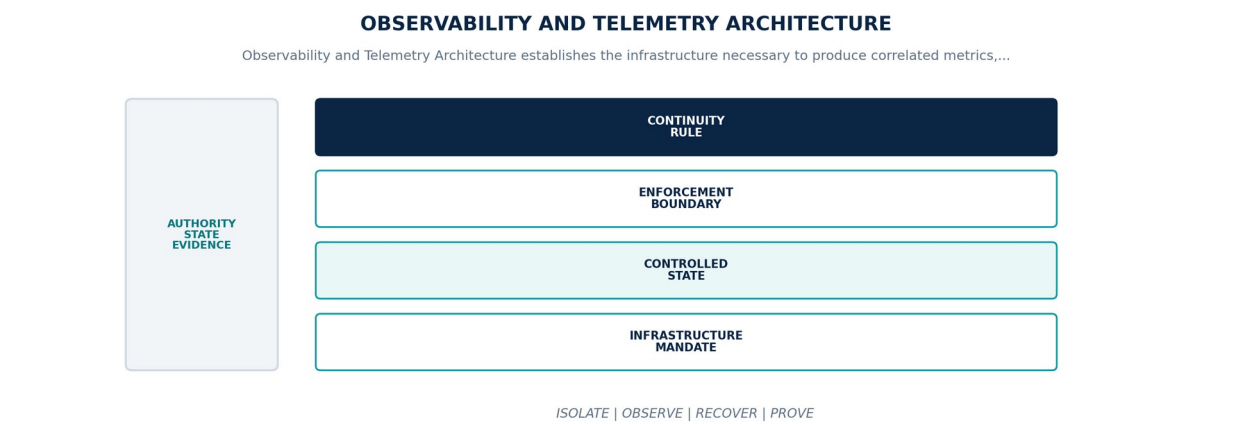


Figure 59. Observability and Telemetry Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Produce correlated metrics, logs, traces, events, profiles, audit references, topology, and user-impact state without leaking protected content.
Controlled state	Telemetry schema, source, identity, tenant and domain scope, correlation, sampling, retention, quality, and consumer.
Enforcement boundary	Operational telemetry supports diagnosis and assurance but cannot replace canonical evidence, workflow state, or protected audit records.
Continuity rule	When telemetry gaps, clock errors, sampling bias, correlation loss, or pipeline overload makes service state indeterminate, the platform shall surface observability degradation, preserve priority signals locally, constrain risky change, and restore complete causal coverage.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-58-001	The infrastructure realization of Observability and Telemetry Architecture SHALL produce correlated metrics, logs, traces, events, profiles, audit references, topology, and user-impact state without leaking protected content.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-58-002	Every production instance SHALL declare and continuously reconcile telemetry schema, source, identity, tenant and domain scope, correlation, sampling, retention, quality, and consumer through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-58-003	The implementation SHALL enforce this boundary: operational telemetry supports diagnosis and assurance but cannot replace canonical evidence, workflow state, or protected audit records.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-58-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Observability and Telemetry Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-58-005	When telemetry gaps, clock errors, sampling bias, correlation loss, or pipeline overload makes service state indeterminate, the platform SHALL surface observability degradation, preserve priority signals locally, constrain risky change, and restore complete causal coverage; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-58-006	Production admission and continued operation SHALL require end-to-end trace fixtures, telemetry-loss and overload tests, clock correlation, privacy review, and incident reconstruction, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Telemetry gaps, clock errors, sampling bias, correlation loss, or pipeline overload makes service state indeterminate	Surface observability degradation, preserve priority signals locally, constrain risky change, and restore complete causal coverage.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared telemetry schema, source, identity, tenant and domain scope, correlation, sampling, retention, quality, and consumer.
- Positive and negative boundary tests prove that operational telemetry supports diagnosis and assurance but cannot replace canonical evidence, workflow state, or protected audit records.
- Fault exercises inject telemetry gaps, clock errors, sampling bias, correlation loss, or pipeline overload makes service state indeterminate and verify that the platform can surface observability degradation, preserve priority signals locally, constrain risky change, and restore complete causal coverage.
- Independent review accepts end-to-end trace fixtures, telemetry-loss and overload tests, clock correlation, privacy review, and incident reconstruction for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Observability and Telemetry Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 28, Ch. 29, Ch. 57, Ch. 58, Ch. 60, Ch. 63. Primary Volume 5 AI anchors: Ch. 29, Ch. 31, Ch. 65, Ch. 68, Ch. 72.

59 Service-Level Management and Error Budgets

Service-Level Management and Error Budgets establishes the infrastructure necessary to govern availability, latency, freshness, durability, recovery, quality, safety, and human-workflow service objectives by consequence class. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

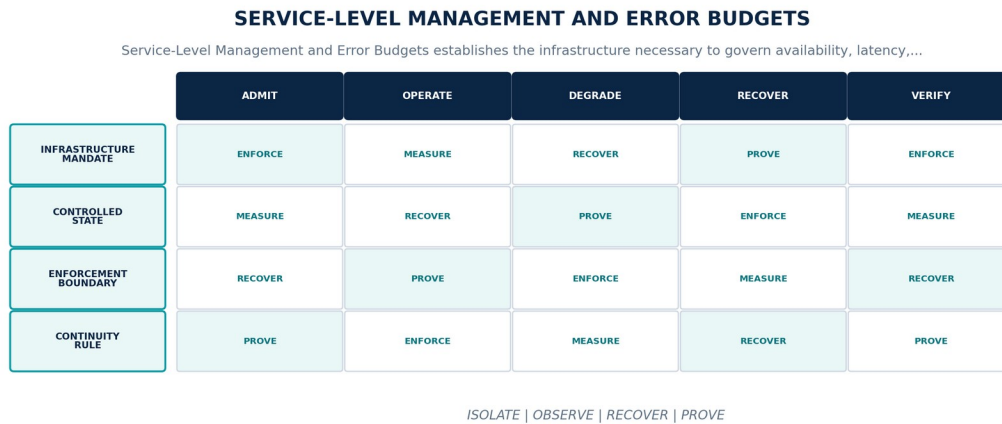


Figure 60. Service-Level Management and Error Budgets. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Govern availability, latency, freshness, durability, recovery, quality, safety, and human-workflow service objectives by consequence class.
Controlled state	Service-level indicator, objective, scope, measurement window, exclusions, error budget, burn state, owner, and action policy.
Enforcement boundary	An SLO is valid only when the measurement covers the user-visible product and explicit partial, stale, denied, and degraded states.
Continuity rule	When an error budget burns beyond policy or measurement quality cannot support a fitness claim, the platform shall halt or restrict change, prioritize recovery and debt, expose affected product state, and require accountable exception.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-59-001	The infrastructure realization of Service-Level Management and Error Budgets SHALL govern availability, latency, freshness, durability, recovery, quality, safety, and human-workflow service objectives by consequence class.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-59-002	Every production instance SHALL declare and continuously reconcile service-level indicator, objective, scope, measurement window, exclusions, error budget, burn state, owner, and action policy through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-59-003	The implementation SHALL enforce this boundary: an SLO is valid only when the measurement covers the user-visible product and explicit partial, stale, denied, and degraded states.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-59-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Service-Level Management and Error Budgets.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-59-005	When an error budget burns beyond policy or measurement quality cannot support a fitness claim, the platform SHALL halt or restrict change, prioritize recovery and debt, expose affected product state, and require accountable exception; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-59-006	Production admission and continued operation SHALL require indicator validation, synthetic and real-journey tests, burn simulations, dependency attribution, and governance review, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
An error budget burns beyond policy or measurement quality cannot support a fitness claim	Halt or restrict change, prioritize recovery and debt, expose affected product state, and require accountable exception.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared service-level indicator, objective, scope, measurement window, exclusions, error budget, burn state, owner, and action policy.
- Positive and negative boundary tests prove that an SLO is valid only when the measurement covers the user-visible product and explicit partial, stale, denied, and degraded states.
- Fault exercises inject an error budget burns beyond policy or measurement quality cannot support a fitness claim and verify that the platform can halt or restrict change, prioritize recovery and debt, expose affected product state, and require accountable exception.
- Independent review accepts indicator validation, synthetic and real-journey tests, burn simulations, dependency attribution, and governance review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Service-Level Management and Error Budgets is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 57, Ch. 58, Ch. 59, Ch. 63. Primary Volume 5 AI anchors: Ch. 63, Ch. 66, Ch. 68, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-035 · C-038 · C-039 · C-040 · C-041

60 Capacity, Performance, and Load Engineering

Capacity, Performance, and Load Engineering establishes the infrastructure necessary to size and validate compute, accelerator, network, storage, messaging, inference, workflow, and human-gate capacity against strategic demand. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

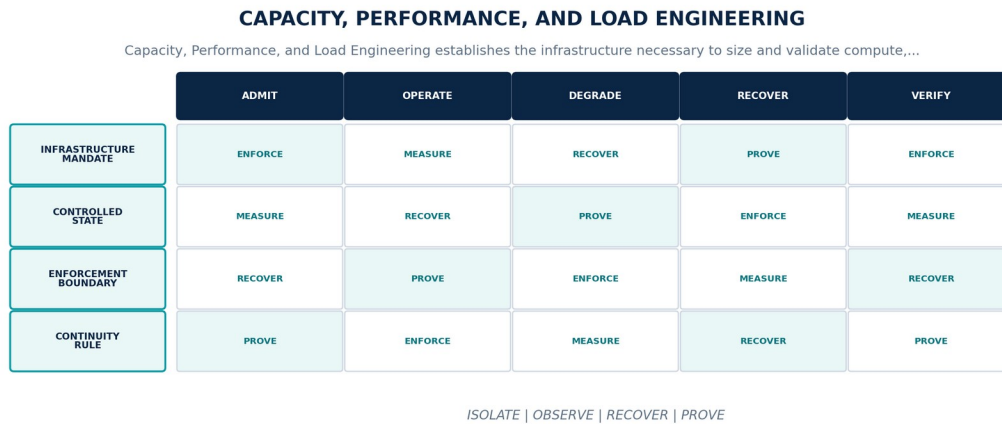


Figure 61. Capacity, Performance, and Load Engineering. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Size and validate compute, accelerator, network, storage, messaging, inference, workflow, and human-gate capacity against strategic demand.
Controlled state	Demand model, workload mix, resource profile, bottleneck, headroom, queue, saturation point, forecast, and scale action.
Enforcement boundary	Capacity models include burst, correlated events, recovery load, reprocessing, model changes, tenant concentration, and operational reserve.
Continuity rule	When measured demand exceeds safe headroom or scaling cannot meet the admitted workload envelope, the platform shall apply admission and priority policy, preserve high-consequence continuity, shed optional work, and publish recovery estimates.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-60-001	The infrastructure realization of Capacity, Performance, and Load Engineering SHALL size and validate compute, accelerator, network, storage, messaging, inference, workflow, and human-gate capacity against strategic demand.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.
IA-60-002	Every production instance SHALL declare and continuously reconcile demand model, workload mix, resource profile, bottleneck, headroom, queue, saturation point, forecast, and scale action through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-60-003	The implementation SHALL enforce this boundary: capacity models include burst, correlated events, recovery load, reprocessing, model changes, tenant concentration, and operational reserve.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-60-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Capacity, Performance, and Load Engineering.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-60-005	When measured demand exceeds safe headroom or scaling cannot meet the admitted workload envelope, the platform SHALL apply admission and priority policy, preserve high-consequence continuity, shed optional work, and publish recovery estimates; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-60-006	Production admission and continued operation SHALL require representative load, stress and soak tests, burst and recovery simulations, bottleneck evidence, and forecast accuracy, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Measured demand exceeds safe headroom or scaling cannot meet the admitted workload envelope	Apply admission and priority policy, preserve high-consequence continuity, shed optional work, and publish recovery estimates.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared demand model, workload mix, resource profile, bottleneck, headroom, queue, saturation point, forecast, and scale action.
- Positive and negative boundary tests prove that capacity models include burst, correlated events, recovery load, reprocessing, model changes, tenant concentration, and operational reserve.
- Fault exercises inject measured demand exceeds safe headroom or scaling cannot meet the admitted workload envelope and verify that the platform can apply admission and priority policy, preserve high-consequence continuity, shed optional work, and publish recovery estimates.
- Independent review accepts representative load, stress and soak tests, burst and recovery simulations, bottleneck evidence, and forecast accuracy for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Capacity, Performance, and Load Engineering is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 9, Ch. 22, Ch. 57, Ch. 58, Ch. 59, Ch. 61. Primary Volume 5 AI anchors: Ch. 19, Ch. 22, Ch. 31, Ch. 63, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-035 · C-038 · C-040 · C-041 · C-045

61 High Availability and Fault Tolerance

High Availability and Fault Tolerance establishes the infrastructure necessary to maintain declared strategic services through component, node, zone, cell, site, and dependency failures without corrupting authority or state. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

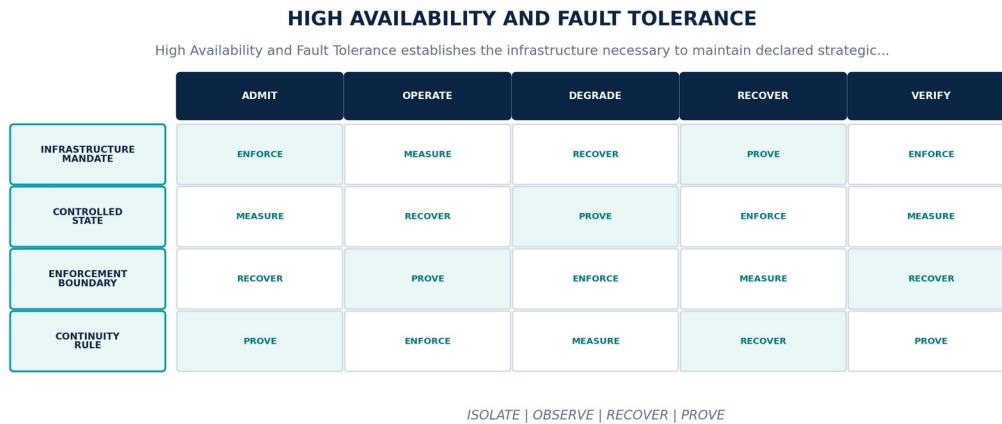


Figure 62. High Availability and Fault Tolerance. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Maintain declared strategic services through component, node, zone, cell, site, and dependency failures without corrupting authority or state.
Controlled state	Availability profile, replica and quorum set, failover policy, health criteria, dependency mode, recovery state, and evidence.
Enforcement boundary	Redundancy is valid only when replicas do not share an undeclared failure, control, key, network, storage, or release domain.
Continuity rule	When a failure exceeds redundancy, quorum is lost, health is ambiguous, or failover would violate consistency or sovereignty, the platform shall prefer safe bounded degradation over false availability, fence unsafe members, and recover under explicit state reconciliation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-61-001	The infrastructure realization of High Availability and Fault Tolerance SHALL maintain declared strategic services through component, node, zone, cell, site, and dependency failures without corrupting authority or state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-61-002	Every production instance SHALL declare and continuously reconcile availability profile, replica and quorum set, failover policy, health criteria, dependency mode, recovery state, and evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-61-003	The implementation SHALL enforce this boundary: redundancy is valid only when replicas do not share an undeclared failure, control, key, network, storage, or release domain.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-61-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for High Availability and Fault Tolerance.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-61-005	When a failure exceeds redundancy, quorum is lost, health is ambiguous, or failover would violate consistency or sovereignty, the platform SHALL prefer safe bounded degradation over false availability, fence unsafe members, and recover under explicit state reconciliation; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-61-006	Production admission and continued operation SHALL require component through site fault injection, quorum and fencing, dependency loss, failback, data reconciliation, and user-impact evidence, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A failure exceeds redundancy, quorum is lost, health is ambiguous, or failover would violate consistency or sovereignty	Prefer safe bounded degradation over false availability, fence unsafe members, and recover under explicit state reconciliation.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared availability profile, replica and quorum set, failover policy, health criteria, dependency mode, recovery state, and evidence.
- Positive and negative boundary tests prove that redundancy is valid only when replicas do not share an undeclared failure, control, key, network, storage, or release domain.
- Fault exercises inject a failure exceeds redundancy, quorum is lost, health is ambiguous, or failover would violate consistency or sovereignty and verify that the platform can prefer safe bounded degradation over false availability, fence unsafe members, and recover under explicit state reconciliation.
- Independent review accepts component through site fault injection, quorum and fencing, dependency loss, failback, data reconciliation, and user-impact evidence for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

High Availability and Fault Tolerance is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 7, Ch. 12, Ch. 22, Ch. 58, Ch. 59, Ch. 61. Primary Volume 5 AI anchors: Ch. 31, Ch. 34, Ch. 38, Ch. 63, Ch. 68, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-034 · C-035 · C-038 · C-041 · C-045

62 Backup, Restore, and State Reconciliation

Backup, Restore, and State Reconciliation establishes the infrastructure necessary to protect every durable state class with policy-matched backup, immutable recovery points, tested restore, integrity validation, and consumer reconciliation. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

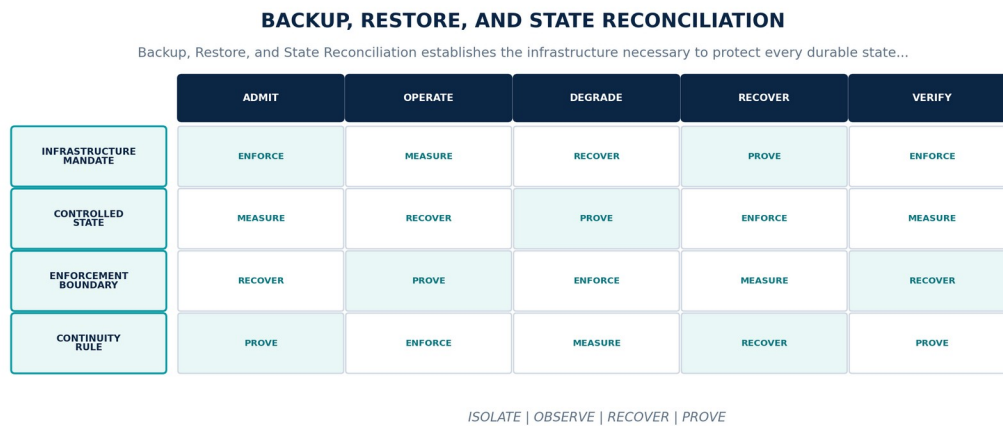


Figure 63. Backup, Restore, and State Reconciliation. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Protect every durable state class with policy-matched backup, immutable recovery points, tested restore, integrity validation, and consumer reconciliation.
Controlled state	Backup set, source revision, consistency point, encryption, locality, retention, legal hold, restore test, and recovery disposition.
Enforcement boundary	Backup coverage follows state ownership and includes metadata, schemas, keys, configurations, and dependency order needed for usable recovery.
Continuity rule	When a backup is missing, corrupt, unverifiable, outside locality, or cannot restore within objective, the platform shall declare the affected recovery posture, protect remaining copies, constrain destructive change, and execute alternate recovery.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-62-001	The infrastructure realization of Backup, Restore, and State Reconciliation SHALL protect every durable state class with policy-matched backup, immutable recovery points, tested restore, integrity validation, and consumer reconciliation.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-62-002	Every production instance SHALL declare and continuously reconcile backup set, source revision, consistency point, encryption, locality, retention, legal hold, restore test, and recovery disposition through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-62-003	The implementation SHALL enforce this boundary: backup coverage follows state ownership and includes metadata, schemas, keys, configurations, and dependency order needed for usable recovery.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-62-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Backup, Restore, and State Reconciliation.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-62-005	When a backup is missing, corrupt, unverifiable, outside locality, or cannot restore within objective, the platform SHALL declare the affected recovery posture, protect remaining copies, constrain destructive change, and execute alternate recovery; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-62-006	Production admission and continued operation SHALL require automated restore verification, corruption and deletion drills, key recovery, point-in-time restore, projection rebuild, and consumer reconciliation, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A backup is missing, corrupt, unverifiable, outside locality, or cannot restore within objective	Declare the affected recovery posture, protect remaining copies, constrain destructive change, and execute alternate recovery.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared backup set, source revision, consistency point, encryption, locality, retention, legal hold, restore test, and recovery disposition.
- Positive and negative boundary tests prove that backup coverage follows state ownership and includes metadata, schemas, keys, configurations, and dependency order needed for usable recovery.
- Fault exercises inject a backup is missing, corrupt, unverifiable, outside locality, or cannot restore within objective and verify that the platform can declare the affected recovery posture, protect remaining copies, constrain destructive change, and execute alternate recovery.
- Independent review accepts automated restore verification, corruption and deletion drills, key recovery, point-in-time restore, projection rebuild, and consumer reconciliation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Backup, Restore, and State Reconciliation is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 30, Ch. 54, Ch. 59, Ch. 62. Primary Volume 5 AI anchors: Ch. 15, Ch. 25, Ch. 26, Ch. 38, Ch. 70, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-012 · C-016 · C-029 · C-033 · C-041

63 Disaster Recovery and Regional Continuity

Disaster Recovery and Regional Continuity establishes the infrastructure necessary to restore strategic service across regional, site, control-plane, dependency, and workforce disasters within declared recovery objectives. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

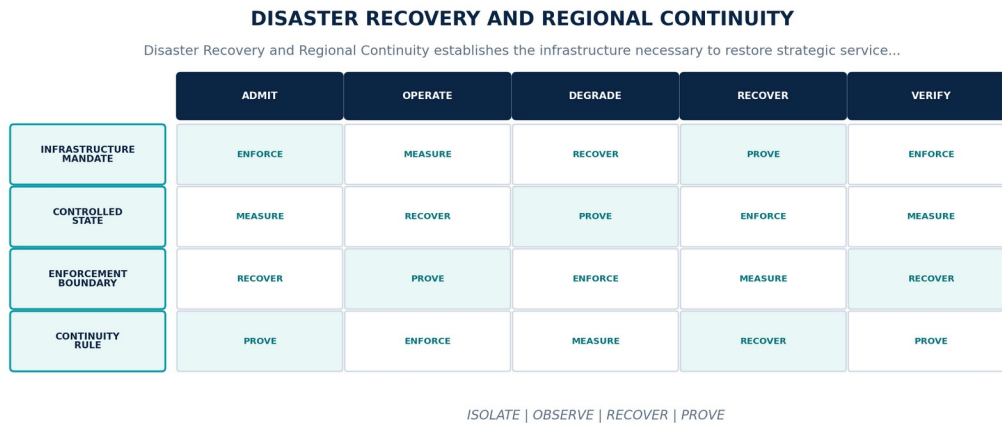


Figure 64. Disaster Recovery and Regional Continuity. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Restore strategic service across regional, site, control-plane, dependency, and workforce disasters within declared recovery objectives.
Controlled state	Recovery unit, dependency order, recovery point and time objective, target location, readiness, authority, exercise, and declaration.
Enforcement boundary	Disaster recovery preserves tenant, sovereignty, key, policy, version, evidence, workflow, and decision semantics; it is not only data copying.
Continuity rule	When the primary domain is unavailable or the recovery target cannot prove readiness and consistency, the platform shall invoke incident authority, choose the certified recovery path, declare data and capability boundaries, and reconcile before reopening.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-63-001	The infrastructure realization of Disaster Recovery and Regional Continuity SHALL restore strategic service across regional, site, control-plane, dependency, and workforce disasters within declared recovery objectives.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-63-002	Every production instance SHALL declare and continuously reconcile recovery unit, dependency order, recovery point and time objective, target location, readiness, authority, exercise, and declaration through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-63-003	The implementation SHALL enforce this boundary: disaster recovery preserves tenant, sovereignty, key, policy, version, evidence, workflow, and decision semantics; it is not only data copying.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-63-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Disaster Recovery and Regional Continuity.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-63-005	When the primary domain is unavailable or the recovery target cannot prove readiness and consistency, the platform SHALL invoke incident authority, choose the certified recovery path, declare data and capability boundaries, and reconcile before reopening; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-63-006	Production admission and continued operation SHALL require regional and site loss exercises, isolated recovery, key and identity continuity, workflow reconciliation, fallback, and customer communication, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
The primary domain is unavailable or the recovery target cannot prove readiness and consistency	Invoke incident authority, choose the certified recovery path, declare data and capability boundaries, and reconcile before reopening.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared recovery unit, dependency order, recovery point and time objective, target location, readiness, authority, exercise, and declaration.
- Positive and negative boundary tests prove that disaster recovery preserves tenant, sovereignty, key, policy, version, evidence, workflow, and decision semantics; it is not only data copying.
- Fault exercises inject the primary domain is unavailable or the recovery target cannot prove readiness and consistency and verify that the platform can invoke incident authority, choose the certified recovery path, declare data and capability boundaries, and reconcile before reopening.
- Independent review accepts regional and site loss exercises, isolated recovery, key and identity continuity, workflow reconciliation, fallback, and customer communication for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Disaster Recovery and Regional Continuity is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 41, Ch. 59, Ch. 61, Ch. 62, Ch. 63. Primary Volume 5 AI anchors: Ch. 38, Ch. 63, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-034 · C-036 · C-038 · C-041 · C-045

64 Incident Command and Operational Response

Incident Command and Operational Response establishes the infrastructure necessary to coordinate detection, triage, authority, containment, continuity, recovery, communication, evidence, decision impact, and learning. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

INCIDENT COMMAND AND OPERATIONAL RESPONSE

Incident Command and Operational Response establishes the infrastructure necessary to coordinate detection, triage,...

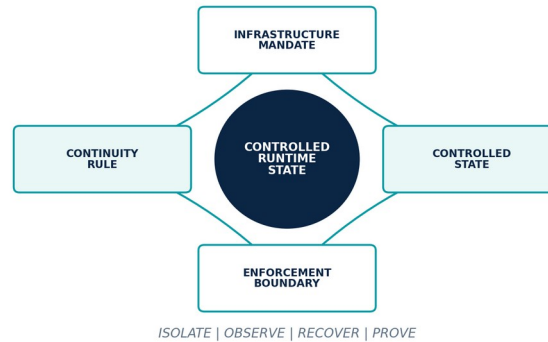


Figure 65. Incident Command and Operational Response. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Coordinate detection, triage, authority, containment, continuity, recovery, communication, evidence, decision impact, and learning.
Controlled state	Incident identity, severity, command roles, affected scope, timeline, actions, evidence, customer state, decision impact, and closure.
Enforcement boundary	Automation may detect and recommend; named humans hold incident command, risk acceptance, customer communication, and return-to-service authority.
Continuity rule	When ownership, severity, scope, or decision impact is uncertain during an active incident, the platform shall assume the safer consequence class, establish command, constrain effects, preserve evidence, and update scope as facts change.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-64-001	The infrastructure realization of Incident Command and Operational Response SHALL coordinate detection, triage, authority, containment, continuity, recovery, communication, evidence, decision impact, and learning.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-64-002	Every production instance SHALL declare and continuously reconcile incident identity, severity, command roles, affected scope, timeline, actions, evidence, customer state, decision impact, and closure through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-64-003	The implementation SHALL enforce this boundary: automation may detect and recommend; named humans hold incident command, risk acceptance, customer communication, and return-to-service authority.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-64-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Incident Command and Operational Response.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-64-005	When ownership, severity, scope, or decision impact is uncertain during an active incident, the platform SHALL assume the safer consequence class, establish command, constrain effects, preserve evidence, and update scope as facts change; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-64-006	Production admission and continued operation SHALL require tabletop and live exercises, command handoff, communications, forensic custody, decision-impact analysis, and closure review, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Ownership, severity, scope, or decision impact is uncertain during an active incident	Assume the safer consequence class, establish command, constrain effects, preserve evidence, and update scope as facts change.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared incident identity, severity, command roles, affected scope, timeline, actions, evidence, customer state, decision impact, and closure.
- Positive and negative boundary tests prove that automation may detect and recommend; named humans hold incident command, risk acceptance, customer communication, and return-to-service authority.
- Fault exercises inject ownership, severity, scope, or decision impact is uncertain during an active incident and verify that the platform can assume the safer consequence class, establish command, constrain effects, preserve evidence, and update scope as facts change.
- Independent review accepts tabletop and live exercises, command handoff, communications, forensic custody, decision-impact analysis, and closure review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Incident Command and Operational Response is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 42, Ch. 55, Ch. 60, Ch. 63. Primary Volume 5 AI anchors: Ch. 3, Ch. 37, Ch. 59, Ch. 65, Ch. 72.

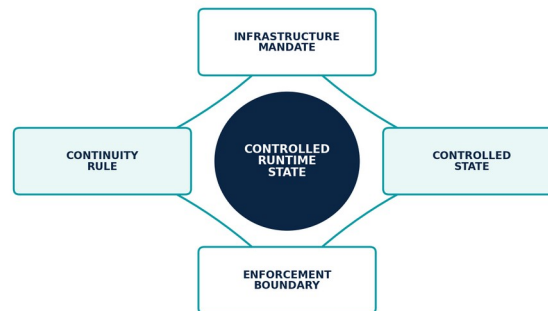
CONSTITUTIONAL INHERITANCE / C-004 · C-028 · C-039 · C-043 · C-049

65 Change, Maintenance, and Lifecycle Operations

Change, Maintenance, and Lifecycle Operations establishes the infrastructure necessary to execute configuration, patch, scale, migration, certificate, hardware, dependency, and maintenance changes through controlled reversible workflows. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

CHANGE, MAINTENANCE, AND LIFECYCLE OPERATIONS

Change, Maintenance, and Lifecycle Operations establishes the infrastructure necessary to execute configuration,...



ISOLATE | OBSERVE | RECOVER | PROVE

Figure 66. Change, Maintenance, and Lifecycle Operations. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Execute configuration, patch, scale, migration, certificate, hardware, dependency, and maintenance changes through controlled reversible workflows.
Controlled state	Change identity, desired diff, affected graph, risk, window, approvers, rollout, observations, rollback, and closure.
Enforcement boundary	Routine automation does not remove peer review, segregation of duties, customer notice, or stop authority for material change.
Continuity rule	When observations diverge from plan, rollback becomes unsafe, or change impact exceeds the approved scope, the platform shall stop the rollout, stabilize the current state, invoke the declared recovery path, and reopen approval with observed evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-65-001	The infrastructure realization of Change, Maintenance, and Lifecycle Operations SHALL execute configuration, patch, scale, migration, certificate, hardware, dependency, and maintenance changes through controlled reversible workflows.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-65-002	Every production instance SHALL declare and continuously reconcile change identity, desired diff, affected graph, risk, window, approvers, rollout, observations, rollback, and closure through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-65-003	The implementation SHALL enforce this boundary: routine automation does not remove peer review, segregation of duties, customer notice, or stop authority for material change.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.
IA-65-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Change, Maintenance, and Lifecycle Operations.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-65-005	When observations diverge from plan, rollback becomes unsafe, or change impact exceeds the approved scope, the platform SHALL stop the rollout, stabilize the current state, invoke the declared recovery path, and reopen approval with observed evidence; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-65-006	Production admission and continued operation SHALL require dry runs, canary and cohort analysis, rollback and roll-forward tests, maintenance-mode behavior, and audit closure, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Observations diverge from plan, rollback becomes unsafe, or change impact exceeds the approved scope	Stop the rollout, stabilize the current state, invoke the declared recovery path, and reopen approval with observed evidence.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared change identity, desired diff, affected graph, risk, window, approvers, rollout, observations, rollback, and closure.
- Positive and negative boundary tests prove that routine automation does not remove peer review, segregation of duties, customer notice, or stop authority for material change.
- Fault exercises inject observations diverge from plan, rollback becomes unsafe, or change impact exceeds the approved scope and verify that the platform can stop the rollout, stabilize the current state, invoke the declared recovery path, and reopen approval with observed evidence.
- Independent review accepts dry runs, canary and cohort analysis, rollback and roll-forward tests, maintenance-mode behavior, and audit closure for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Change, Maintenance, and Lifecycle Operations is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 30, Ch. 55, Ch. 63, Ch. 66, Ch. 68, Ch. 69. Primary Volume 5 AI anchors: Ch. 15, Ch. 18, Ch. 68, Ch. 69, Ch. 70, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-003 · C-008 · C-035 · C-043 · C-050

66 Degraded Modes and Continuity of Strategic Operations

Degraded Modes and Continuity of Strategic Operations establishes the infrastructure necessary to preserve the safest useful strategic capability when dependencies, data, models, networks, control services, or sites are impaired. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

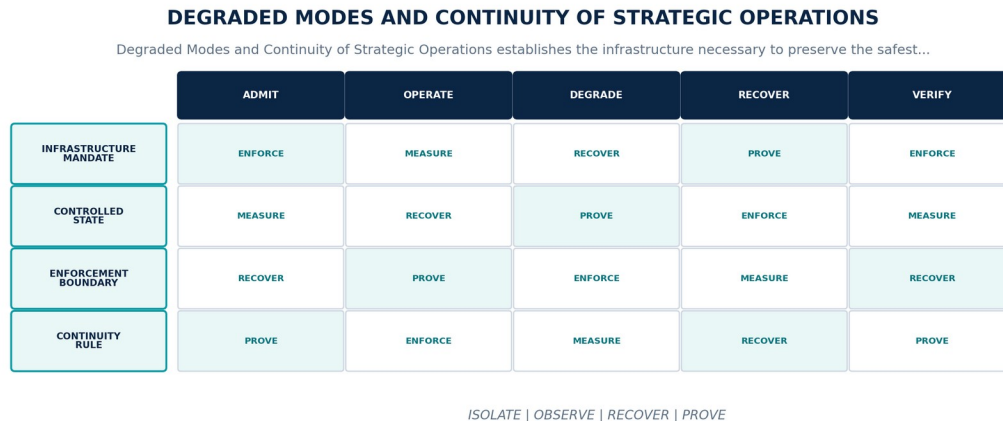


Figure 67. Degraded Modes and Continuity of Strategic Operations. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Preserve the safest useful strategic capability when dependencies, data, models, networks, control services, or sites are impaired.
Controlled state	Degraded-mode identity, trigger, available capability, unavailable capability, trust and freshness limits, owner, entry, and exit evidence.
Enforcement boundary	Degradation is explicit product state and may never fabricate completeness, freshness, confidence, approval, external effect, or successful learning.
Continuity rule	When normal service cannot be maintained or the declared degraded mode itself becomes unsafe, the platform shall enter the next safer mode, suspend consequential functions where required, maintain human awareness, and recover through verified exit criteria.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-66-001	The infrastructure realization of Degraded Modes and Continuity of Strategic Operations SHALL preserve the safest useful strategic capability when dependencies, data, models, networks, control services, or sites are impaired.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-66-002	Every production instance SHALL declare and continuously reconcile degraded-mode identity, trigger, available capability, unavailable capability, trust and freshness limits, owner, entry, and exit evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-66-003	The implementation SHALL enforce this boundary: degradation is explicit product state and may never fabricate completeness, freshness, confidence, approval, external effect, or successful learning.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-66-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Degraded Modes and Continuity of Strategic Operations.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-66-005	When normal service cannot be maintained or the declared degraded mode itself becomes unsafe, the platform SHALL enter the next safer mode, suspend consequential functions where required, maintain human awareness, and recover through verified exit criteria; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-66-006	Production admission and continued operation SHALL require dependency-loss matrix, degraded journey tests, authority and warning behavior, duration limits, recovery, and user comprehension, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Normal service cannot be maintained or the declared degraded mode itself becomes unsafe	Enter the next safer mode, suspend consequential functions where required, maintain human awareness, and recover through verified exit criteria.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared degraded-mode identity, trigger, available capability, unavailable capability, trust and freshness limits, owner, entry, and exit evidence.
- Positive and negative boundary tests prove that degradation is explicit product state and may never fabricate completeness, freshness, confidence, approval, external effect, or successful learning.
- Fault exercises inject normal service cannot be maintained or the declared degraded mode itself becomes unsafe and verify that the platform can enter the next safer mode, suspend consequential functions where required, maintain human awareness, and recover through verified exit criteria.
- Independent review accepts dependency-loss matrix, degraded journey tests, authority and warning behavior, duration limits, recovery, and user comprehension for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Degraded Modes and Continuity of Strategic Operations is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 41, Ch. 58, Ch. 61, Ch. 63, Ch. 64. Primary Volume 5 AI anchors: Ch. 3, Ch. 20, Ch. 31, Ch. 37, Ch. 63, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-035 · C-041 · C-049

PART IX

Delivery, Economics, and Governance

Infrastructure-as-code, release, fleet security, economics, sustainability, acceptance, and baseline governance.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

67 Infrastructure-as-Code and Configuration Architecture

Infrastructure-as-Code and Configuration Architecture establishes the infrastructure necessary to represent infrastructure desired state, policy, topology, deployment profiles, and exceptions as reviewed, versioned, testable code and data. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

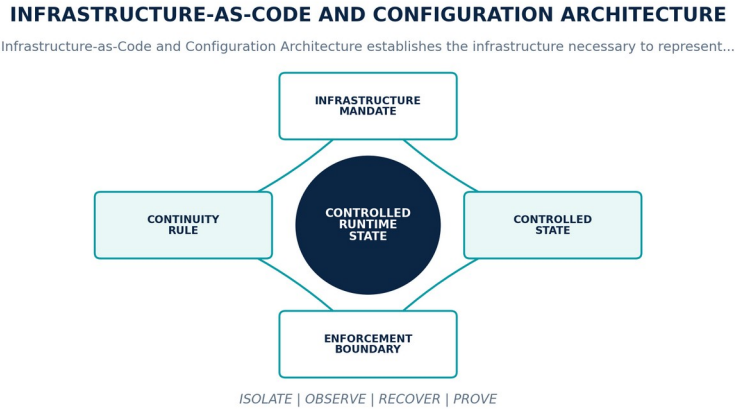


Figure 68. Infrastructure-as-Code and Configuration Architecture. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Represent infrastructure desired state, policy, topology, deployment profiles, and exceptions as reviewed, versioned, testable code and data.
Controlled state	Module and configuration identity, source revision, input schema, plan, policy result, state backend, applied revision, drift, and owner.
Enforcement boundary	Declarative automation controls infrastructure state but cannot bypass customer authority, runtime safety checks, or protected manual ceremonies.
Continuity rule	When desired and actual state drift, state locks fail, plans become stale, or an apply partially succeeds, the platform shall stop further mutation, preserve the exact plan and partial effects, reconcile under change control, and restore a trusted state backend.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-67-001	The infrastructure realization of Infrastructure-as-Code and Configuration Architecture SHALL represent infrastructure desired state, policy, topology, deployment profiles, and exceptions as reviewed, versioned, testable code and data.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-67-002	Every production instance SHALL declare and continuously reconcile module and configuration identity, source revision, input schema, plan, policy result, state backend, applied revision, drift, and owner through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-67-003	The implementation SHALL enforce this boundary: declarative automation controls infrastructure state but cannot bypass customer authority, runtime safety checks, or protected manual ceremonies.	System test + review	End-to-end result plus named customer or institutional authority, exact scope, rationale, conditions, and expiry.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-67-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Infrastructure-as-Code and Configuration Architecture.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-67-005	When desired and actual state drift, state locks fail, plans become stale, or an apply partially succeeds, the platform SHALL stop further mutation, preserve the exact plan and partial effects, reconcile under change control, and restore a trusted state backend; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-67-006	Production admission and continued operation SHALL require module tests, policy checks, plan review, drift detection, partial-apply recovery, state-backend restore, and parity fixtures, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Desired and actual state drift, state locks fail, plans become stale, or an apply partially succeeds	Stop further mutation, preserve the exact plan and partial effects, reconcile under change control, and restore a trusted state backend.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared module and configuration identity, source revision, input schema, plan, policy result, state backend, applied revision, drift, and owner.
- Positive and negative boundary tests prove that declarative automation controls infrastructure state but cannot bypass customer authority, runtime safety checks, or protected manual ceremonies.
- Fault exercises inject desired and actual state drift, state locks fail, plans become stale, or an apply partially succeeds and verify that the platform can stop further mutation, preserve the exact plan and partial effects, reconcile under change control, and restore a trusted state backend.
- Independent review accepts module tests, policy checks, plan review, drift detection, partial-apply recovery, state-backend restore, and parity fixtures for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Infrastructure-as-Code and Configuration Architecture is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 6, Ch. 13, Ch. 65, Ch. 66, Ch. 68, Ch. 72. Primary Volume 5 AI anchors: Ch. 15, Ch. 18, Ch. 70, Ch. 71.

68 Build, Release, Promotion, and Rollback Infrastructure

Build, Release, Promotion, and Rollback Infrastructure establishes the infrastructure necessary to assemble dependency-closed infrastructure releases and promote immutable configurations through evidence-gated environments and cohorts. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

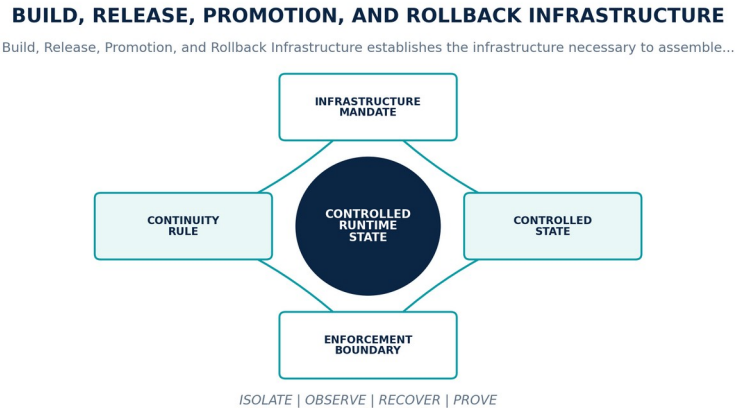


Figure 69. Build, Release, Promotion, and Rollback Infrastructure. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Assemble dependency-closed infrastructure releases and promote immutable configurations through evidence-gated environments and cohorts.
Controlled state	Release identity, artifacts, configuration, schemas, policies, models, tests, approvals, cohort, promotion, and rollback target.
Enforcement boundary	A release is the complete resolved system configuration; mutable tags, floating dependencies, and out-of-band production edits are non-conformant.
Continuity rule	When promotion evidence fails, rollout indicators degrade, dependencies drift, or rollback compatibility is uncertain, the platform shall halt promotion, contain the cohort, rollback or roll forward through an approved target, and preserve all decision evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-68-001	The infrastructure realization of Build, Release, Promotion, and Rollback Infrastructure SHALL assemble dependency-closed infrastructure releases and promote immutable configurations through evidence-gated environments and cohorts.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-68-002	Every production instance SHALL declare and continuously reconcile release identity, artifacts, configuration, schemas, policies, models, tests, approvals, cohort, promotion, and rollback target through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-68-003	The implementation SHALL enforce this boundary: a release is the complete resolved system configuration; mutable tags, floating dependencies, and out-of-band production edits are non-conformant.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-68-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Build, Release, Promotion, and Rollback Infrastructure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-68-005	When promotion evidence fails, rollout indicators degrade, dependencies drift, or rollback compatibility is uncertain, the platform SHALL halt promotion, contain the cohort, rollback or roll forward through an approved target, and preserve all decision evidence; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-68-006	Production admission and continued operation SHALL require dependency closure, reproducibility, staged promotion, canary analysis, rollback and migration, and cross-deployment acceptance, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Promotion evidence fails, rollout indicators degrade, dependencies drift, or rollback compatibility is uncertain	Halt promotion, contain the cohort, rollback or roll forward through an approved target, and preserve all decision evidence.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared release identity, artifacts, configuration, schemas, policies, models, tests, approvals, cohort, promotion, and rollback target.
- Positive and negative boundary tests prove that a release is the complete resolved system configuration; mutable tags, floating dependencies, and out-of-band production edits are non-conformant.
- Fault exercises inject promotion evidence fails, rollout indicators degrade, dependencies drift, or rollback compatibility is uncertain and verify that the platform can halt promotion, contain the cohort, rollback or roll forward through an approved target, and preserve all decision evidence.
- Independent review accepts dependency closure, reproducibility, staged promotion, canary analysis, rollback and migration, and cross-deployment acceptance for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Build, Release, Promotion, and Rollback Infrastructure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 30, Ch. 66, Ch. 68, Ch. 70, Ch. 71. Primary Volume 5 AI anchors: Ch. 15, Ch. 18, Ch. 66, Ch. 67, Ch. 70, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-003 · C-008 · C-035 · C-043 · C-050

69 Fleet, Patch, and Vulnerability Management

Fleet, Patch, and Vulnerability Management establishes the infrastructure necessary to maintain complete asset inventory, configuration posture, vulnerability state, patch eligibility, exceptions, and evidence across connected and disconnected fleets. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

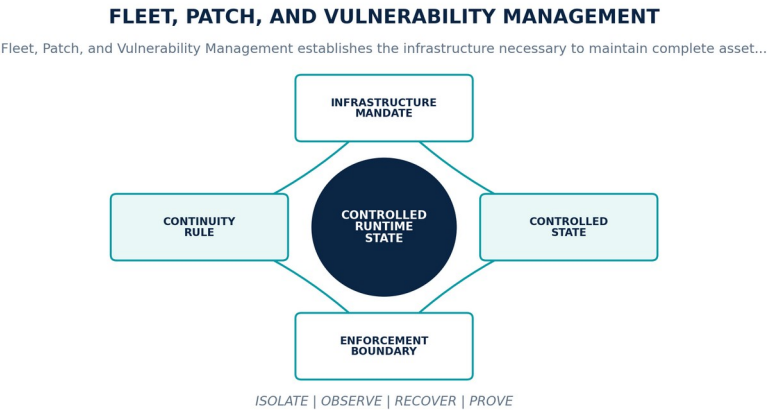


Figure 70. Fleet, Patch, and Vulnerability Management. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Maintain complete asset inventory, configuration posture, vulnerability state, patch eligibility, exceptions, and evidence across connected and disconnected fleets.
Controlled state	Asset identity, hardware and software bill of materials, posture, vulnerability, exploitability, patch, cohort, exception, and retirement.
Enforcement boundary	Severity alone does not determine action; exploitability, exposure, consequence, compatibility, compensating controls, and recovery are evaluated together.
Continuity rule	When a critical exposure emerges, inventory is incomplete, patching fails, or support expires, the platform shall contain reachable risk, revoke or isolate affected assets, deploy the certified remedy, and track exceptions to closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-69-001	The infrastructure realization of Fleet, Patch, and Vulnerability Management SHALL maintain complete asset inventory, configuration posture, vulnerability state, patch eligibility, exceptions, and evidence across connected and disconnected fleets.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-69-002	Every production instance SHALL declare and continuously reconcile asset identity, hardware and software bill of materials, posture, vulnerability, exploitability, patch, cohort, exception, and retirement through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-69-003	The implementation SHALL enforce this boundary: severity alone does not determine action; exploitability, exposure, consequence, compatibility, compensating controls, and recovery are evaluated together.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-69-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Fleet, Patch, and Vulnerability Management.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-69-005	When a critical exposure emerges, inventory is incomplete, patching fails, or support expires, the platform SHALL contain reachable risk, revoke or isolate affected assets, deploy the certified remedy, and track exceptions to closure; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-69-006	Production admission and continued operation SHALL require inventory reconciliation, vulnerability and exploit simulation, emergency patching, rollback, offline fleet update, and retirement proofs, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A critical exposure emerges, inventory is incomplete, patching fails, or support expires	Contain reachable risk, revoke or isolate affected assets, deploy the certified remedy, and track exceptions to closure.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared asset identity, hardware and software bill of materials, posture, vulnerability, exploitability, patch, cohort, exception, and retirement.
- Positive and negative boundary tests prove that severity alone does not determine action; exploitability, exposure, consequence, compatibility, compensating controls, and recovery are evaluated together.
- Fault exercises inject a critical exposure emerges, inventory is incomplete, patching fails, or support expires and verify that the platform can contain reachable risk, revoke or isolate affected assets, deploy the certified remedy, and track exceptions to closure.
- Independent review accepts inventory reconciliation, vulnerability and exploit simulation, emergency patching, rollback, offline fleet update, and retirement proofs for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Fleet, Patch, and Vulnerability Management is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 53, Ch. 55, Ch. 56, Ch. 63, Ch. 69. Primary Volume 5 AI anchors: Ch. 15, Ch. 18, Ch. 62, Ch. 67, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-007 · C-039 · C-043 · C-048 · C-050

70 Cost, FinOps, Sustainability, and Unit Economics

Cost, FinOps, Sustainability, and Unit Economics establishes the infrastructure necessary to make infrastructure cost, capacity, energy, carbon, license, provider, and operational efficiency visible by tenant, capability, product, and deployment profile. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

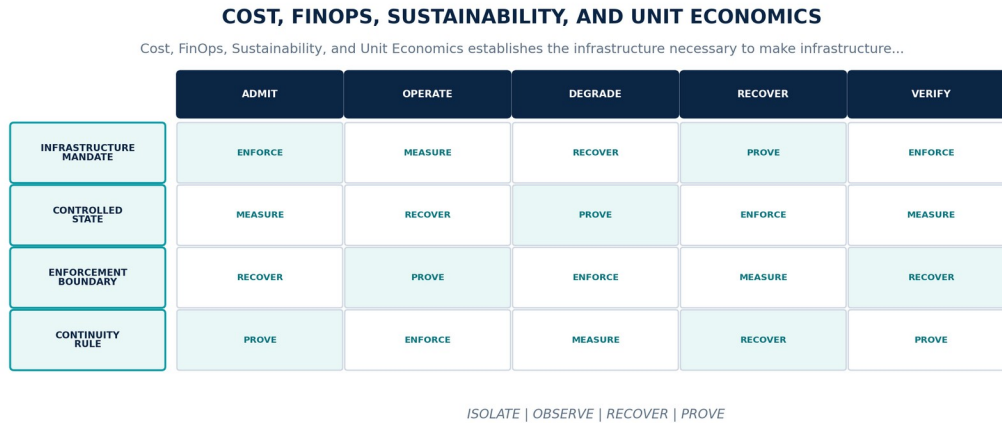


Figure 71. Cost, FinOps, Sustainability, and Unit Economics. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Make infrastructure cost, capacity, energy, carbon, license, provider, and operational efficiency visible by tenant, capability, product, and deployment profile.
Controlled state	Cost and resource ledger, allocation key, rate, reservation, budget, forecast, anomaly, energy, carbon, and optimization decision.
Enforcement boundary	Economic optimization may not weaken isolation, resilience, sovereignty, quality, evidence, accessibility, or human authority.
Continuity rule	When cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a control or SLO, the platform shall stop the unsafe optimization, expose trade-offs, apply approved demand controls, and route material variance to accountable owners.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-70-001	The infrastructure realization of Cost, FinOps, Sustainability, and Unit Economics SHALL make infrastructure cost, capacity, energy, carbon, license, provider, and operational efficiency visible by tenant, capability, product, and deployment profile.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.
IA-70-002	Every production instance SHALL declare and continuously reconcile cost and resource ledger, allocation key, rate, reservation, budget, forecast, anomaly, energy, carbon, and optimization decision through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-70-003	The implementation SHALL enforce this boundary: economic optimization may not weaken isolation, resilience, sovereignty, quality, evidence, accessibility, or human authority.	Security / sovereignty test	Signed control result bound to identity, tenant, data class, topology, deployment, policy, artifacts, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-70-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Cost, FinOps, Sustainability, and Unit Economics.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-70-005	When cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a control or SLO, the platform SHALL stop the unsafe optimization, expose trade-offs, apply approved demand controls, and route material variance to accountable owners; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-70-006	Production admission and continued operation SHALL require allocation reconciliation, unit-cost benchmarks, budget anomaly tests, capacity and energy models, and control-preserving optimization review, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a control or SLO	Stop the unsafe optimization, expose trade-offs, apply approved demand controls, and route material variance to accountable owners.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared cost and resource ledger, allocation key, rate, reservation, budget, forecast, anomaly, energy, carbon, and optimization decision.
- Positive and negative boundary tests prove that economic optimization may not weaken isolation, resilience, sovereignty, quality, evidence, accessibility, or human authority.
- Fault exercises inject cost or energy exceeds budget, allocation is unreliable, or an optimization would breach a control or SLO and verify that the platform can stop the unsafe optimization, expose trade-offs, apply approved demand controls, and route material variance to accountable owners.
- Independent review accepts allocation reconciliation, unit-cost benchmarks, budget anomaly tests, capacity and energy models, and control-preserving optimization review for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Cost, FinOps, Sustainability, and Unit Economics is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 9, Ch. 14, Ch. 58, Ch. 59, Ch. 71, Ch. 72. Primary Volume 5 AI anchors: Ch. 22, Ch. 68, Ch. 70, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-006 · C-035 · C-038 · C-045 · C-050

71 Deployment Acceptance and Operational Readiness

Deployment Acceptance and Operational Readiness establishes the infrastructure necessary to prove that a specific customer deployment can operate, fail, recover, secure, explain, support, upgrade, and retire the contracted product scope. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.

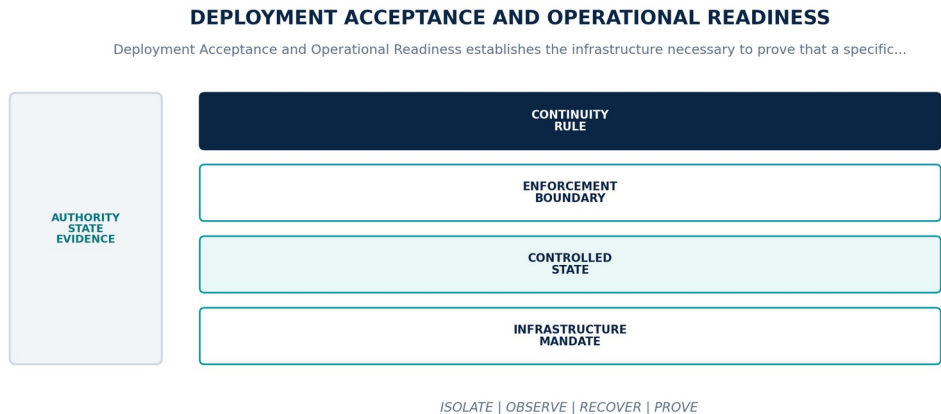


Figure 72. Deployment Acceptance and Operational Readiness. The diagram connects infrastructure mandate, controlled state, enforcement boundary, continuity rule as one controlled infrastructure contract across admission, operation, degradation, recovery, and assurance.

Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Prove that a specific customer deployment can operate, fail, recover, secure, explain, support, upgrade, and retire the contracted product scope.
Controlled state	Acceptance scope, deployment profile, topology, release, tests, open conditions, operators, customer controls, evidence, and expiry.
Enforcement boundary	Acceptance is scoped to exact artifacts, environment, integrations, data classes, workloads, consequence levels, and operating responsibilities.
Continuity rule	When a mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required procedures, the platform shall withhold production authority or restrict the accepted scope, record conditions, and retest after verified remediation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-71-001	The infrastructure realization of Deployment Acceptance and Operational Readiness SHALL prove that a specific customer deployment can operate, fail, recover, secure, explain, support, upgrade, and retire the contracted product scope.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-71-002	Every production instance SHALL declare and continuously reconcile acceptance scope, deployment profile, topology, release, tests, open conditions, operators, customer controls, evidence, and expiry through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-71-003	The implementation SHALL enforce this boundary: acceptance is scoped to exact artifacts, environment, integrations, data classes, workloads, consequence levels, and operating responsibilities.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-71-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Deployment Acceptance and Operational Readiness.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-71-005	When a mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required procedures, the platform SHALL withhold production authority or restrict the accepted scope, record conditions, and retest after verified remediation; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-71-006	Production admission and continued operation SHALL require site and workload qualification, resilience and recovery, security, parity, operational drills, documentation, and signed joint acceptance, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
A mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required procedures	Withhold production authority or restrict the accepted scope, record conditions, and retest after verified remediation.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared acceptance scope, deployment profile, topology, release, tests, open conditions, operators, customer controls, evidence, and expiry.
- Positive and negative boundary tests prove that acceptance is scoped to exact artifacts, environment, integrations, data classes, workloads, consequence levels, and operating responsibilities.
- Fault exercises inject a mandatory test fails, evidence is stale, an open condition exceeds tolerance, or operators cannot execute required procedures and verify that the platform can withhold production authority or restrict the accepted scope, record conditions, and retest after verified remediation.
- Independent review accepts site and workload qualification, resilience and recovery, security, parity, operational drills, documentation, and signed joint acceptance for each supported deployment profile.

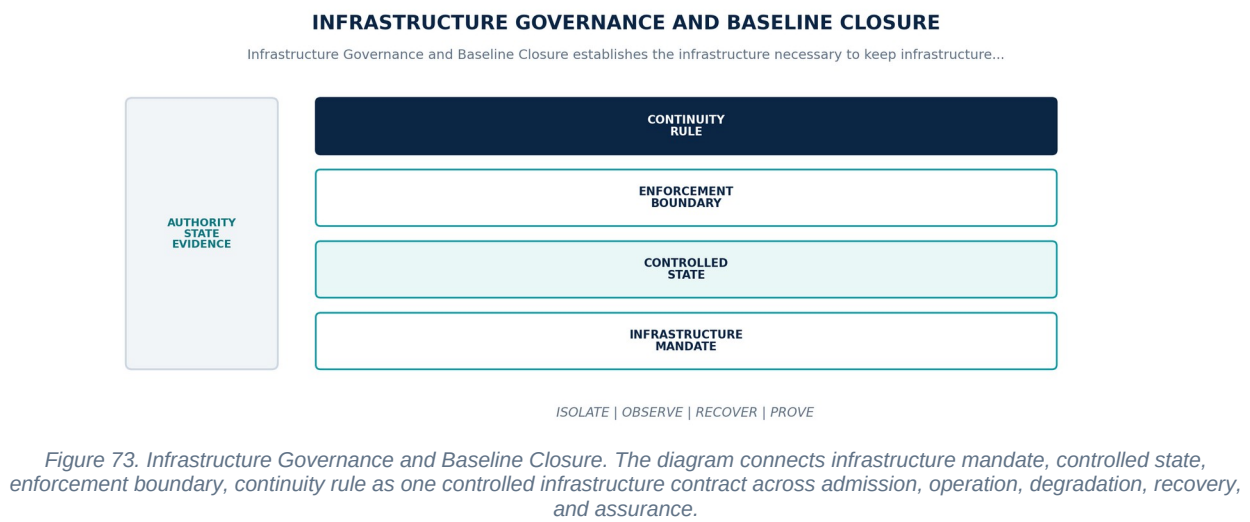
CHAPTER ACCEPTANCE AND INHERITANCE

Deployment Acceptance and Operational Readiness is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 57, Ch. 61, Ch. 63, Ch. 71, Ch. 72. Primary Volume 5 AI anchors: Ch. 66, Ch. 67, Ch. 68, Ch. 70, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-003 · C-006 · C-008 · C-049 · C-051

72 Infrastructure Governance and Baseline Closure

Infrastructure Governance and Baseline Closure establishes the infrastructure necessary to keep infrastructure architecture coherent through ownership, exceptions, architecture decisions, lifecycle evidence, metrics, reviews, and controlled amendment. It fixes ownership, state, boundaries, degraded behavior, and assurance while preserving one Strategic Intelligence Operating System across SaaS, Private Cloud, and On-Premise.



Infrastructure architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Infrastructure mandate	Keep infrastructure architecture coherent through ownership, exceptions, architecture decisions, lifecycle evidence, metrics, reviews, and controlled amendment.
Controlled state	Architecture decision, standard, exception, risk acceptance, review cadence, conformance state, amendment, and closure evidence.
Enforcement boundary	Technology and provider choices may evolve behind frozen contracts; local implementation cannot redefine the Strategic Intelligence Operating System.
Continuity rule	When material divergence, unmanaged debt, repeated exception, obsolete control, or unresolved responsibility threatens the baseline, the platform shall constrain affected change, assign accountable remediation, elevate semantic conflicts, and amend only through the authority chain.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-72-001	The infrastructure realization of Infrastructure Governance and Baseline Closure SHALL keep infrastructure architecture coherent through ownership, exceptions, architecture decisions, lifecycle evidence, metrics, reviews, and controlled amendment.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.
IA-72-002	Every production instance SHALL declare and continuously reconcile architecture decision, standard, exception, risk acceptance, review cadence, conformance state, amendment, and closure evidence through versioned desired state, observed state, ownership, and audit evidence.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-72-003	The implementation SHALL enforce this boundary: technology and provider choices may evolve behind frozen contracts; local implementation cannot redefine the Strategic Intelligence Operating System.	Automated conformance test	Versioned fixture and signed result for exact manifests, artifacts, configuration, topology, workload, and environment.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
IA-72-004	SaaS, Private Cloud, and On-Premise SHALL preserve the same authority, state, isolation, failure, recovery, evidence, replay, and human-decision semantics for Infrastructure Governance and Baseline Closure.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-72-005	When material divergence, unmanaged debt, repeated exception, obsolete control, or unresolved responsibility threatens the baseline, the platform SHALL constrain affected change, assign accountable remediation, elevate semantic conflicts, and amend only through the authority chain; it SHALL NOT infer healthy, complete, current, authorized, or recovered state.	Fault / recovery exercise	Fault trace, blast radius, state boundary, degraded mode, recovery, reconciliation, user impact, and accountable closure.
IA-72-006	Production admission and continued operation SHALL require architecture review, exception aging, conformance trends, decision replay, retirement evidence, and executive baseline attestation, bound to the exact artifacts, configuration, topology, customer scope, deployment profile, and period under review.	Load / service-level test	Representative workload, demand model, distributions, thresholds, bottlenecks, headroom, and admitted service decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Material divergence, unmanaged debt, repeated exception, obsolete control, or unresolved responsibility threatens the baseline	Constrain affected change, assign accountable remediation, elevate semantic conflicts, and amend only through the authority chain.
Control evidence is absent, stale, or bound to another release	Deny promotion or enter the declared restricted state; never infer conformance from successful process execution.
Deployment-profile behavior diverges from the common contract	Suspend the affected capability in that profile until shared fixtures pass or a narrow non-semantic exception closes.

Assurance evidence

- A versioned manifest and runtime inventory prove the declared architecture decision, standard, exception, risk acceptance, review cadence, conformance state, amendment, and closure evidence.
- Positive and negative boundary tests prove that technology and provider choices may evolve behind frozen contracts; local implementation cannot redefine the Strategic Intelligence Operating System.
- Fault exercises inject material divergence, unmanaged debt, repeated exception, obsolete control, or unresolved responsibility threatens the baseline and verify that the platform can constrain affected change, assign accountable remediation, elevate semantic conflicts, and amend only through the authority chain.
- Independent review accepts architecture review, exception aging, conformance trends, decision replay, retirement evidence, and executive baseline attestation for each supported deployment profile.

CHAPTER ACCEPTANCE AND INHERITANCE

Infrastructure Governance and Baseline Closure is conformant only when every IA requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 1, Ch. 2, Ch. 6, Ch. 64, Ch. 72. Primary Volume 5 AI anchors: Ch. 1, Ch. 2, Ch. 6, Ch. 70, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-004 · C-049 · C-051 · C-052

PART APP

Infrastructure Architecture Reference Appendices

Normative component, deployment, runtime, network, state, messaging, compute, manifest, service-level, failure, security, recovery, traceability, decision, and terminology catalogs.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

A Infrastructure Component Catalog

The 92 stable infrastructure component identities realize deployment, compute, network, state, platform, trust, reliability, delivery, and governance responsibilities. Technology products may implement these identities but do not replace their contracts.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
INF-DP-01	Deployment	Deployment profile registry	Registry	Management plane	Resolve certified SaaS, Private Cloud, On-Premise, disconnected, and air-gapped profiles.
INF-DP-02	Deployment	Environment and site registry	Registry	Management plane	Own regions, zones, sites, environments, and installed instance identity.
INF-DP-03	Deployment	Intelligence cell controller	Control	Cell	Reconcile cell composition, placement, isolation, and lifecycle.
INF-DP-04	Deployment	Tenant placement service	Control	Cell	Apply locality, capacity, sovereignty, and isolation constraints to tenant placement.
INF-DP-05	Deployment	Topology controller	Control	Region / site	Reconcile regional, zonal, site, and dependency topology.
INF-DP-06	Deployment	Disconnected operations controller	Control	Site	Coordinate offline continuity, imports, exports, and later reconciliation.
INF-DP-07	Deployment	Management-plane gateway	Control	Management plane	Mediate signed fleet commands and bounded support operations.
INF-DP-08	Deployment	Customer control bridge	Control	Customer boundary	Expose customer-held controls, approvals, evidence, and responsibility state.
INF-DP-09	Deployment	Fleet inventory service	Registry	Management plane	Track physical and virtual assets, versions, posture, and ownership.
INF-DP-10	Deployment	Deployment conformance service	Evidence	Release cohort	Run common semantic and infrastructure conformance suites.
INF-CP-01	Compute	General compute pool	Runtime	Node pool	Host portable service, policy, workflow, and analytical workloads.
INF-CP-02	Compute	Memory-optimized compute pool	Runtime	Node pool	Host graph, cache, index, and memory-intensive workloads.
INF-CP-03	Compute	Compute-intensive pool	Runtime	Node pool	Host simulation, transformation, and deterministic analytical workloads.
INF-CP-04	Compute	Accelerator pool	Runtime	Accelerator island	Host approved inference, simulation, and AI processing workloads.
INF-CP-05	Compute	Accelerator partition manager	Control	Device	Enforce device partitions, tenancy, topology, and memory isolation.
INF-CP-06	Compute	Model serving runtime	Runtime	Serving cell	Execute immutable model and adapter artifacts behind task contracts.
INF-CP-07	Compute	Agent worker pool	Runtime	Worker cell	Execute identity-bound agent tasks inside controlled sandboxes.
INF-CP-08	Compute	Batch worker pool	Runtime	Processing cell	Execute bounded data, graph, evaluation, and index workloads.
INF-CP-09	Compute	Stream processor pool	Runtime	Processing cell	Execute continuous event and signal processing with checkpoints.
INF-CP-10	Compute	Edge runtime	Runtime	Remote site	Collect, buffer, filter, and transfer evidence under constrained connectivity.
INF-CP-11	Compute	Scheduler and admission controller	Control	Cluster / cell	Apply fit, quota, priority, fairness, and disruption policy.
INF-CP-12	Compute	Resource metering service	Ledger	Management plane	Record resource, cost, energy, quota, and budget consumption.
INF-NW-01	Network	External access edge	Runtime	Region / site	Terminate authenticated user, API, and system access.
INF-NW-02	Network	API gateway	Runtime	Region / site	Enforce API contracts, identity context, rate, and request evidence.
INF-NW-03	Network	Service connectivity fabric	Runtime	Cluster / cell	Provide mutually authenticated, policy-governed service connectivity.
INF-NW-04	Network	Network policy controller	Control	Trust domain	Reconcile declared reachability, segmentation, and deny-by-default state.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
INF-NW-05	Network	Egress gateway	Runtime	Region / site	Mediate destination, purpose, data class, and transfer controls.
INF-NW-06	Network	Connector network enclave	Runtime	Connector cell	Isolate external source and enterprise-system connectivity.
INF-NW-07	Network	Private connectivity gateway	Runtime	Customer boundary	Terminate customer tunnels, private links, and controlled routes.
INF-NW-08	Network	DNS authority	Stateful service	Region / site	Serve authoritative internal and deployment-local naming.
INF-NW-09	Network	Service discovery service	Stateful service	Cluster / cell	Resolve healthy endpoints and version-aware service identity.
INF-NW-10	Network	Trusted time service	Stateful service	Region / site	Provide attested time and clock-quality evidence.
INF-NW-11	Network	Network telemetry collector	Telemetry	Region / site	Measure flow, path, latency, loss, reachability, and change.
INF-NW-12	Network	DDoS and protocol protection	Protection	Access edge	Contain volumetric, protocol, and abuse conditions at ingress.
INF-ST-01	Storage and State	Transactional state platform	Authoritative state	Database cell	Host consistency-sensitive operational state by owning service.
INF-ST-02	Storage and State	Evidence object store	Authoritative evidence	Storage cell	Retain versioned source, media, document, and evidence objects.
INF-ST-03	Storage and State	Immutable audit store	Protected evidence	Audit domain	Retain append-only audit evidence and integrity anchors.
INF-ST-04	Storage and State	Knowledge graph store	Canonical state	Graph cell	Persist temporal graph, ontology, identity, evidence, and correction state.
INF-ST-05	Storage and State	Graph projection and index store	Derived state	Graph cell	Serve rebuildable traversal and analytical projections.
INF-ST-06	Storage and State	Search index platform	Derived state	Search cell	Serve lexical, structured, temporal, and geospatial indexes.
INF-ST-07	Storage and State	Vector index platform	Derived state	Search cell	Serve versioned embedding and multimodal retrieval projections.
INF-ST-08	Storage and State	Digital twin state store	Canonical state	Twin cell	Persist definitions, snapshots, branches, and intervention state.
INF-ST-09	Storage and State	Simulation result store	Product state	Simulation cell	Persist run inputs, checkpoints, outputs, and comparisons.
INF-ST-10	Storage and State	Event streaming platform	Durable log	Messaging cell	Deliver ordered partitioned events with replay.
INF-ST-11	Storage and State	Durable queue platform	Durable work state	Messaging cell	Deliver work, retry, delay, and dead-letter state.
INF-ST-12	Storage and State	Distributed cache platform	Derived temporary state	Cell	Accelerate reads under explicit source revision and invalidation.
INF-ST-13	Storage and State	Checkpoint store	Recoverable execution state	Workflow cell	Persist agent, workflow, stream, and simulation recovery points.
INF-ST-14	Storage and State	Backup and recovery vault	Recovery state	Recovery domain	Retain immutable encrypted recovery sets and restore evidence.
INF-PF-01	Platform Runtime	Container orchestration control	Control	Cluster	Reconcile workloads, placement, identity, health, and disruption.
INF-PF-02	Platform Runtime	Runtime admission service	Control	Cluster / cell	Verify identity, artifact, policy, resource, and placement before execution.
INF-PF-03	Platform Runtime	Internal developer platform	Control	Management plane	Provide governed service, state, delivery, and operations patterns.
INF-PF-04	Platform Runtime	Service catalog	Registry	Management plane	Own service, owner, contract, dependency, SLO, and support registration.
INF-PF-05	Platform Runtime	Workflow engine	Durable runtime	Workflow cell	Execute versioned workflows, timers, tasks, gates, and compensation.
INF-PF-06	Platform Runtime	Agent sandbox controller	Control	Agent cell	Create and destroy capability-bounded agent execution environments.
INF-PF-07	Platform Runtime	Tool capability bridge	Control	Agent cell	Enforce typed tool calls, side-effect classes, receipts, and revocation.
INF-PF-08	Platform Runtime	Model gateway	Runtime	Inference cell	Resolve approved model routes, policies, budgets, and provider adapters.
INF-PF-09	Platform Runtime	Retrieval gateway	Runtime	Retrieval cell	Mediate hybrid retrieval and context-manifest construction.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
INF-PF-10	Platform Runtime	Pipeline orchestrator	Durable runtime	Processing cell	Execute governed data, graph, model, index, and evaluation pipelines.
INF-PF-11	Platform Runtime	Experience delivery platform	Runtime	Experience cell	Serve executive surfaces, briefings, approvals, and collaboration.
INF-PF-12	Platform Runtime	Connector runtime	Runtime	Connector cell	Execute isolated source and enterprise connectors.
INF-TR-01	Trust and Security	Human identity federation	Trust state	Trust domain	Federate and authenticate human principals and assurance context.
INF-TR-02	Trust and Security	Workload identity issuer	Trust state	Trust domain	Issue short-lived attested service and workload identities.
INF-TR-03	Trust and Security	Privileged access manager	Control	Administrative domain	Approve, broker, record, expire, and revoke privilege.
INF-TR-04	Trust and Security	Policy decision service	Control	Policy cell	Evaluate versioned authorization, purpose, data, locality, and effect policy.
INF-TR-05	Trust and Security	Policy enforcement sidecar	Control	Workload boundary	Enforce decisions and obligations at workload boundaries.
INF-TR-06	Trust and Security	Key management service	Trust state	Key domain	Generate, wrap, rotate, revoke, and audit cryptographic keys.
INF-TR-07	Trust and Security	Hardware security module	Trust anchor	Key domain	Protect root keys, signing operations, and recovery ceremonies.
INF-TR-08	Trust and Security	Secret issuance service	Trust state	Trust domain	Issue identity-bound short-lived secrets and certificates.
INF-TR-09	Trust and Security	Artifact trust service	Trust state	Supply-chain domain	Verify provenance, signatures, bills of materials, and revocation.
INF-TR-10	Trust and Security	Confidential execution attestor	Trust state	Compute domain	Verify protected execution identity and measurement.
INF-TR-11	Trust and Security	Security detection platform	Security evidence	Security domain	Correlate threats, misuse, control failures, and response state.
INF-TR-12	Trust and Security	Sovereignty policy controller	Control	Jurisdiction domain	Enforce locality, administrator, provider, key, and transfer boundaries.
INF-OP-01	Reliability and Operations	Telemetry collection fabric	Telemetry	Region / site	Collect and route metrics, logs, traces, events, and profiles.
INF-OP-02	Reliability and Operations	Observability store	Operational evidence	Operations domain	Retain correlated telemetry under tenant and content controls.
INF-OP-03	Reliability and Operations	Service-level management service	Control	Operations domain	Compute indicators, objectives, budgets, burn, and action state.
INF-OP-04	Reliability and Operations	Capacity planning service	Planning state	Operations domain	Model demand, headroom, bottlenecks, and scale actions.
INF-OP-05	Reliability and Operations	Availability controller	Control	Service domain	Coordinate health, fencing, routing, failover, and reconciliation.
INF-OP-06	Reliability and Operations	Backup controller	Control	Recovery domain	Schedule, verify, retain, and expire recovery sets.
INF-OP-07	Reliability and Operations	Restore orchestrator	Durable runtime	Recovery domain	Restore state in dependency order and reconcile consumers.
INF-OP-08	Reliability and Operations	Disaster recovery controller	Control	Recovery site	Coordinate regional or site recovery and failback.
INF-OP-09	Reliability and Operations	Incident command platform	Case state	Operations domain	Manage command roles, timeline, actions, evidence, and communication.
INF-OP-10	Reliability and Operations	Change orchestration service	Durable runtime	Management plane	Execute approved changes, cohorts, observations, and rollback.
INF-OP-11	Reliability and Operations	Degraded-mode controller	Control	Service domain	Declare, enforce, communicate, and exit strategic degraded modes.
INF-OP-12	Reliability and Operations	Operational evidence vault	Protected evidence	Operations domain	Retain acceptance, exercise, incident, change, and readiness evidence.
INF-DL-01	Delivery and Governance	Infrastructure source registry	Registry	Delivery domain	Own infrastructure modules, configuration schemas, and source identity.
INF-DL-02	Delivery and Governance	Plan and policy engine	Control	Delivery domain	Produce deterministic plans and policy decisions before mutation.
INF-DL-03	Delivery and Governance	Build and provenance service	Evidence	Build domain	Build reproducible artifacts and attest source and dependencies.
INF-DL-04	Delivery and Governance	Release registry	Registry	Delivery domain	Own dependency-closed infrastructure release identity and lifecycle.

ID	DOMAIN	COMPONENT	PRIMARY STATE	FAILURE DOMAIN	RESPONSIBILITY
INF-DL-05	Delivery and Governance	Promotion controller	Durable runtime	Delivery domain	Promote cohorts through evidence gates and rollback boundaries.
INF-DL-06	Delivery and Governance	Vulnerability and patch service	Control	Fleet domain	Resolve exposure, patch eligibility, exceptions, and closure.
INF-DL-07	Delivery and Governance	Cost and sustainability ledger	Ledger	Management plane	Allocate cost, energy, carbon, license, and efficiency measures.
INF-DL-08	Delivery and Governance	Infrastructure governance registry	Registry	Governance domain	Own architecture decisions, exceptions, conformance, review, and amendment.

Component conformance rules

RULE	BINDING APPLICATION
Identity precedes product	A vendor service, appliance, managed offering, open-source package, or internal implementation becomes conformant only through a stable INF component identity and approved manifest.
Ownership remains singular	Each component has one accountable infrastructure owner even where operation, hosting, support, key control, or incident duties are shared with a customer or provider.
State is explicit	Authoritative, derived, temporary, checkpoint, evidence, telemetry, registry, control, and recovery state are declared separately and cannot be inferred from the selected technology.
Failure is bounded	Every component identifies its containing failure domain, maximum blast radius, degraded behavior, recovery order, reconciliation obligations, and terminal escalation.
Evidence follows release	Conformance, security, capacity, service-level, resilience, recovery, deployment-parity, and retirement evidence resolves to the exact component and released configuration.

CATALOG AUTHORITY

The component catalog freezes infrastructure responsibilities, not products. Technology may be substituted when the replacement passes the same state, boundary, isolation, failure, recovery, evidence, and deployment-parity contracts.

B Deployment Profile Matrix

The deployment profiles move physical responsibility and locality while preserving product semantics, human authority, canonical state, evidence, failure, correction, replay, and lifecycle behavior.

CAPABILITY	SAAS	PRIVATE CLOUD	ON-PREMISE	PARITY EVIDENCE
Product semantics	Common canonical contracts	Common canonical contracts	Common canonical contracts	Identical conformance fixtures
Primary control boundary	The Eye operated boundary	Customer cloud boundary	Customer site boundary	Attested topology and responsibility
Tenancy	Cell-isolated multi-tenant	Single or bounded multi-tenant	Customer-selected domain model	Tenant and domain isolation tests
Control plane	Managed regional service	Mediated customer-cloud control	Local or mediated site control	Plane separation and outage tests
Data plane	Regional intelligence cells	Customer-cloud intelligence cells	Local intelligence cells	Canonical object and workflow fixtures
Identity	Federated plus workload trust	Customer federation plus workload trust	Local federation plus workload trust	Authentication and delegation tests
Keys	Platform or customer-held	Customer or delegated key authority	Local customer key authority	Rotation and recovery ceremony
Network ingress	Managed access edge	Customer landing-zone edge	Customer site edge	Protocol and policy conformance
Egress	Registered mediated destinations	Customer-controlled routes	Local allowlists or no egress	Destination and data-policy tests
External models	Approved provider or local	Approved private route or local	Local unless explicitly connected	Task and privacy parity tests
Object storage	Managed regional service	Customer-cloud certified service	Certified local service	Integrity and restore fixtures
Graph and indexes	Managed graph and search cells	Customer-cloud graph and search cells	Local graph and search cells	Temporal and policy query fixtures
Messaging	Managed regional messaging	Certified cloud messaging	Certified local messaging	Delivery and replay fixtures
Observability	Managed with tenant controls	Customer-integrated or local	Local with optional export	Causal trace and privacy tests
Security operations	The Eye-led shared model	Joint or customer-led model	Customer-led with support option	Incident and evidence exercises
Updates	Continuous cohorts	Controlled customer windows	Signed online or offline packages	Promotion and rollback tests
Disconnected operation	Regional continuity	Optional restricted mode	Supported profile	Zero-dependency and reconciliation tests
Air-gap	Not applicable	Special isolated profile	Certified isolated profile	Zero-egress acceptance
Backup	Managed recovery domains	Customer locality and keys	Local customer-controlled vault	Automated restore evidence
Disaster recovery	Cross-region within policy	Customer-selected recovery region	Customer recovery site or isolated restore	RTO, RPO, sovereignty, and reconciliation
Support access	Recorded privileged access	Customer-approved mediated access	Customer-approved local or offline support	Privilege ceremony and revocation
Sovereignty	Regional placement and provider policy	Customer cloud and administrator policy	Local jurisdiction and administrator policy	End-to-end locality proof
Customer export	Canonical export service	Canonical export service	Canonical local export service	Integrity and re-import fixtures
Retirement	Managed tenant and artifact closure	Joint environment closure	Local removal and evidence handoff	Deletion, export, and historical replay evidence

C Runtime Unit Catalog

Runtime unit classes define the repeatable execution patterns from which deployable infrastructure is composed. Every workload manifest selects one class and refines its state, scaling, disruption, failure, and recovery contract.

ID	RUNTIME UNIT	SCALE KEY	STATE / REPLICATION	FAILURE AND RECOVERY
RU-01	Stateless service	Request or event	Replicated; disposable	Drain and reschedule
RU-02	Stateful service	Partition or shard	Quorum or owner-aware	Fence, fail over, reconcile
RU-03	Control-plane controller	Desired-state object	Leader or quorum	Freeze change; restore control
RU-04	Gateway	Route and policy key	Horizontally replicated	Fail closed or bounded failover
RU-05	Durable workflow worker	Task lease	Disposable with checkpoint	Lease expiry and reassignment
RU-06	Human task worker	Gate and case	Durable task state	Escalate and preserve deadline
RU-07	Agent sandbox	Execution identity	Ephemeral isolated runtime	Terminate, revoke, restore checkpoint
RU-08	Model server	Model revision and endpoint	Replicated or partitioned	Drain, revoke, approved fallback
RU-09	Batch worker	Job and partition	Checkpointed	Retry from committed boundary
RU-10	Stream processor	Topic partition	Checkpointed offsets	Replay and reconcile
RU-11	Connector worker	Source and cursor	Isolated, lease-based	Suspend and preserve custody
RU-12	Edge collector	Device and stream	Local buffered	Store-and-forward
RU-13	Search shard	Index shard	Replicated derived state	Rebuild or policy-equivalent fallback
RU-14	Graph partition	Entity or domain partition	Replicated temporal state	Constrain traversals and replay
RU-15	Queue broker	Partition	Durable replicated log	Elect, replay, apply backpressure
RU-16	Cache node	Key partition	Disposable derived state	Bypass and rebuild
RU-17	Telemetry collector	Source and signal class	Buffered	Prioritize and forward later
RU-18	Security sensor	Asset and signal class	Locally buffered	Enter security-degraded mode
RU-19	Policy decision point	Policy and tenant	Replicated signed bundles	Safe cached decision or deny
RU-20	Identity issuer	Trust domain	Quorum and hardware roots	Stop issuance; preserve existing bounded leases
RU-21	Key service	Key domain	Quorum and hardware roots	Continuity keys or stop protected operations
RU-22	Build worker	Build identity	Ephemeral isolated runtime	Discard and rerun reproducibly
RU-23	Release controller	Release and cohort	Durable workflow	Stop cohort and rollback
RU-24	Backup worker	State class and recovery set	Idempotent durable task	Retry and verify integrity
RU-25	Restore worker	Recovery unit	Dependency-ordered task	Stop, preserve point, reconcile
RU-26	Executive experience node	Session and tenant	Replicated with durable records	Alternate channel and last-known state

D Network Zone and Trust Boundary Catalog

Network zones express reachability, not identity. Every permitted flow remains authenticated, purpose-bound, policy-authorized, tenant-scoped, observable, and revocable.

ID	ZONE	TRUST POSTURE	PERMITTED REACHABILITY	REQUIRED ENFORCEMENT
NZ-01	Public access edge	Untrusted external	Authenticated ingress only	Gateway and protection policy
NZ-02	Customer private edge	Customer network	Registered private services	Mutual trust and route policy
NZ-03	External source enclave	Approved sources	Connector runtime only	Destination, content, and custody controls
NZ-04	External model enclave	Approved providers	Model gateway only	Data, task, provider, and egress policy
NZ-05	User experience zone	Authenticated users	Experience APIs	Identity, session, tenant, and rate controls
NZ-06	API service zone	Registered clients	Canonical service APIs	Workload identity and object policy
NZ-07	Intelligence cell zone	Cell workloads	Cell-local services and mediated shared services	Tenant and domain policy
NZ-08	Agent sandbox zone	Untrusted execution	Capability bridge only	Deny-by-default process, file, network, and secret policy
NZ-09	Connector sandbox zone	External protocol code	Source gateway and ingestion boundary	Isolation, egress, and content inspection
NZ-10	Model serving zone	Approved inference workloads	Model gateway and local state	Artifact, context, and accelerator isolation
NZ-11	Canonical state zone	Owning services	Typed state interfaces	No direct cross-service storage access
NZ-12	Evidence storage zone	Custody-controlled services	Object and evidence APIs	Classification, retention, and integrity
NZ-13	Messaging zone	Registered producers and consumers	Topics and queues	Schema, tenant, delivery, and rate policy
NZ-14	Policy and identity zone	Trusted control services	Authenticated decision and issuance APIs	Hardware roots and short-lived authority
NZ-15	Key management zone	Key services and hardware roots	Cryptographic operations only	Dual control and purpose separation
NZ-16	Management plane zone	Fleet and platform operators	Signed control commands and minimal telemetry	Privileged mediation and plane separation
NZ-17	Security operations zone	Security operators and sensors	Security telemetry and response	Evidence custody and privileged controls
NZ-18	Build and release zone	Controlled delivery workloads	Artifact registries and promotion APIs	Supply-chain isolation and signatures
NZ-19	Backup and recovery zone	Recovery services	State-class backup and restore paths	Immutable copies and separate authority
NZ-20	Air-gap transfer zone	Approved transfer stations	Signed import and export packages	Two-person custody and content inspection

E Storage and State Class Catalog

State classes fix authority, consistency, durability, derivation, locality, protection, lifecycle, and recovery before storage technology is selected.

ID	STATE CLASS	OWNER	CONSISTENCY / AUTHORITY	LIFECYCLE AND RECOVERY
SC-01	Canonical strategic state	Owning canonical layer	Strong or contract-defined	Versioned, corrected, never silently overwritten
SC-02	Operational authoritative state	Owning service	Transactional	Recoverable to declared point
SC-03	Temporal graph state	Knowledge Graph	Temporal and revision-aware	Replayable changes and deterministic projection rebuild
SC-04	Enterprise Memory state	Enterprise Memory	Versioned temporal	Correction and retention aware
SC-05	Digital twin state	Digital Twins	Snapshot and branch aware	Branch isolation and checkpoint restore
SC-06	Decision record state	Decision Intelligence	Immutable versioned	Preserve authority, rationale, evidence, and outcome links
SC-07	Evidence object	Producing or custody component	Immutable versioned bytes	Digest, custody, retention, and legal hold
SC-08	Audit evidence	Trust and provenance control	Append-only	Integrity anchored and separately protected
SC-09	Workflow state	Workflow engine	Durable transition log	Resume, compensate, or terminally escalate
SC-10	Agent execution state	Agent runtime	Checkpointed envelope	Resume only under compatible release and authority
SC-11	Messaging log	Messaging platform	Partition-ordered durable log	Replay within retention
SC-12	Durable work queue	Queue platform	At-least-once delivery	Retry, dead-letter, and effect reconciliation
SC-13	Search projection	Search platform	Eventually consistent derived state	Rebuild from authoritative revisions
SC-14	Vector projection	Vector platform	Versioned derived state	Re-embed and rebuild with compatibility tracking
SC-15	Graph projection	Graph platform	Revision-pinned derived state	Replay and deterministic rebuild
SC-16	Cache	Consuming service	Bounded stale derived state	Expire, invalidate, bypass, and rebuild
SC-17	Session state	Experience service	Short-lived tenant-scoped	Expire without losing decision records
SC-18	Checkpoint	Producing runtime	Immutable recovery point	Compatibility checked before resume
SC-19	Temporary processing state	Producing job	Ephemeral encrypted	Delete at terminal state or expiry
SC-20	Configuration desired state	Management plane	Versioned declarative	Plan, apply, drift, and rollback evidence
SC-21	Registry state	Registry owner	Transactional versioned	Historical resolution and revocation
SC-22	Telemetry state	Observability owner	Append-oriented sampled	Privacy-controlled and quality-marked
SC-23	Backup set	Recovery owner	Immutable encrypted copy	Automated integrity and restore verification
SC-24	Export package	Owning export service	Immutable signed bundle	Custody, completeness, and re-import evidence

F Messaging and Delivery Semantics Catalog

Messaging semantics are contract properties. Broker acknowledgements do not prove canonical admission, human approval, or exactly-once institutional effects.

ID	MESSAGE CLASS	DELIVERY	ORDERING SCOPE	CONSUMER OBLIGATION
MS-01	Domain event	At least once	Aggregate or partition	Consumer idempotency and replay
MS-02	Integration event	At least once	Source and entity key	Schema registry and consumer compatibility
MS-03	Command	At least once	Command target	Idempotency key and one authoritative effect
MS-04	Work item	At least once with lease	Queue priority	Lease expiry, retry, and terminal escalation
MS-05	Human task	Durable until terminal	Case and gate	Named authority, deadline, and escalation
MS-06	Correction event	Durable at least once	Corrected object	Impact propagation and consumer acknowledgement
MS-07	Revocation event	Durable prioritized	Artifact or authority	Bounded propagation and enforcement evidence
MS-08	Security signal	Durable prioritized	Asset and identity	Evidence custody and response correlation
MS-09	Telemetry signal	Best effort or buffered	Source and time	Quality, sampling, and loss visibility
MS-10	Audit record	Durable append-only	Causation chain	Integrity anchor and non-repudiation
MS-11	Collection observation	At least once	Source and cursor	Custody, deduplication, and gap state
MS-12	Workflow transition	Exactly-once committed state	Workflow instance	Transactional history and replay
MS-13	Model inference request	Bounded request or durable task	Task and execution	No retry across effect boundary without idempotency
MS-14	Agent message	At least once	Workflow and execution	Sequence, sender identity, budget, and checkpoint
MS-15	Notification	At least once with status	Recipient and subscription	Delivered, acknowledged, expired, or failed
MS-16	Early warning	Durable prioritized	Risk or signal	Escalation, acknowledgement, and correction
MS-17	Export transfer	Exactly-once package registration	Export identity	Digest, custody, receipt, and retransfer control
MS-18	Backup completion	At least once	Recovery set	Integrity and restore-verification state
MS-19	Deployment command	Exactly-once desired-state mutation	Release and target	Approval, plan digest, and rollout evidence
MS-20	Incident action	Durable ordered case event	Incident	Actor, authority, reason, effect, and timestamp

G Compute and Accelerator Profiles

Compute profiles define portable workload intent. Exact instance types, processors, devices, and providers resolve behind certified CPU, memory, accelerator, edge, isolation, topology, and evidence requirements.

ID	PROFILE	PRIMARY USE	RESOURCE ENVELOPE	QUALIFICATION
CPU-GEN-S	General CPU small	Control and light services	2-4 vCPU; 4-16 GiB	Portable amd64/arm64
CPU-GEN-M	General CPU medium	APIs and orchestration	4-16 vCPU; 16-64 GiB	Portable amd64/arm64
CPU-GEN-L	General CPU large	High-throughput services	16-64 vCPU; 64-256 GiB	NUMA-aware where required
CPU-MEM-M	Memory optimized medium	Search, graph, caches	8-32 vCPU; 64-256 GiB	Memory bandwidth qualified
CPU-MEM-L	Memory optimized large	Graph and index nodes	32-96 vCPU; 256-1024 GiB	NUMA and page profile qualified
CPU-CMP-M	Compute optimized medium	Batch and analytics	16-64 vCPU; 32-128 GiB	Vector instruction profile declared
CPU-CMP-L	Compute optimized large	Simulation and transformation	64-192 vCPU; 128-512 GiB	Vector and topology qualified
GPU-INF-S	Inference accelerator small	Low-latency model serving	1 device or secure partition	Runtime and model family qualified
GPU-INF-M	Inference accelerator medium	High-throughput inference	2-4 devices	Interconnect and batching qualified
GPU-INF-L	Inference accelerator large	Large-model inference	4-8 devices	Tensor and pipeline topology qualified
GPU-SIM-M	Simulation accelerator	Parallel simulation	1-4 devices	Solver precision and determinism declared
GPU-EVAL-M	Evaluation accelerator	Model and safety evaluation	1-4 devices	Reproducible evaluation image
ACC-CONF	Confidential accelerator	Protected inference	Attested device and runtime	Confidential-compute evidence required
EDGE-TINY	Constrained edge	Sensor collection and buffering	Sub-2 CPU; sub-4 GiB	Offline buffer and signed firmware
EDGE-STD	Standard edge	Collection, filtering, local analytics	2-8 CPU; 4-32 GiB	Remote update and custody controls
EDGE-ACC	Accelerated edge	Local vision or signal inference	CPU plus edge accelerator	Model and device attestation
BUILD-ISO	Isolated build worker	Software and artifact builds	Ephemeral fit-to-build	No production credentials
AGENT-SBX	Agent sandbox worker	Bounded agent execution	Per-task CPU, memory, and time	Deny-by-default capabilities
RESTORE-IO	Recovery I/O worker	Backup verification and restore	High network and storage bandwidth	Separate recovery authority
CTRL-QUORUM	Control service node	Identity, policy, registry, orchestration	Low-latency durable profile	Quorum and dedicated reserve

H Infrastructure Manifest Field Dictionary

The infrastructure manifest defines a stable component as a controlled deployment, state, boundary, dependency, control, failure, recovery, release, and lifecycle contract.

FIELD	BINDING DEFINITION	PRESENCE RULE
infrastructure_component_id	Stable identity from the infrastructure component catalog	Required unless contractually inapplicable
component_version	Immutable component contract version	Required unless contractually inapplicable
owning_domain	Accountable infrastructure domain	Required unless contractually inapplicable
accountable_owner	Named technical and operational owner	Required unless contractually inapplicable
product_layer_bindings	Canonical layers supported without ownership transfer	Required unless contractually inapplicable
engineering_anchors	Controlling Volume 4 chapters and requirements	Required unless contractually inapplicable
ai_workload_anchors	Controlling Volume 5 runtime and AI obligations	Required unless contractually inapplicable
deployment_profiles	Certified SaaS, Private Cloud, On-Premise, disconnected, or air-gap profiles	Required unless contractually inapplicable
tenant_and_domain_model	Isolation and administrative scope	Required unless contractually inapplicable
desired_state_schema	Versioned declarative configuration contract	Required unless contractually inapplicable
observed_state_schema	Runtime inventory and reconciliation contract	Required unless contractually inapplicable
runtime_units	Deployable and stateful runtime identities	Required unless contractually inapplicable
state_classes	Owned, consumed, cached, checkpoint, evidence, and recovery state	Required unless contractually inapplicable
interfaces	APIs, commands, queries, events, and operational interfaces	Required unless contractually inapplicable
dependencies	Required infrastructure and service dependencies	Required unless contractually inapplicable
failure_domains	Node, pool, cell, zone, region, site, provider, and control domains	Required unless contractually inapplicable
blast_radius	Maximum declared affected scope	Required unless contractually inapplicable
placement_constraints	Locality, hardware, topology, tenancy, and sovereignty rules	Required unless contractually inapplicable
capacity_profile	Demand, headroom, limits, reservations, and scale policy	Required unless contractually inapplicable
slo_profile	Indicators, objectives, windows, budgets, and action policy	Required unless contractually inapplicable
degraded_modes	Triggers, available capability, limitations, and exit criteria	Required unless contractually inapplicable
recovery_profile	Retry, failover, restore, reconciliation, and terminal behavior	Required unless contractually inapplicable
backup_profile	Coverage, consistency point, retention, locality, and restore tests	Required unless contractually inapplicable
disaster_recovery_profile	Recovery units, objectives, target, authority, and exercises	Required unless contractually inapplicable
network_zones	Permitted attachments and flow policies	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
identity_profile	Human, workload, device, service, and support trust	Required unless contractually inapplicable
policy_enforcement	Decision and enforcement points, bundles, failure behavior	Required unless contractually inapplicable
cryptographic_profile	Algorithms, keys, certificates, roots, rotation, and recovery	Required unless contractually inapplicable
secret_profile	Dynamic secret and certificate issuance and revocation	Required unless contractually inapplicable
data_protection_profile	In-transit, at-rest, in-use, backup, and telemetry protection	Required unless contractually inapplicable
sovereignty_profile	Data, processing, keys, administrators, providers, and support locality	Required unless contractually inapplicable
supply_chain_profile	Source, build, dependency, provenance, signature, and revocation	Required unless contractually inapplicable
telemetry_profile	Metrics, logs, traces, events, audit links, privacy, and retention	Required unless contractually inapplicable
security_monitoring_profile	Signals, detectors, thresholds, containment, and response	Required unless contractually inapplicable
change_profile	Approval, plan, cohort, observation, stop, rollback, and closure	Required unless contractually inapplicable
release_manifest	Dependency-closed artifacts and configuration	Required unless contractually inapplicable
acceptance_profile	Required site, workload, security, resilience, and operations evidence	Required unless contractually inapplicable
exceptions	Narrow scope, owner, risk, controls, expiry, and closure	Required unless contractually inapplicable
effective_and_expiry_time	Period of permitted production resolution	Required unless contractually inapplicable
retirement_profile	Export, migration, revocation, deletion, historical evidence, and closure	Required unless contractually inapplicable

I Workload Runtime Manifest

The workload manifest binds an executable service, agent, model, job, stream, connector, controller, or experience unit to exact identity, artifacts, resources, placement, connectivity, state, controls, service levels, failure, and lifecycle.

FIELD	BINDING DEFINITION	PRESENCE RULE
workload_id	Stable deployable workload identity	Required unless contractually inapplicable
workload_version	Immutable executable contract version	Required unless contractually inapplicable
artifact_digest	Signed image, package, model, or binary digest	Required unless contractually inapplicable
release_manifest_id	Dependency-closed released configuration	Required unless contractually inapplicable
owning_component	Logical and infrastructure component owner	Required unless contractually inapplicable
canonical_layer	Owning or served product layer	Required unless contractually inapplicable
task_and_product_contracts	Permitted input, output, and state semantics	Required unless contractually inapplicable
tenant_and_domain_scope	Isolation and policy context	Required unless contractually inapplicable
purpose_scope	Approved institutional purposes	Required unless contractually inapplicable
consequence_class	Effective control and human-gate class	Required unless contractually inapplicable
runtime_unit_class	Runtime pattern from the controlled catalog	Required unless contractually inapplicable
compute_profile	CPU, memory, accelerator, architecture, and topology requirements	Required unless contractually inapplicable
resource_requests_and_limits	Reserved and maximum resources	Required unless contractually inapplicable
quota_and_priority_class	Admission, fairness, preemption, and budget behavior	Required unless contractually inapplicable
placement_constraints	Locality, tenancy, topology, data, and sovereignty requirements	Required unless contractually inapplicable
disruption_budget	Permitted concurrent or temporal unavailability	Required unless contractually inapplicable
replica_and_quorum_profile	Replication, quorum, fencing, and failover semantics	Required unless contractually inapplicable
network_attachments	Permitted network zones and destinations	Required unless contractually inapplicable
ingress_contracts	Accepted protocols, routes, identity, and rate classes	Required unless contractually inapplicable
egress_contracts	Registered destinations, purpose, data classes, and receipts	Required unless contractually inapplicable
workload_identity	Issuer, trust domain, claims, and lease	Required unless contractually inapplicable
capability_grants	Permitted infrastructure and tool capabilities	Required unless contractually inapplicable
secret_bindings	Identity-bound secrets and certificates	Required unless contractually inapplicable
key_bindings	Cryptographic purposes and key domains	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
state_bindings	Authoritative, derived, cache, temporary, checkpoint, and evidence stores	Required unless contractually inapplicable
messaging_bindings	Topics, queues, schemas, delivery, ordering, and replay	Required unless contractually inapplicable
health_contract	Startup, readiness, liveness, safety, and dependency health	Required unless contractually inapplicable
slo_bindings	User-visible indicators, objectives, and error budgets	Required unless contractually inapplicable
telemetry_contract	Required metrics, logs, traces, events, and correlation	Required unless contractually inapplicable
audit_contract	Protected actions and evidence references	Required unless contractually inapplicable
data_protection	Classification, minimization, retention, locality, and encryption	Required unless contractually inapplicable
sandbox_profile	Process, filesystem, network, secret, and resource controls	Required unless contractually inapplicable
scaling_profile	Scale signals, bounds, stabilization, and cold-start behavior	Required unless contractually inapplicable
checkpoint_profile	Committed recovery state and compatibility	Required unless contractually inapplicable
failure_policy	Retry, queue, failover, compensate, abstain, and terminal behavior	Required unless contractually inapplicable
degraded_mode	Allowed capability and explicit trust limitations	Required unless contractually inapplicable
rollout_profile	Cohort, canary, observations, gates, and stop policy	Required unless contractually inapplicable
rollback_target	Known compatible system configuration	Required unless contractually inapplicable
monitoring_and_response	Runtime thresholds, owners, playbooks, and incidents	Required unless contractually inapplicable
retirement_and_revocation	Drain, migrate, revoke, delete, and preserve history	Required unless contractually inapplicable

J SLO and Capacity Metric Catalog

Metrics establish a common operating vocabulary for user journeys, control services, state, evidence, security, continuity, capacity, economics, and deployment parity. Each SLO profile declares scope, population, measurement, exclusions, threshold, budget, action, owner, and evidence.

ID	METRIC	MEASURED MEANING	MANDATORY SCOPE
IM-001	Availability	Proportion of eligible user journeys completing in an acceptable terminal state	Service + tenant + deployment + workload + window
IM-002	Decision-path availability	Availability of the complete evidence-to-human-decision path	Service + tenant + deployment + workload + window
IM-003	Request latency	Distribution of accepted request completion time	Service + tenant + deployment + workload + window
IM-004	Workflow duration	Distribution from admitted intent to explicit terminal state	Service + tenant + deployment + workload + window
IM-005	Human-gate latency	Time awaiting and processing required human authority	Service + tenant + deployment + workload + window
IM-006	Inference latency	Model gateway time by task, model, provider, and route	Service + tenant + deployment + workload + window
IM-007	Retrieval latency	Context and evidence retrieval time by method and scope	Service + tenant + deployment + workload + window
IM-008	Freshness	Age of decision-active state relative to its source and contract	Service + tenant + deployment + workload + window
IM-009	Collection delay	Time from source observation to custody and admission	Service + tenant + deployment + workload + window
IM-010	Index lag	Distance between authoritative revision and searchable projection	Service + tenant + deployment + workload + window
IM-011	Graph projection lag	Distance between graph log and queryable projection	Service + tenant + deployment + workload + window
IM-012	Event lag	Producer-to-consumer offset delay	Service + tenant + deployment + workload + window
IM-013	Queue age	Age distribution of unprocessed durable work	Service + tenant + deployment + workload + window
IM-014	Deadline attainment	Share of work reaching terminal state before contractual deadline	Service + tenant + deployment + workload + window
IM-015	Durability	Probability committed state survives declared failures	Service + tenant + deployment + workload + window
IM-016	Write correctness	Share of writes meeting consistency and integrity contracts	Service + tenant + deployment + workload + window
IM-017	Read correctness	Share of reads meeting consistency, revision, and authorization contracts	Service + tenant + deployment + workload + window
IM-018	Evidence completeness	Share of material outputs with complete provenance and custody	Service + tenant + deployment + workload + window
IM-019	Trace completeness	Share of material journeys with end-to-end causal telemetry	Service + tenant + deployment + workload + window
IM-020	Policy decision availability	Availability of correct current policy evaluation	Service + tenant + deployment + workload + window
IM-021	Identity issuance availability	Availability of trust-preserving short-lived identities	Service + tenant + deployment + workload + window
IM-022	Key operation availability	Availability of authorized cryptographic operations	Service + tenant + deployment + workload + window
IM-023	Isolation violation rate	Confirmed cross-tenant, domain, or workload isolation failures	Service + tenant + deployment + workload + window
IM-024	Security detection delay	Time from observable malicious activity to controlled detection	Service + tenant + deployment + workload + window

ID	METRIC	MEASURED MEANING	MANDATORY SCOPE
IM-025	Containment delay	Time from confirmed incident to bounded reachability and authority	Service + tenant + deployment + workload + window
IM-026	Recovery point attainment	Share of recoveries meeting declared maximum data loss	Service + tenant + deployment + workload + window
IM-027	Recovery time attainment	Share of recoveries meeting declared restoration time	Service + tenant + deployment + workload + window
IM-028	Restore success	Share of automated recovery sets restoring and reconciling successfully	Service + tenant + deployment + workload + window
IM-029	Failover success	Share of exercised failovers preserving state, policy, and user semantics	Service + tenant + deployment + workload + window
IM-030	Capacity headroom	Safe available resource relative to modeled peak and recovery demand	Service + tenant + deployment + workload + window
IM-031	Admission rejection	Work rejected by reason, class, tenant, and resource	Service + tenant + deployment + workload + window
IM-032	Preemption impact	Eligible work interrupted and successfully resumed after preemption	Service + tenant + deployment + workload + window
IM-033	Resource efficiency	Useful work per compute, accelerator, storage, or network unit	Service + tenant + deployment + workload + window
IM-034	Unit cost	Infrastructure cost per governed product, workflow, model task, or tenant unit	Service + tenant + deployment + workload + window
IM-035	Energy intensity	Energy consumed per controlled workload or product unit	Service + tenant + deployment + workload + window
IM-036	Deployment parity	Share of common conformance fixtures passing in every supported profile	Service + tenant + deployment + workload + window
IM-037	Change failure rate	Material changes requiring rollback, repair, or incident response	Service + tenant + deployment + workload + window
IM-038	Time to safe rollback	Time from stop decision to verified compatible state	Service + tenant + deployment + workload + window
IM-039	Observability loss	Share of expected signals absent, late, corrupt, or uncorrelated	Service + tenant + deployment + workload + window
IM-040	Degraded-mode duration	Time spent in each explicit restricted operating mode	Service + tenant + deployment + workload + window

K Failure Mode and Degraded State Matrix

The baseline failure matrix makes containment, degradation, recovery, reconciliation, and evidence explicit. Service-specific failure analyses may add detail but cannot remove these control families.

ID	FAILURE	PRIMARY DOMAIN	REQUIRED RESPONSE	MINIMUM EXERCISE
FM-01	Compute node loss	Node pool	Reschedule disposable work; restore stateful ownership	Node-loss exercise
FM-02	Accelerator fault	Device / serving pool	Drain, invalidate work, restore checkpoint	Device qualification and fault injection
FM-03	Scheduler partition	Cluster / cell	Freeze unsafe placement; preserve running work	Control partition exercise
FM-04	Region loss	Region	Recover within sovereignty and consistency policy	Regional continuity exercise
FM-05	Site isolation	On-premise site	Enter disconnected mode and reconcile later	WAN-loss exercise
FM-06	Cell failure	Intelligence cell	Contain tenant and domain scope; restore cell	Cell evacuation exercise
FM-07	Management-plane outage	Fleet	Freeze change; continue approved local control	Plane outage exercise
FM-08	Identity issuer outage	Trust domain	Honor bounded leases; stop new protected issuance	Trust continuity exercise
FM-09	Policy service outage	Protected operations	Safe cache or deny according to contract	Fail-closed test
FM-10	Key service outage	Key domain	Use approved continuity key path or stop	Key recovery ceremony
FM-11	Secret exposure	Credential scope	Revoke, rotate, contain, prove cleanup	Leak response exercise
FM-12	Network segmentation drift	Trust zone	Deny suspect flows and reconcile state	Reachability drift test
FM-13	DNS or discovery failure	Region / site	Use bounded cached state; block unsafe freshness	Naming partition exercise
FM-14	Trusted time loss	Region / site	Block time-sensitive authority and expose uncertainty	Clock drift and step test
FM-15	Egress destination compromise	Connector or provider path	Block, revoke, quarantine, assess effects	Destination substitution test
FM-16	Database quorum loss	State cluster	Fence writes and recover consistency	Quorum and failover test
FM-17	Storage corruption	State class	Quarantine, restore, reconcile consumers	Corruption and restore drill
FM-18	Object digest mismatch	Evidence object	Deny use and preserve custody investigation	Integrity test
FM-19	Graph partition divergence	Graph cell	Constrain traversals and replay changes	Partition recovery test
FM-20	Index staleness	Search cell	Label degraded or use policy-equivalent fallback	Staleness and rebuild test
FM-21	Broker partition	Messaging cell	Backpressure, elect, replay, reconcile	Broker-loss test
FM-22	Poison message	Consumer group	Quarantine, dead-letter, preserve sequence	Schema and poison test
FM-23	Workflow engine failure	Workflow cell	Resume from committed transition	Engine restart and replay
FM-24	Agent sandbox breach	Agent cell	Terminate, revoke, quarantine, assess effects	Escape and forensic test
FM-25	Model provider failure	Inference route	Abstain or approved equivalent fallback	Provider failover test
FM-26	Telemetry pipeline loss	Operations domain	Prioritize local signals and constrain risky change	Telemetry-loss exercise
FM-27	Backup failure	Recovery domain	Protect remaining copies and constrain destructive change	Backup failure drill
FM-28	Restore incompatibility	Recovery unit	Stop recovery and select compatible release	Cross-version restore test
FM-29	Partial infrastructure apply	Change scope	Stop mutation and reconcile actual state	Partial-apply recovery
FM-30	Supply-chain revocation	Fleet / release	Block, revoke, replace or rollback	Artifact revocation exercise
FM-31	Capacity exhaustion	Service or cell	Admission control and explicit shedding	Stress and overload test

ID	FAILURE	PRIMARY DOMAIN	REQUIRED RESPONSE	MINIMUM EXERCISE
FM-32	Operator error	Administrative scope	Stop, recover through controlled workflow, preserve evidence	Privileged operations exercise

Degraded-state command rules

COMMAND RULE	REQUIRED OPERATING BEHAVIOR
Declare before use	Every degraded mode identifies its trigger, authority, available and unavailable capabilities, trust and freshness limits, customer impact, maximum duration, and exit evidence.
Protect strategic meaning	The system does not represent partial infrastructure, stale data, failed inference, incomplete evidence, lost authority, or unreconciled recovery as a complete strategic product.
Constrain effects	External and authoritative side effects stop when their identity, policy, idempotency, human gate, destination, or effect-reconciliation boundary cannot be established.
Preserve evidence	Entry, decisions, automation, human actions, suppressed work, lost signals, fallback use, recovery, and exit remain correlated to the incident and affected product state.
Exit through proof	Normal service resumes only after controlling dependencies, state, policy, topology, security, capacity, and user-visible semantics pass their declared reconciliation and acceptance criteria.

FAILURE DOCTRINE

Availability is never manufactured by hiding degradation. When the safe boundary is uncertain, The Eye selects the more restrictive consequence class, preserves human awareness, and restores service through tested state.

L Security Control and Enforcement Matrix

Security controls are enforced through independent infrastructure boundaries. Application, agent, model, provider, or operator intent cannot waive their authority.

ID	CONTROL	ENFORCEMENT BOUNDARY	REQUIRED EVIDENCE
IC-01	Human strong authentication	Access edge and identity provider	Authentication logs and assurance claims
IC-02	Workload identity	Runtime and service fabric	Attested short-lived identity records
IC-03	Device identity	Edge and fleet	Firmware and device attestation
IC-04	Privileged access mediation	Administrative boundary	Approval, session record, expiry, and revocation
IC-05	Policy decision enforcement	API, service, data, tool, and release boundaries	Versioned decision traces
IC-06	Tenant and domain isolation	Compute, network, state, key, and telemetry	Negative isolation tests
IC-07	Network deny by default	Every trust zone	Reachability matrix and continuous probes
IC-08	Egress allowlisting	Connector, model, and effect paths	Destination and transfer receipts
IC-09	Encryption in transit	All protected connections	Certificate and protocol posture
IC-10	Encryption at rest	Every protected state class	Key bindings and storage posture
IC-11	Data-in-use protection	Selected protected workloads	Execution attestation and memory controls
IC-12	Key purpose separation	Key domains	Key policy and dual-control evidence
IC-13	Dynamic secret issuance	Workload boundaries	Lease, delivery, use, and revocation evidence
IC-14	Artifact signature enforcement	Build, registry, and runtime admission	Provenance and verification records
IC-15	Dependency bill of materials	Software, models, firmware, modules	Resolved bill and vulnerability state
IC-16	Reproducible build	Controlled build domain	Source-to-artifact comparison
IC-17	Sandbox isolation	Agent, connector, and build runtimes	Escape, file, network, and process tests
IC-18	Capability-based tool access	Agent and automation boundaries	Grant, call, receipt, and revocation
IC-19	Content isolation	External and untrusted sources	Malware, parsing, and instruction-boundary tests
IC-20	Data minimization	Collection, retrieval, inference, telemetry	Field and content reviews
IC-21	Residency enforcement	Placement, storage, backup, provider, support	End-to-end locality evidence
IC-22	Immutable audit	Material authority and state change	Integrity anchors and custody
IC-23	Security telemetry	Every critical plane and component	Coverage and loss state
IC-24	Detection validation	Security monitoring	Attack simulations and detector metrics
IC-25	Automated containment	Pre-approved reversible controls	Action authority and effect evidence
IC-26	Vulnerability management	Complete fleet and artifacts	Exposure, patch, exception, and closure
IC-27	Configuration drift control	Infrastructure and policy	Desired-to-observed reconciliation
IC-28	Backup immutability	Recovery domains	Protection and delete-resistance tests
IC-29	Break-glass governance	Critical operations	Two-person approval, reason, session, and review
IC-30	Security incident command	Security domain	Case, containment, decision impact, and closure

M Backup, Restore, and Disaster Recovery Matrix

Recovery objectives are defined per state class and recovery unit. A copied byte set is not a recovered service until identity, keys, policies, versions, workflows, projections, consumers, sovereignty, and user-visible state reconcile.

ID	RECOVERY UNIT	RPO	RTO	RECOVERY BASIS	MINIMUM EVIDENCE
DR-01	Identity and trust roots	Zero or bounded lease state	Minutes to hours	Separate root and recovery ceremony	Trust continuity exercise
DR-02	Policy bundles	Last approved revision	Minutes	Signed local copies	Fail-closed and restore test
DR-03	Key management	No committed key loss	Minutes to hours	Separate hardware-backed recovery	Key recovery ceremony
DR-04	Configuration desired state	Last committed revision	Minutes	Separate state backend copy	State-backend restore
DR-05	Release registry	No released identity loss	Minutes to hours	Immutable replicated registry	Registry recovery
DR-06	Transactional operational state	Contract-specific seconds to hours	Minutes to hours	Consistent logs and snapshots	Point-in-time restore
DR-07	Knowledge Graph	Declared graph log position	Hours	Change log plus snapshots	Graph replay and index rebuild
DR-08	Enterprise Memory	Declared object revision	Hours	Canonical objects plus indexes	Temporal restore and reconciliation
DR-09	Digital Twins	Last consistent snapshot and branch log	Hours	Snapshots, events, and checkpoints	Twin restore and branch test
DR-10	Evidence object store	No acknowledged object loss where required	Hours	Immutable cross-domain copies	Large-scale restore and digest check
DR-11	Audit evidence	No committed audit loss	Hours	Separate immutable copies and anchors	Audit restore and integrity verification
DR-12	Messaging logs	Declared committed offset	Minutes to hours	Replicated logs and consumer checkpoints	Broker recovery and replay
DR-13	Workflow state	Last committed transition	Minutes	Durable history and task leases	Workflow engine recovery
DR-14	Agent execution checkpoints	Last compatible checkpoint	Minutes to hours	Checkpoint store and release manifest	Agent resume and replay
DR-15	Search and vector indexes	Rebuildable	Hours to days	Authoritative sources and definitions	Deterministic rebuild
DR-16	Telemetry	Bounded accepted loss	Hours	Local buffers and regional copies	Observability continuity
DR-17	Security cases	No acknowledged case loss	Minutes to hours	Separate protected case store	Security operations recovery
DR-18	Executive decisions and briefings	No committed decision loss	Hours	Canonical records and evidence links	Decision-path recovery
DR-19	On-premise site	Per state-class objectives	Hours to days	Local recovery vault or alternate site	Isolated site recovery
DR-20	Air-gapped deployment	Per local recovery policy	Hours to days	Offline signed recovery media	No-network recovery exercise

N Requirement, Constitutional, Engineering, and AI Traceability

Traceability resolves each stable IA requirement range upward to constitutional invariants and across the controlled engineering and AI baselines. Individual requirements inherit every anchor listed for their chapter.

C H.	INFRASTRUCTURE SUBJECT	REQUIREMENT RANGE	CO UN T	CONSTITUTION	V4 ANCHORS	V5 ANCHORS
1	Infrastructure Architecture Authority and Scope	IA-01-001-IA-01-006	6	C-001, C-003, C-004, C-008, C-051, C-052	V4:1, V4:4, V4:6, V4:7, V4:72	V5:1, V5:2, V5:6, V5:71, V5:72
2	Specification Hierarchy and Conformance	IA-02-001-IA-02-006	6	C-003, C-008, C-050, C-051, C-052	V4:2, V4:3, V4:6, V4:23, V4:72	V5:2, V5:6, V5:70, V5:71, V5:72
3	Infrastructure Principles and Semantic Parity	IA-03-001-IA-03-006	6	C-001, C-004, C-036, C-045, C-051	V4:5, V4:14, V4:23, V4:64, V4:71	V5:7, V5:9, V5:20, V5:71
4	Shared Responsibility and Customer Control	IA-04-001-IA-04-006	6	C-004, C-006, C-028, C-036, C-049	V4:2, V4:10, V4:49, V4:63, V4:72	V5:3, V5:37, V5:65, V5:71, V5:72
5	Failure-Domain and Blast-Radius Doctrine	IA-05-001-IA-05-006	6	C-008, C-034, C-038, C-041, C-049	V4:7, V4:9, V4:11, V4:59, V4:61	V5:31, V5:34, V5:38, V5:68, V5:72
6	Infrastructure Traceability and Documentation Inheritance	IA-06-001-IA-06-006	6	C-003, C-008, C-043, C-050, C-052	V4:6, V4:23, V4:66, V4:69, V4:72	V5:6, V5:15, V5:30, V5:70, V5:72
7	System Context and Deployment Models	IA-07-001-IA-07-006	6	C-001, C-007, C-036, C-045, C-049	V4:7, V4:8, V4:10, V4:14, V4:71	V5:7, V5:8, V5:9, V5:71
8	SaaS Reference Architecture	IA-08-001-IA-08-006	6	C-006, C-034, C-036, C-038, C-045	V4:7, V4:8, V4:9, V4:10, V4:14, V4:61	V5:7, V5:10, V5:12, V5:68, V5:71
9	Private Cloud Reference Architecture	IA-09-001-IA-09-006	6	C-006, C-036, C-045, C-046, C-049	V4:8, V4:10, V4:13, V4:14, V4:71	V5:7, V5:17, V5:21, V5:71, V5:72
10	On-Premise Reference Architecture	IA-10-001-IA-10-006	6	C-006, C-036, C-045, C-046, C-047	V4:8, V4:14, V4:56, V4:64, V4:71	V5:7, V5:17, V5:22, V5:70, V5:71
11	Disconnected and Air-Gapped Operations	IA-11-001-IA-11-006	6	C-007, C-017, C-036, C-046, C-047	V4:14, V4:29, V4:54, V4:56, V4:62	V5:21, V5:26, V5:60, V5:71, V5:72
12	Regional, Zonal, Site, and Cell Topology	IA-12-001-IA-12-006	6	C-034, C-036, C-038, C-041, C-045	V4:7, V4:9, V4:12, V4:59, V4:61	V5:12, V5:19, V5:22, V5:71
13	Tenant, Domain, and Intelligence Cell Isolation	IA-13-001-IA-13-006	6	C-006, C-007, C-030, C-034, C-049	V4:8, V4:9, V4:13, V4:25, V4:51	V5:7, V5:12, V5:19, V5:61
14	Shared Control Services and Management Plane	IA-14-001-IA-14-006	6	C-007, C-020, C-030, C-036, C-049	V4:10, V4:13, V4:49, V4:54, V4:63	V5:10, V5:11, V5:14, V5:65, V5:71
15	Compute Platform Architecture	IA-15-001-IA-15-006	6	C-008, C-034, C-038, C-041, C-045	V4:7, V4:9, V4:11, V4:58, V4:59	V5:22, V5:31, V5:68, V5:71
16	CPU Workload Architecture	IA-16-001-IA-16-006	6	C-008, C-035, C-038, C-041	V4:11, V4:41, V4:58, V4:59	V5:22, V5:34, V5:66, V5:68
17	GPU and Accelerator Architecture	IA-17-001-IA-17-006	6	C-007, C-034, C-038, C-045, C-048	V4:11, V4:53, V4:56, V4:58, V4:59	V5:15, V5:16, V5:22, V5:62, V5:70
18	Model Serving Infrastructure	IA-18-001-IA-18-006	6	C-004, C-005, C-024, C-035, C-048	V4:43, V4:44, V4:52, V4:58, V4:61	V5:15, V5:16, V5:17, V5:20, V5:22, V5:70
19	Agent and Workflow Runtime Infrastructure	IA-19-001-IA-19-006	6	C-004, C-005, C-024, C-028, C-035	V4:41, V4:42, V4:43, V4:49, V4:59	V5:30, V5:31, V5:33, V5:34, V5:36, V5:38
20	Batch and Stream Processing Infrastructure	IA-20-001-IA-20-006	6	C-011, C-016, C-035, C-037, C-041	V4:19, V4:22, V4:28, V4:41, V4:57	V5:19, V5:23, V5:26, V5:66, V5:68
21	Edge, IoT, and Remote Collection Infrastructure	IA-21-001-IA-21-006	6	C-007, C-010, C-017, C-036, C-048	V4:19, V4:25, V4:26, V4:28, V4:56	V5:7, V5:21, V5:26, V5:39, V5:60
22	Resource Scheduling, Quotas, and Admission	IA-22-001-IA-22-006	6	C-004, C-006, C-035, C-038, C-041	V4:9, V4:13, V4:21, V4:58, V4:59	V5:19, V5:22, V5:31, V5:63, V5:68
23	Network Architecture and Segmentation	IA-23-001-IA-23-006	6	C-007, C-030, C-034, C-036, C-049	V4:13, V4:49, V4:53, V4:54, V4:60	V5:21, V5:36, V5:60, V5:65
24	Ingress, API, and Access Edge	IA-24-001-IA-24-006	6	C-007, C-020, C-030, C-035, C-049	V4:15, V4:16, V4:20, V4:49, V4:60	V5:7, V5:16, V5:20, V5:60
25	Service-to-Service Connectivity	IA-25-001-IA-25-006	6	C-007, C-020, C-030, C-035, C-049	V4:11, V4:15, V4:23, V4:49, V4:53	V5:16, V5:17, V5:21, V5:36

C H.	INFRASTRUCTURE SUBJECT	REQUIREMENT RANGE	CO UN T	CONSTITUTION	V4 ANCHORS	V5 ANCHORS
26	Data Plane and Control Plane Separation	IA-26-001-IA-26-006	6	C-007, C-020, C-030, C-036, C-049	V4:10, V4:13, V4:29, V4:49, V4:54	V5:10, V5:11, V5:12, V5:21, V5:65
27	DNS, Naming, Discovery, and Time	IA-27-001-IA-27-006	6	C-007, C-016, C-017, C-035, C-049	V4:18, V4:25, V4:26, V4:53, V4:59	V5:15, V5:26, V5:29, V5:65
28	Egress, External Integration, and Source Connectivity	IA-28-001-IA-28-006	6	C-007, C-018, C-029, C-030, C-049	V4:13, V4:19, V4:29, V4:49, V4:60	V5:21, V5:23, V5:36, V5:39, V5:59, V5:60
29	Private Connectivity and Customer Networks	IA-29-001-IA-29-006	6	C-006, C-007, C-036, C-038, C-049	V4:8, V4:10, V4:14, V4:53, V4:63	V5:7, V5:12, V5:21, V5:71
30	Network Resilience, Performance, and Observability	IA-30-001-IA-30-006	6	C-035, C-038, C-041, C-045, C-049	V4:19, V4:22, V4:58, V4:59, V4:61	V5:22, V5:31, V5:68, V5:72
31	Storage Architecture and State Classes	IA-31-001-IA-31-006	6	C-011, C-012, C-016, C-033, C-043	V4:12, V4:24, V4:28, V4:29, V4:30, V4:62	V5:9, V5:13, V5:19, V5:25, V5:38
32	Transactional and Operational Datastores	IA-32-001-IA-32-006	6	C-011, C-012, C-016, C-035, C-041	V4:12, V4:18, V4:22, V4:24, V4:30, V4:62	V5:15, V5:20, V5:31, V5:34, V5:70
33	Object and Evidence Storage	IA-33-001-IA-33-006	6	C-010, C-013, C-018, C-029, C-043	V4:12, V4:28, V4:29, V4:30, V4:54, V4:62	V5:15, V5:21, V5:29, V5:38, V5:65
34	Knowledge Graph Storage Infrastructure	IA-34-001-IA-34-006	6	C-014, C-015, C-016, C-033, C-041	V4:12, V4:26, V4:27, V4:28, V4:30, V4:34	V5:13, V5:24, V5:27, V5:43, V5:44, V5:45
35	Vector, Search, and Index Infrastructure	IA-35-001-IA-35-006	6	C-011, C-016, C-027, C-033, C-035	V4:12, V4:18, V4:23, V4:28, V4:30, V4:57	V5:19, V5:23, V5:26, V5:27, V5:29, V5:40
36	Digital Twin and Simulation State Infrastructure	IA-36-001-IA-36-006	6	C-019, C-021, C-022, C-023, C-033	V4:12, V4:26, V4:27, V4:30, V4:44, V4:62	V5:13, V5:28, V5:49, V5:50, V5:63
37	Event Streaming, Messaging, and Durable Queues	IA-37-001-IA-37-006	6	C-011, C-016, C-035, C-037, C-041	V4:19, V4:20, V4:22, V4:23, V4:41, V4:61	V5:31, V5:34, V5:35, V5:38, V5:68
38	Cache, Temporary State, and Checkpoint Infrastructure	IA-38-001-IA-38-006	6	C-011, C-016, C-029, C-033, C-043	V4:12, V4:21, V4:22, V4:29, V4:30, V4:41	V5:19, V5:26, V5:31, V5:38, V5:69
39	Container, Orchestration, and Runtime Platform	IA-39-001-IA-39-006	6	C-007, C-008, C-034, C-038, C-048	V4:7, V4:9, V4:11, V4:53, V4:59, V4:61	V5:15, V5:18, V5:30, V5:68, V5:70
40	Service Platform and Internal Developer Platform	IA-40-001-IA-40-006	6	C-003, C-008, C-035, C-048, C-051	V4:11, V4:15, V4:23, V4:65, V4:66, V4:72	V5:15, V5:17, V5:30, V5:66, V5:70
41	Workflow Engine Infrastructure	IA-41-001-IA-41-006	6	C-004, C-011, C-024, C-035, C-041	V4:22, V4:41, V4:42, V4:44, V4:61, V4:62	V5:31, V5:34, V5:37, V5:38, V5:72
42	Agent Execution Sandbox Infrastructure	IA-42-001-IA-42-006	6	C-004, C-005, C-007, C-024, C-030	V4:43, V4:49, V4:51, V4:54, V4:60	V5:30, V5:31, V5:36, V5:59, V5:60, V5:62
43	Model Gateway and Inference Fabric	IA-43-001-IA-43-006	6	C-004, C-005, C-007, C-024, C-035	V4:15, V4:20, V4:43, V4:52, V4:58, V4:60	V5:10, V5:15, V5:16, V5:17, V5:20, V5:21, V5:22
44	Retrieval and Context Infrastructure	IA-44-001-IA-44-006	6	C-013, C-016, C-024, C-027, C-029	V4:18, V4:27, V4:28, V4:29, V4:57	V5:19, V5:23, V5:24, V5:25, V5:26, V5:27, V5:28, V5:29
45	Data and AI Pipeline Runtime	IA-45-001-IA-45-006	6	C-008, C-011, C-024, C-033, C-044	V4:28, V4:30, V4:41, V4:57, V4:68, V4:70	V5:15, V5:20, V5:66, V5:68, V5:69, V5:70
46	Executive Experience and Collaboration Infrastructure	IA-46-001-IA-46-006	6	C-004, C-005, C-024, C-032, C-035	V4:15, V4:18, V4:21, V4:42, V4:46, V4:58	V5:9, V5:37, V5:56, V5:57, V5:64
47	Integration Runtime and Connector Isolation	IA-47-001-IA-47-006	6	C-007, C-010, C-018, C-029, C-030	V4:19, V4:23, V4:25, V4:28, V4:49, V4:56	V5:21, V5:36, V5:39, V5:40, V5:60
48	Platform APIs, Events, and Contract Enforcement	IA-48-001-IA-48-006	6	C-003, C-011, C-035, C-037, C-043	V4:15, V4:19, V4:20, V4:21, V4:22, V4:23	V5:7, V5:9, V5:20, V5:31, V5:35
49	Identity, Workload Trust, and Privileged Access	IA-49-001-IA-49-006	6	C-004, C-006, C-007, C-020, C-028	V4:25, V4:49, V4:50, V4:55, V4:60	V5:3, V5:36, V5:37, V5:59, V5:65
50	Policy Enforcement Infrastructure	IA-50-001-IA-50-006	6	C-004, C-007, C-020, C-024, C-030	V4:13, V4:29, V4:49, V4:50, V4:60	V5:11, V5:12, V5:21, V5:36, V5:59, V5:65
51	Cryptography, Key Management, and Trust Anchors	IA-51-001-IA-51-006	6	C-006, C-007, C-030, C-036, C-049	V4:51, V4:53, V4:54, V4:55, V4:62	V5:15, V5:21, V5:60, V5:65, V5:71
52	Secrets, Certificates, and Credential Lifecycle	IA-52-001-IA-52-006	6	C-007, C-020, C-030, C-043, C-049	V4:49, V4:51, V4:53, V4:55, V4:60	V5:21, V5:36, V5:60, V5:65, V5:70
53	Data Protection and Confidential Computing	IA-53-001-IA-53-006	6	C-007, C-018, C-029, C-030, C-036	V4:29, V4:51, V4:53, V4:54, V4:60	V5:21, V5:23, V5:29, V5:61, V5:65, V5:71
54	Tenant, Domain, and Workload Isolation	IA-54-001-IA-54-006	6	C-006, C-007, C-030, C-034, C-049	V4:8, V4:9, V4:13, V4:50, V4:54, V4:60	V5:12, V5:21, V5:36, V5:59, V5:61

C H.	INFRASTRUCTURE SUBJECT	REQUIREMENT RANGE	CO UN T	CONSTITUTION	V4 ANCHORS	V5 ANCHORS
55	Software Supply Chain and Artifact Integrity	IA-55-001-IA-55-006	6	C-003, C-007, C-043, C-048, C-050	V4:53, V4:56, V4:65, V4:66, V4:68, V4:69	V5:15, V5:18, V5:30, V5:62, V5:67, V5:70
56	Security Monitoring, Detection, and Response	IA-56-001-IA-56-006	6	C-007, C-030, C-039, C-043, C-049	V4:49, V4:54, V4:55, V4:56, V4:60, V4:63	V5:58, V5:59, V5:60, V5:61, V5:62, V5:65, V5:72
57	Sovereignty, Residency, and Administrative Control	IA-57-001-IA-57-006	6	C-006, C-018, C-029, C-036, C-046	V4:14, V4:29, V4:49, V4:51, V4:54, V4:62	V5:21, V5:60, V5:61, V5:65, V5:71
58	Observability and Telemetry Architecture	IA-58-001-IA-58-006	6	C-016, C-029, C-035, C-039, C-049	V4:28, V4:29, V4:57, V4:58, V4:60, V4:63	V5:29, V5:31, V5:65, V5:68, V5:72
59	Service-Level Management and Error Budgets	IA-59-001-IA-59-006	6	C-035, C-038, C-039, C-040, C-041	V4:21, V4:57, V4:58, V4:59, V4:63	V5:63, V5:66, V5:68, V5:72
60	Capacity, Performance, and Load Engineering	IA-60-001-IA-60-006	6	C-035, C-038, C-040, C-041, C-045	V4:9, V4:22, V4:57, V4:58, V4:59, V4:61	V5:19, V5:22, V5:31, V5:63, V5:68
61	High Availability and Fault Tolerance	IA-61-001-IA-61-006	6	C-034, C-035, C-038, C-041, C-045	V4:7, V4:12, V4:22, V4:58, V4:59, V4:61	V5:31, V5:34, V5:38, V5:63, V5:68, V5:72
62	Backup, Restore, and State Reconciliation	IA-62-001-IA-62-006	6	C-012, C-016, C-029, C-033, C-041	V4:12, V4:30, V4:54, V4:59, V4:62	V5:15, V5:25, V5:26, V5:38, V5:70, V5:72
63	Disaster Recovery and Regional Continuity	IA-63-001-IA-63-006	6	C-034, C-036, C-038, C-041, C-045	V4:14, V4:41, V4:59, V4:61, V4:62, V4:63	V5:38, V5:63, V5:71, V5:72
64	Incident Command and Operational Response	IA-64-001-IA-64-006	6	C-004, C-028, C-039, C-043, C-049	V4:21, V4:42, V4:55, V4:60, V4:63	V5:3, V5:37, V5:59, V5:65, V5:72
65	Change, Maintenance, and Lifecycle Operations	IA-65-001-IA-65-006	6	C-003, C-008, C-035, C-043, C-050	V4:23, V4:30, V4:55, V4:63, V4:66, V4:68, V4:69	V5:15, V5:18, V5:68, V5:69, V5:70, V5:72
66	Degraded Modes and Continuity of Strategic Operations	IA-66-001-IA-66-006	6	C-004, C-005, C-035, C-041, C-049	V4:21, V4:41, V4:58, V4:61, V4:63, V4:64	V5:3, V5:20, V5:31, V5:37, V5:63, V5:72
67	Infrastructure-as-Code and Configuration Architecture	IA-67-001-IA-67-006	6	C-003, C-008, C-043, C-048, C-050	V4:6, V4:13, V4:65, V4:66, V4:68, V4:72	V5:15, V5:18, V5:70, V5:71
68	Build, Release, Promotion, and Rollback Infrastructure	IA-68-001-IA-68-006	6	C-003, C-008, C-035, C-043, C-050	V4:23, V4:30, V4:66, V4:68, V4:70, V4:71	V5:15, V5:18, V5:66, V5:67, V5:70, V5:71
69	Fleet, Patch, and Vulnerability Management	IA-69-001-IA-69-006	6	C-007, C-039, C-043, C-048, C-050	V4:53, V4:55, V4:56, V4:63, V4:69	V5:15, V5:18, V5:62, V5:67, V5:70
70	Cost, FinOps, Sustainability, and Unit Economics	IA-70-001-IA-70-006	6	C-006, C-035, C-038, C-045, C-050	V4:9, V4:14, V4:58, V4:59, V4:71, V4:72	V5:22, V5:68, V5:70, V5:71
71	Deployment Acceptance and Operational Readiness	IA-71-001-IA-71-006	6	C-003, C-006, C-008, C-049, C-051	V4:14, V4:57, V4:61, V4:63, V4:71, V4:72	V5:66, V5:67, V5:68, V5:70, V5:71, V5:72
72	Infrastructure Governance and Baseline Closure	IA-72-001-IA-72-006	6	C-001, C-003, C-004, C-049, C-051, C-052	V4:1, V4:2, V4:6, V4:64, V4:72	V5:1, V5:2, V5:6, V5:70, V5:71, V5:72

Constitutional infrastructure architecture matrix

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 6 ANCHORS
C-001	Category integrity	The Eye shall be conceived, designed, sold, and operated as a Strategic Intelligence Operating System. It shall not be reduced to a chatbot, dashboard, BI tool, copilot, or reporting product.	Ch. 1, Ch. 3, Ch. 7, Ch. 72, Ch. 4, Ch. 46, Ch. 49
C-002	Closed strategic loop	The product shall implement one continuous operating loop: Observe, Understand, Predict, Simulate, Decide, and Learn. No stage may be treated as a disconnected feature island.	Ch. 1, Ch. 3, Ch. 4, Ch. 7, Ch. 46, Ch. 49, Ch. 72
C-003	Product boundary	Interfaces such as chat, dashboards, reports, and APIs may exist, but they are access modes to the operating system rather than the product definition.	Ch. 1, Ch. 2, Ch. 6, Ch. 40, Ch. 48, Ch. 55, Ch. 65, Ch. 67, Ch. 68, Ch. 71, Ch. 72, Ch. 3, Ch. 4, Ch. 7, Ch. 46, Ch. 49
C-004	Human decision sovereignty	A named human authority shall make every final strategic or high-impact decision. The system may advise, prepare, and execute approved workflows, but it shall not silently assume decision rights.	Ch. 1, Ch. 3, Ch. 4, Ch. 18, Ch. 19, Ch. 22, Ch. 41, Ch. 42, Ch. 43, Ch. 46, Ch. 49, Ch. 50, Ch. 64, Ch. 66, Ch. 72, Ch. 7
C-005	Explain every recommendation	Every material forecast, scenario, simulation result, risk, opportunity, and recommendation shall expose evidence, assumptions, uncertainty, lineage, alternatives, and accountable system versions.	Ch. 18, Ch. 19, Ch. 42, Ch. 43, Ch. 46, Ch. 66, Ch. 1, Ch. 3, Ch. 4, Ch. 7, Ch. 49, Ch. 72

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 6 ANCHORS
C-006	Enterprise operating standard	The common platform shall satisfy enterprise requirements for governance, security, scale, auditability, reliability, and lifecycle management from its first production architecture.	Ch. 4, Ch. 8, Ch. 9, Ch. 10, Ch. 13, Ch. 22, Ch. 29, Ch. 49, Ch. 51, Ch. 54, Ch. 57, Ch. 70, Ch. 71, Ch. 1, Ch. 3, Ch. 7, Ch. 46, Ch. 72
C-007	Universal core, governed specialization	A single domain-neutral core shall serve all customer classes. Domain ontologies, models, workflows, agents, and policies may extend the core without forking its constitutional semantics.	Ch. 7, Ch. 11, Ch. 13, Ch. 14, Ch. 17, Ch. 21, Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 28, Ch. 29, Ch. 39, Ch. 42, Ch. 43, Ch. 47, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 55, Ch. 56, Ch. 69, Ch. 31, Ch. 33
C-008	Ten-layer architecture	The ten named architecture layers are the canonical product structure. Cross-cutting planes may support them but shall not replace, collapse, or rename them without constitutional amendment.	Ch. 1, Ch. 2, Ch. 5, Ch. 6, Ch. 15, Ch. 16, Ch. 39, Ch. 40, Ch. 45, Ch. 65, Ch. 67, Ch. 68, Ch. 71, Ch. 13, Ch. 23, Ch. 28, Ch. 31, Ch. 33, Ch. 49, Ch. 50, Ch. 51, Ch. 54, Ch. 55
C-009	Knowledge Graph centrality	The Knowledge Graph shall be the connected semantic heart of The Eye and the shared context substrate for memory, twins, reasoning, prediction, scenarios, simulations, and decisions.	Ch. 13, Ch. 23, Ch. 28, Ch. 31, Ch. 33, Ch. 49, Ch. 50, Ch. 51, Ch. 54, Ch. 55
C-010	Truth-state separation	Observed evidence, extracted claims, inferred relationships, model outputs, human judgments, and synthetic scenario data shall remain explicitly distinguishable throughout the system.	Ch. 21, Ch. 33, Ch. 47, Ch. 13, Ch. 23, Ch. 28, Ch. 31, Ch. 49, Ch. 50, Ch. 51, Ch. 54, Ch. 55
C-011	Temporal truth	Material facts and relationships shall preserve event time, observation time, validity time, and system-record time so that past world states can be reconstructed without hindsight contamination.	Ch. 20, Ch. 31, Ch. 32, Ch. 35, Ch. 37, Ch. 38, Ch. 41, Ch. 45, Ch. 48, Ch. 13, Ch. 23, Ch. 28, Ch. 33, Ch. 49, Ch. 50, Ch. 51, Ch. 54, Ch. 55
C-012	End-to-end provenance	Every material intelligence object shall carry source, chain of custody, transformation history, ownership, rights, quality, and version lineage back to its evidence.	Ch. 31, Ch. 32, Ch. 62, Ch. 13, Ch. 23, Ch. 28, Ch. 33, Ch. 49, Ch. 50, Ch. 51, Ch. 54, Ch. 55
C-013	Observation breadth	The World Observation Layer shall support the full canonical source universe and allow governed custom sources without architectural rework.	Ch. 33, Ch. 44, Ch. 13, Ch. 23, Ch. 28, Ch. 31, Ch. 49, Ch. 50, Ch. 51, Ch. 54, Ch. 55
C-014	Freshness and quality contracts	Sources, derived data, twins, models, and intelligence products shall declare measurable freshness, completeness, quality, and expiry expectations.	Ch. 34, Ch. 27, Ch. 31, Ch. 35, Ch. 36, Ch. 44
C-015	Durable institutional memory	The Eye shall preserve institutional context, decisions, assumptions, evidence, outcomes, and lessons as durable memory, subject to lawful retention, deletion, and customer policy.	Ch. 34, Ch. 27, Ch. 31, Ch. 35, Ch. 36, Ch. 44
C-016	Memory governance	Memory shall be permission-aware, tenant-isolated, versioned, retrievable, correctable, and protected from unverified model output becoming accepted organizational truth.	Ch. 20, Ch. 27, Ch. 31, Ch. 32, Ch. 34, Ch. 35, Ch. 37, Ch. 38, Ch. 44, Ch. 58, Ch. 62, Ch. 36
C-017	Strategy Graph	Objectives, hypotheses, decisions, initiatives, capabilities, resources, indicators, risks, opportunities, and outcomes shall be connected in a governed Strategy Graph anchored to the Knowledge Graph.	Ch. 11, Ch. 21, Ch. 27, Ch. 31, Ch. 34, Ch. 35, Ch. 36, Ch. 44
C-018	Grounded Digital Twins	Every Digital Twin shall be grounded in evidence, graph entities, temporal state, declared assumptions, and calibrated models; it shall never present synthetic state as observed fact.	Ch. 28, Ch. 33, Ch. 47, Ch. 53, Ch. 57, Ch. 27, Ch. 31, Ch. 34, Ch. 35, Ch. 36, Ch. 44
C-019	Twin versionability	Digital Twins shall support time travel, branching, comparison, replay, reconciliation, and versioned evolution across actual and simulated states.	Ch. 36, Ch. 27, Ch. 31, Ch. 34, Ch. 35, Ch. 44
C-020	Canonical prediction horizons	The Prediction Engine shall support 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons with horizon-appropriate methods and uncertainty.	Ch. 14, Ch. 24, Ch. 25, Ch. 26, Ch. 49, Ch. 50, Ch. 52, Ch. 19, Ch. 41, Ch. 42, Ch. 43
C-021	Probabilistic forecasting	Forecasts shall communicate distributions, confidence, calibration, drivers, sensitivity, and failure conditions rather than false precision or unsupported certainty.	Ch. 36, Ch. 19, Ch. 41, Ch. 42, Ch. 43, Ch. 49, Ch. 50
C-022	Scenario plurality	Scenario Intelligence shall maintain multiple coherent futures, including baseline, upside, downside, disruption, and user-defined alternatives; no single forecast shall masquerade as destiny.	Ch. 36, Ch. 19, Ch. 41, Ch. 42, Ch. 43, Ch. 49, Ch. 50
C-023	Reproducible simulation	Every simulation shall declare initial state, assumptions, interventions, model versions, constraints, random seeds where applicable, outputs, and sensitivity so results can be reproduced and challenged.	Ch. 36, Ch. 19, Ch. 41, Ch. 42, Ch. 43, Ch. 49, Ch. 50
C-024	Decision completeness	Decision Intelligence shall compare viable options against explicit objectives, constraints, risks, opportunities, trade-offs, second-order effects, reversibility, and information value.	Ch. 18, Ch. 19, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 45, Ch. 46, Ch. 50, Ch. 49
C-025	Decision Replay	The Eye shall preserve the state of knowledge available at decision time and support replay against later evidence and outcomes without rewriting the historical record.	Ch. 31, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 38, Ch. 44, Ch. 53, Ch. 57

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 6 ANCHORS
C-026	Objective alignment	Recommendations shall be evaluated against declared objectives, stakeholder obligations, risk appetite, policy, and strategic constraints rather than an opaque global score.	Ch. 31, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 38, Ch. 44, Ch. 53, Ch. 57
C-027	Executive Operating System	The executive surface shall organize strategic attention, decisions, scenarios, evidence, objectives, commitments, and outcomes. It shall not degrade into a passive dashboard collection.	Ch. 35, Ch. 44, Ch. 31, Ch. 33, Ch. 34, Ch. 36, Ch. 38, Ch. 53, Ch. 57
C-028	Bounded agent authority	Every agent shall have a declared purpose, tools, data scope, permissions, budgets, escalation rules, prohibited actions, evaluation suite, and accountable owner.	Ch. 4, Ch. 19, Ch. 49, Ch. 64, Ch. 31, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 38, Ch. 44, Ch. 53, Ch. 57
C-029	Supervised orchestration	Planner, Supervisor, and Workflow agents shall enforce policy, separation of duties, checkpoints, timeouts, retries, human gates, and complete execution traces across multi-agent work.	Ch. 28, Ch. 33, Ch. 38, Ch. 44, Ch. 47, Ch. 53, Ch. 57, Ch. 58, Ch. 62, Ch. 31, Ch. 34, Ch. 35, Ch. 36
C-030	Model pluralism	The platform shall support multiple models and reasoning approaches behind governed interfaces so customers retain control over cost, sovereignty, performance, and vendor dependence.	Ch. 13, Ch. 14, Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 28, Ch. 42, Ch. 47, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 56, Ch. 31, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 38, Ch. 44, Ch. 57
C-031	No silent self-modification	Continuous learning shall never permit unreviewed production models, agents, prompts, policies, ontologies, or decision logic to modify themselves silently.	Ch. 46, Ch. 56, Ch. 58, Ch. 59, Ch. 61, Ch. 64, Ch. 66
C-032	Proactive intelligence	The system shall surface material changes, weak signals, risks, and opportunities before users ask, while respecting relevance, authority, attention, and notification policy.	Ch. 46, Ch. 56, Ch. 58, Ch. 59, Ch. 61, Ch. 64, Ch. 66
C-033	Early-warning discipline	Warnings shall be deduplicated, prioritized, evidence-backed, confidence-aware, time-bounded, and linked to affected objectives, assets, decisions, and response playbooks.	Ch. 31, Ch. 34, Ch. 35, Ch. 36, Ch. 38, Ch. 45, Ch. 62, Ch. 46, Ch. 56, Ch. 58, Ch. 59, Ch. 61, Ch. 64, Ch. 66
C-034	Decomposable Strategic Health Score	Any Strategic Health Score shall expose component measures, weights, evidence, confidence, trend, and sensitivity. The score shall never replace the underlying analysis.	Ch. 5, Ch. 8, Ch. 12, Ch. 13, Ch. 15, Ch. 17, Ch. 23, Ch. 39, Ch. 54, Ch. 61, Ch. 63, Ch. 46, Ch. 56, Ch. 58, Ch. 59, Ch. 64, Ch. 66
C-035	Trust as a system property	Trust, provenance, identity, authorization, policy, safety, and audit shall operate across every layer and intelligence object, not as a downstream compliance add-on.	Ch. 16, Ch. 18, Ch. 19, Ch. 20, Ch. 22, Ch. 24, Ch. 25, Ch. 27, Ch. 30, Ch. 32, Ch. 35, Ch. 37, Ch. 40, Ch. 41, Ch. 43, Ch. 46, Ch. 48, Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 65, Ch. 66, Ch. 68, Ch. 70, Ch. 56, Ch. 64
C-036	Non-repudiable accountability	Material access, transformations, agent actions, approvals, decisions, overrides, releases, and exports shall produce tamper-evident audit records attributable to identities and versions.	Ch. 3, Ch. 4, Ch. 7, Ch. 8, Ch. 9, Ch. 10, Ch. 11, Ch. 12, Ch. 14, Ch. 21, Ch. 23, Ch. 26, Ch. 29, Ch. 51, Ch. 53, Ch. 57, Ch. 63, Ch. 60, Ch. 61, Ch. 62
C-037	Customer data control	Customers retain governance over their data, models, twins, policies, retention, residency, encryption, exports, and deletion within the chosen deployment and contractual boundary.	Ch. 20, Ch. 37, Ch. 48, Ch. 8, Ch. 9, Ch. 10, Ch. 11, Ch. 12, Ch. 14, Ch. 60, Ch. 61, Ch. 62, Ch. 63
C-038	Open interoperability	The Eye shall integrate through governed APIs, events, connectors, common data formats, and export paths; core customer knowledge shall not be trapped behind proprietary-only access.	Ch. 5, Ch. 8, Ch. 12, Ch. 15, Ch. 16, Ch. 17, Ch. 22, Ch. 29, Ch. 30, Ch. 39, Ch. 59, Ch. 60, Ch. 61, Ch. 63, Ch. 70, Ch. 9, Ch. 10, Ch. 11, Ch. 14, Ch. 62
C-039	Zero-trust security	Identity, device, workload, agent, model, source, and data access shall be continuously authenticated, authorized, scoped, logged, and re-evaluated under least privilege.	Ch. 56, Ch. 58, Ch. 59, Ch. 64, Ch. 69, Ch. 8, Ch. 9, Ch. 10, Ch. 11, Ch. 12, Ch. 14, Ch. 60, Ch. 61, Ch. 62, Ch. 63
C-040	Privacy and minimization	Collection and use shall be purpose-bound, proportionate, policy-enforced, and minimized, with support for consent, legal basis, access, correction, retention, and deletion obligations.	Ch. 59, Ch. 60, Ch. 8, Ch. 9, Ch. 10, Ch. 11, Ch. 12, Ch. 14, Ch. 61, Ch. 62, Ch. 63
C-041	Sovereignty by design	Data residency, key ownership, network isolation, local inference, model choice, administrative control, and disconnected operation shall be supported where customer policy requires them.	Ch. 5, Ch. 12, Ch. 15, Ch. 16, Ch. 20, Ch. 22, Ch. 30, Ch. 32, Ch. 34, Ch. 37, Ch. 41, Ch. 59, Ch. 60, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 8, Ch. 9, Ch. 10, Ch. 11, Ch. 14
C-042	Deployment semantic parity	SaaS, Private Cloud, and On-Premise deployments shall preserve the same core semantics, governance model, APIs, artifacts, and intelligence lifecycle, with documented capability differences only where unavoidable.	Ch. 55, Ch. 56, Ch. 58, Ch. 62, Ch. 64, Ch. 65, Ch. 67, Ch. 68, Ch. 69, Ch. 71

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 6 ANCHORS
C-043	Resilient operation	The platform shall fail safely, degrade intelligibly, recover predictably, preserve evidence and audit continuity, and expose when data, models, or dependencies are unavailable or stale.	Ch. 6, Ch. 31, Ch. 33, Ch. 38, Ch. 48, Ch. 52, Ch. 55, Ch. 56, Ch. 64, Ch. 65, Ch. 67, Ch. 68, Ch. 69, Ch. 58, Ch. 62, Ch. 71
C-044	Governed learning loop	Learning from outcomes, feedback, overrides, and new evidence shall pass through evaluation, approval, release, monitoring, rollback, and tenant-isolation controls.	Ch. 45, Ch. 55, Ch. 56, Ch. 58, Ch. 62, Ch. 64, Ch. 65, Ch. 67, Ch. 68, Ch. 69, Ch. 71
C-045	Continuous evaluation	Models, agents, forecasts, extractors, ontologies, simulations, and recommendations shall be evaluated for accuracy, calibration, robustness, drift, bias, safety, cost, and operational fitness.	Ch. 3, Ch. 7, Ch. 8, Ch. 9, Ch. 10, Ch. 12, Ch. 15, Ch. 17, Ch. 30, Ch. 60, Ch. 61, Ch. 63, Ch. 70, Ch. 55, Ch. 56, Ch. 58, Ch. 62, Ch. 64, Ch. 65, Ch. 67, Ch. 68, Ch. 69, Ch. 71
C-046	Governed marketplaces	Scenario and Agent Marketplace packages shall be signed, versioned, inspectable, permission-scoped, licensed, evaluated, sandboxed, monitored, revocable, and traceable to publishers.	Ch. 9, Ch. 10, Ch. 11, Ch. 57, Ch. 55, Ch. 56, Ch. 58, Ch. 62, Ch. 64, Ch. 65, Ch. 67, Ch. 68, Ch. 69, Ch. 71
C-047	Experience integrity	The product experience shall expose context, uncertainty, provenance, decision state, and consequence at the point of use; critical meaning shall not be hidden behind ornamental visualization.	Ch. 10, Ch. 11, Ch. 55, Ch. 56, Ch. 58, Ch. 62, Ch. 64, Ch. 65, Ch. 67, Ch. 68, Ch. 69, Ch. 71
C-048	Role and accessibility clarity	Experiences shall be accessible and role-appropriate for executives, analysts, operators, domain experts, administrators, and governance teams without creating separate truths.	Ch. 17, Ch. 18, Ch. 21, Ch. 39, Ch. 40, Ch. 55, Ch. 67, Ch. 69, Ch. 56, Ch. 58, Ch. 62, Ch. 64, Ch. 65, Ch. 68, Ch. 71
C-049	Operational quality contract	Reliability, performance, scale, freshness, accuracy, calibration, explainability, reproducibility, security, accessibility, and cost shall be measured as product behavior, not assumed as implementation detail.	Ch. 4, Ch. 5, Ch. 7, Ch. 9, Ch. 13, Ch. 14, Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 28, Ch. 29, Ch. 30, Ch. 51, Ch. 52, Ch. 54, Ch. 56, Ch. 58, Ch. 64, Ch. 66, Ch. 71, Ch. 72, Ch. 55, Ch. 62, Ch. 65, Ch. 67, Ch. 68, Ch. 69
C-050	Responsible-use boundary	The Eye shall not autonomously execute decisions that materially determine life, liberty, legal status, access to essential services, employment, credit, or equivalent high-impact rights and interests.	Ch. 2, Ch. 6, Ch. 55, Ch. 65, Ch. 67, Ch. 68, Ch. 69, Ch. 70, Ch. 1, Ch. 40, Ch. 71, Ch. 72
C-051	Constitutional change control	Changes to mission, product category, human authority, principles, canonical layers, core semantics, or constitutional invariants require explicit amendment, impact analysis, ratification, and versioned publication.	Ch. 1, Ch. 2, Ch. 3, Ch. 40, Ch. 71, Ch. 72, Ch. 6, Ch. 67, Ch. 68
C-052	Documentation inheritance	Volumes 1 through 10 shall inherit this constitution, use its canonical terms, trace material requirements to invariant IDs, and identify rather than conceal any unresolved conflict.	Ch. 1, Ch. 2, Ch. 6, Ch. 72, Ch. 40, Ch. 67, Ch. 68, Ch. 71

O Infrastructure Architecture Decision Baseline

These decisions freeze the governing infrastructure architecture beneath Volumes 0, 3, 4, and 5. Implementations may change providers and technologies behind them but cannot treat a local exception as authority to change their semantics.

ADR	DECISION	BINDING CONSEQUENCE	IMPLEMENTATION OBLIGATION
IADR-001	One infrastructure contract across deployment profiles	Physical implementations vary; strategic semantics and conformance do not.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-002	Cell-oriented isolation and failure containment	Tenant and domain workloads execute in bounded intelligence cells.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-003	Management plane separated from customer data planes	Fleet control cannot become an unrestricted customer-data path.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-004	Workload identity precedes network trust	Network location alone never authorizes service access.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-005	Policy enforcement external to models and agents	AI behavior cannot expand infrastructure authority.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-006	State-class architecture precedes storage technology	Authority, consistency, locality, recovery, and lifecycle select the store.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-007	Canonical state separated from indexes and caches	Derived acceleration remains rebuildable and visibly stale.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-008	Durable execution for material multi-step work	Workflow and agent progress survives process and infrastructure failure.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-009	Explicit messaging delivery semantics	Retries, ordering, replay, and institutional effects remain testable.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-010	Accelerators are governed shared infrastructure	Artifact, tenancy, topology, capacity, and attestation are controlled.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-011	Runtime sandboxes enforce agent and connector reachability	Instructions cannot grant file, network, secret, or side-effect access.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-012	Registered egress instead of generic internet access	Destination, purpose, data, credentials, and receipts are explicit.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-013	Customer-held key and administrative options	Deployment profiles preserve customer control and sovereignty.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-014	Artifact integrity closes over software, models, firmware, and configuration	Production admission resolves the complete supply chain.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-015	Service levels include partial and degraded states	Process uptime cannot conceal unusable strategic service.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-016	Error budgets govern change	Reliability evidence can stop promotion and redirect capacity.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.

ADR	DECISION	BINDING CONSEQUENCE	IMPLEMENTATION OBLIGATION
IADR-017	Backups are untrusted until restored and reconciled	Automated recovery evidence is a production control.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-018	Disaster recovery preserves authority and sovereignty	Recovery is not only copying data into another location.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-019	Degraded modes are explicit product states	The platform never fabricates freshness, completeness, confidence, or approval.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-020	Infrastructure changes execute as durable controlled workflows	Plans, approvals, observations, stop, rollback, and closure are retained.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-021	Immutable dependency-closed releases	Floating dependencies and out-of-band production edits are non-conformant.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-022	Operational evidence is a first-class release and acceptance output	Exact configuration and period are bound to assurance.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-023	Disconnected and air-gapped operation is designed, not improvised	External dependencies, transfer, time, update, and reconciliation are explicit.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.
IADR-024	Humans retain incident command and return-to-service authority	Automation recommends and executes only pre-authorized reversible controls.	Implement through owned components, manifests, external controls, conformance evidence, immutable releases, monitoring, recovery, and audit.

P Infrastructure Glossary

These terms are canonical for deployment, compute, network, state, runtime, trust, reliability, recovery, delivery, operations, economics, and governance. Domain designs may add narrower terms but may not redefine the concepts below.

TERM	INFRASTRUCTURE ARCHITECTURE DEFINITION
Admission	A controlled decision that a workload, request, release, artifact, or state transition may enter a defined runtime scope.
Air-gapped deployment	A deployment profile with no direct connectivity to external networks and controlled physical or mediated transfer.
Artifact attestation	Evidence that a specific executable, model, firmware, or configuration artifact has approved identity and integrity.
Availability domain	A location or infrastructure grouping designed to fail independently within a declared dependency model.
Blast radius	The maximum actual or potential scope affected by a fault, compromise, change, or resource condition.
Cell	A bounded execution and state topology that contains tenant, domain, capacity, release, and failure scope.
Checkpoint	An immutable compatible recovery point for a workflow, agent, stream, job, simulation, or other durable execution.
Cold start	The latency and resource state required to activate a previously inactive runtime or artifact.
Confidential computing	Attested hardware and runtime protection intended to reduce exposure of data while it is processed.
Consistency model	The declared visibility, ordering, concurrency, and failure behavior of reads and writes.
Control plane	Infrastructure that governs identity, policy, configuration, registry, release, fleet, or operational desired state.
Control service	A service that issues, evaluates, or reconciles governing state rather than owning strategic product truth.
Customer control	Authority retained by a customer over identity, keys, networks, administrators, data, updates, operations, or acceptance.
Data plane	Infrastructure that processes customer inputs and strategic intelligence objects under externally enforced control.
Degraded mode	An explicit restricted operating state with declared triggers, capabilities, limitations, authority, and exit evidence.
Deployment profile	A certified physical and responsibility realization of the common product contract.
Desired state	The approved declarative infrastructure configuration a controller is required to realize.
Disconnected operation	Operation with denied or intermittent external connectivity and planned local continuity and reconciliation.
Disruption budget	The maximum permitted concurrent or temporal workload unavailability during controlled operations.
Egress	A mediated outbound transfer from a workload or trust zone to an external destination.
Error budget	The permitted measured unreliability inside an SLO window and the policy that governs its consumption.
Evidence vault	Protected storage for release, audit, acceptance, recovery, incident, and operational assurance evidence.
Failure domain	A bounded set of components expected to share a causal infrastructure failure.
Failover	Controlled transfer of service or state ownership to a certified alternate after failure.
Failback	Controlled return from a recovery target to the intended primary topology after reconciliation.
Fencing	A mechanism preventing an unsafe or stale member from performing authoritative operations.
Fleet	The complete set of managed physical and virtual infrastructure assets in a defined scope.
Headroom	Capacity reserved above measured demand for burst, failure, recovery, change, and uncertainty.
Immutable release	A dependency-closed configuration whose referenced artifacts and policies do not change after publication.

TERM	INFRASTRUCTURE ARCHITECTURE DEFINITION
Infrastructure component	A stable accountable infrastructure responsibility that may be realized by one or more technologies.
Infrastructure manifest	The versioned contract for a component's state, dependencies, controls, failure, recovery, and evidence.
Infrastructure-as-Code	Versioned declarative source used to plan, review, apply, and reconcile infrastructure state.
Ingress	A controlled inbound access path from a user, customer system, source, or external service.
Intelligence cell	A cell hosting bounded Strategic Intelligence Operating System workloads and their scoped dependencies.
Locality	The physical, cloud, regional, site, jurisdictional, or hardware location constraint on data or processing.
Management plane	The shared infrastructure that manages fleets, releases, configuration, policy distribution, and operations.
Noisy neighbor	A shared-resource workload whose consumption impairs another tenant, domain, or workload contract.
Observed state	Measured infrastructure state reconciled against approved desired state.
Operational reserve	Capacity and authority deliberately retained for failure, recovery, incidents, and change.
Parity	Preservation of product, authority, state, failure, evidence, replay, and lifecycle semantics across deployment profiles.
Placement	The controlled assignment of a workload or state replica to infrastructure satisfying all declared constraints.
Private Cloud	The deployment profile operated inside a customer-controlled cloud boundary using certified dependencies.
Quorum	The minimum set of members required to authorize a consistent distributed operation.
Recovery point objective	The maximum accepted loss of committed state measured from the recovery point.
Recovery time objective	The maximum accepted time to restore a declared recovery unit to its usable state.
Reconciliation	The controlled comparison and repair of desired, observed, authoritative, derived, or recovered state.
Residency	The approved physical or jurisdictional location of data, processing, keys, backups, telemetry, or support access.
Restore	Reconstruction of usable state from a recovery set followed by integrity and consumer reconciliation.
Runtime unit	A deployable or executable infrastructure pattern with explicit state, scaling, failure, and recovery behavior.
SaaS	The managed multi-region deployment profile operated by The Eye under a shared-responsibility model.
Semantic parity	Identity of strategic meaning and authority despite permitted physical or operational implementation differences.
Service-level indicator	A governed measure of a user-visible or control-critical service property.
Service-level objective	A target for an indicator over a declared scope, population, and time window.
Side effect	An operation that changes external, institutional, customer, or authoritative state beyond the executing runtime.
Site	A customer or provider physical location containing a bounded deployment topology.
Sovereignty	Control over where and under whose authority data, processing, keys, administration, providers, and support operate.
State class	A controlled category defining ownership, authority, consistency, durability, locality, protection, and recovery.
Supply-chain provenance	Traceable evidence from source and builder through dependencies to a released artifact.
Tenant	A customer isolation and administrative scope preserved across identity, policy, compute, network, state, keys, telemetry, and operations.
Topology	The declared graph of locations, cells, components, dependencies, routes, failure domains, and recovery relationships.
Trust anchor	A protected root used to establish cryptographic, identity, artifact, or execution trust.
Unit economics	Cost and resource consumption attributed to a controlled customer, capability, workflow, product, or workload unit.
Workload identity	A short-lived attested identity bound to a specific executable runtime and trust domain.

TERM	INFRASTRUCTURE ARCHITECTURE DEFINITION
Workload manifest	The deployable contract for identity, artifacts, resources, placement, connectivity, state, controls, failure, and lifecycle.
Zone	A network, availability, trust, or locality boundary with explicit membership and permitted flows.

Baseline closure

Volume 6 closes when every production capability can identify its deployment profile, customer and platform responsibility, topology, cell, runtime units, state classes, network zones, workload manifests, identities, policies, keys, artifacts, capacity, service levels, failure domains, degraded modes, backup, recovery, incident path, change path, release, cost, acceptance, evidence, exception, and retirement state.

END STATE

The Eye is deployable here as one enterprise Strategic Intelligence Operating System: physically adaptable, semantically invariant, sovereign by design, isolated by default, accelerated where justified, evidence-bound in operation, explicit in degradation, recoverable through tested state, and always subordinate to accountable human authority.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-004 · C-008 · C-035 · C-036 · C-041 · C-043 · C-049 · C-051 · C-052